

**INFORMATION AND COMMUNICATION
TECHNOLOGY**

S2

STUDENT BOOK

EXPERIMENTAL VERSION

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EXPERIMENTAL VERSION

FOREWORD

Dear Student,

Rwanda Basic Education Board is honored to present to you this ICT Student's Book for Senior Two which serves as a guide to competence-based teaching and learning to ensure consistency and coherence in the learning of ICT. The Rwandan educational philosophy is to ensure that you achieve full potential at every level of education which will prepare you to be well integrated in society and exploit employment opportunities.

The government of Rwanda emphasizes the importance of aligning teaching and learning materials with the syllabus to facilitate your learning process. Many factors influence what you learn, how well you learn and the competences you acquire. Those factors include the instructional materials available among others. In this book, special attention was paid to the activities that facilitate the learning process in which you can develop your ideas and make new discoveries during concrete activities carried out individually or with peers.

In competence-based curriculum, learning is considered as a process of active building by the learner where concepts are mainly introduced by an activity, a situation or a scenario that helps the learner to construct knowledge, develop skills and acquire positive attitudes and values.

For effective use of this textbook, your role is to:

- Work on given activities which lead to the development of knowledge and skills;
- Share relevant information with other learners through presentations, discussions, group work and other active learning techniques such as field visit, laboratory work, investigation, role play, case studies and research in the library, from the internet or from your community;
- Participate and take responsibility for your own learning;
- Draw conclusions based on the findings from the learning activities.

I wish to sincerely extend my appreciation to REB staff who organized the editing process of this book. Special gratitude goes to the lecturers, teachers, illustrators and designers who diligently worked to successful completion of this book. Any comment or contribution would be welcome for the improvement of this textbook for the next edition.



Dr. MBARUSHIMANA Nelson
Director General, REB

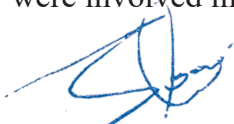
EXPERIMENTAL VERSION

ACKNOWLEDGEMENT

I wish to express my appreciation to all the people who played a major role in editing process of ICT Student's Book for Senior Two. It would not have been successful without their active participation.

Special thanks are given to those who gave their time to read and refine this textbook to meet the needs of competence-based curriculum. I owe gratitude to different Universities and secondary schools in Rwanda that allowed their staff to work with REB to edit this book. I therefore, wish to extend my sincere gratitude to lecturers, teachers, illustrators, designers and all other individuals whose efforts in one way or the other contributed to the completion of this edition.

Finally, my word of gratitude goes to the Rwanda Basic Education Board staff particularly those from Curriculum, Teaching and Learning Resources Department who were involved in the whole process of editorial work.



Joan MURUNGI,

Head of Department of Curriculum, Teaching and Learning Resources

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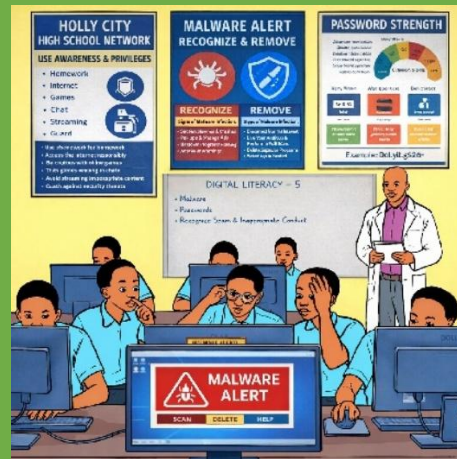
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UNIT 1: DATA PROTECTION



Key Unit Competency: Use computers safely and securely to ensure that data is protected

Learning Objectives

At the end of this unit you should be able to:

Knowledge and understanding

- Explain the types of computer user accounts, their privileges, and the importance of passwords.
- Demonstrate knowledge of harmful programs, their sources, and the role of antivirus
- Explain preventive and protective measures used to secure computers and user data.

Skills

- Create, manage, and secure user accounts using passwords and access privileges.
- Detect and handle harmful programs through scanning, installing, updating, and using antivirus tools.
- Apply preventive procedures to protect computers and removable devices from security threats.

Attitudes and values

- Exercise responsible and ethical behavior when managing personal accounts and data
- Show awareness and concern about computer security and vulnerability.
- Demonstrate cooperation and teamwork when safely and ethically sharing devices or resources

1.0. Introduction

Data refers to a collection of numbers, characters, and symbols which are held in a computer. Data are usually represented in units of eight bits, which are called bytes.

Data protection is the process of safeguarding data from corruption, loss, or unauthorized access. All forms of data are considered assets for an organization or an institution. In this unit you will learn what you must do to keep your data safe.

Introductory activity

One day in the computer lab, David found a link to download a free racing game that normally costs money. He saved the file on an old USB. When he tried to open the game, it did not work, so he ignored it and went home. The next day the computers started showing strange pop-ups, became very slow, and displayed unknown programs. Even the teacher's browser changed by itself. Teachers became worried that important school data was no longer safe.

Later, David confesses what he had done, and they discovered that the file he downloaded had brought a harmful program into the computer.

- 1) What action did David take that caused the computers in the lab to get infected?
- 2) What signs showed that the computers had been infected by a virus?
- 3) Why were the teachers worried about the safety of the school's data after the computers started acting strangely?

1.1. Protecting the computer

Activity 1.1

Irasubiza recently purchased a new computer for work. At the workplace, his colleagues sometimes need to use the computer to complete shared tasks and access common files. Irasubiza wants to make sure that his confidential documents remain private while still allowing authorized colleagues to use the computer when necessary.

- 1) What Irasubiza should do to protect his documents while still allowing his colleagues to use the computer
- 2) Explain other measures that can be done to protect a computer.

In any organization, it is always necessary to think about the security of computers, as well as the data and information stored in them. Security should be ensured for both computer software and hardware.

It is essential to take measures to protect the computer from threats such as unauthorized access, trojans, virus, toolkits and much more. People who gain unauthorized access to a computer system pose a great threat to any institution.

A computer system can be protected from security threats by assigning every authorized person user privileges or passwords in order to access it. In this case, every authorized user would have to log into the system. Every user's history of activities can be tracked if

the system is carefully inspected.

1.1.1 User privilege

User privilege is the permission given to a user to perform an action. It is used to regulate who can view or use the resources in a computer. This ensures that personal data or information in the computer is secured. Every user of the computer can have an account created for their use, each account with different privileges.

There are also other privileges namely system privilege and object privilege

- **System privileges:** These are rights provided to users to allow them to create, alter or drop data.
- **Object privileges:** These allow user to execute insert, update or delete data

In order to access a computer, the user requires some form of identification and authentication.

Identification refers to the process of a proving one's identity while authentication refers to the process of verifying the identity of a user. For the identification process, the computer system first verifies that the user has a valid user account. The process then requires a confirmation of the validity by use of a password if one had been set.

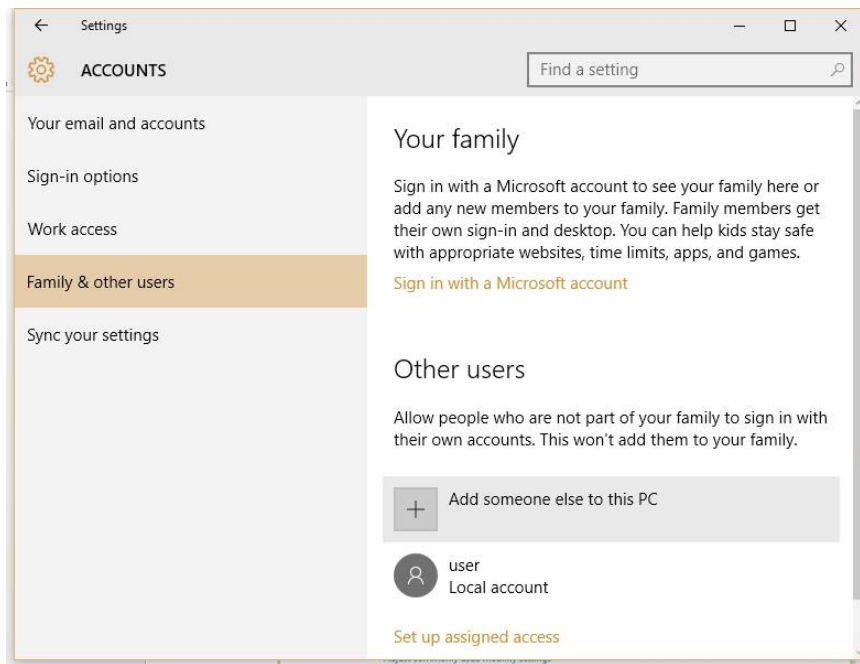
a) Types of accounts

An account can either be an **Administrator** or **Standard user (Guest)** in Windows operating system.

- **Administrator:** This is a user that has superior privileges. Administrators can make changes to the accounts of other users such as changing the security settings, installing software and hardware, as well as accessing and modifying all files on the computer.
- **Standard user (Guest):** This is a user account that allows people to have temporary or limited access to a computer. People with this type of account can use most of the software installed. They can also change the system settings that do not affect other users or the security of the computer. Standard users cannot, however install software or hardware or create passwords.

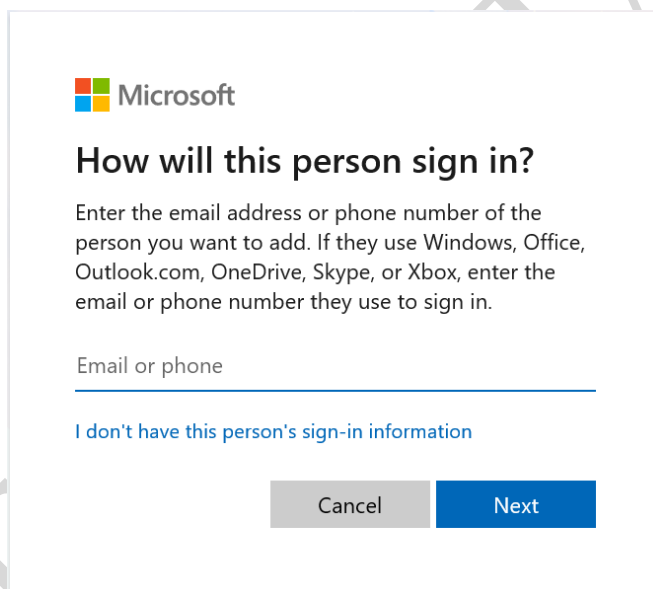
b) Procedure for creating a user account

- Click the **Start** button then click **All Apps** from the Start menu.
- Click on **Settings** then select **Accounts** icon. A dialog box appears.
- Click on **Family & other users** in the left pane.



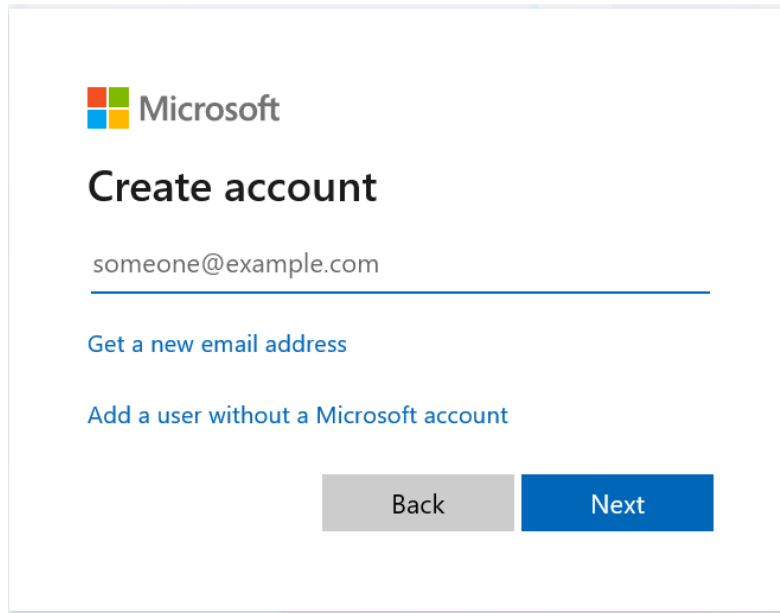
Picture 1. 1: Account windows

- Select **Add someone else to this PC** in the right pane. Then you get the window below:



Picture 1. 2: process of signing in

- Click on **I don't have this person's sign-in information**



The image shows a Microsoft account creation form. At the top is the Microsoft logo. Below it is the heading "Create account". There is a text input field containing "someone@example.com". Below the input field are two links: "Get a new email address" and "Add a user without a Microsoft account". At the bottom are two buttons: "Back" (disabled) and "Next" (active).

Microsoft

Create account

someone@example.com

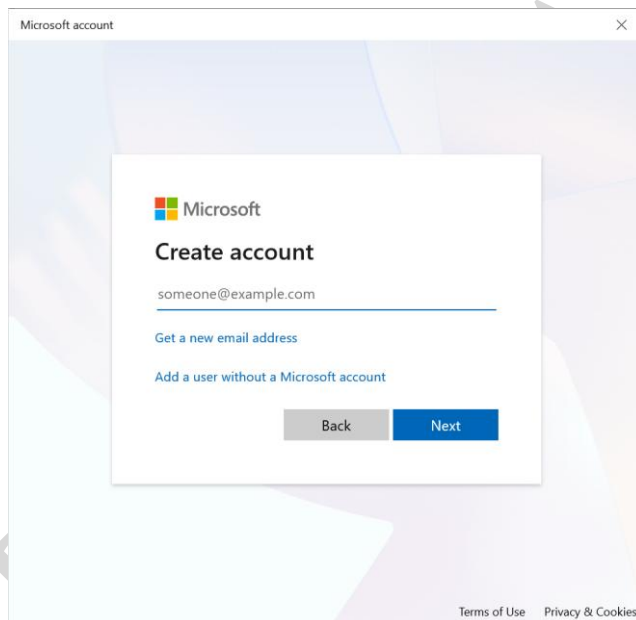
[Get a new email address](#)

[Add a user without a Microsoft account](#)

Back Next

Picture 1. 3: Account creation without Microsoft

- Click on Add user without a Microsoft account



The image shows the same Microsoft account creation form as in Picture 1.3, but it is displayed within a window titled "Microsoft account". The form contains the Microsoft logo, the heading "Create account", the email address "someone@example.com", the links "Get a new email address" and "Add a user without a Microsoft account", and the "Back" and "Next" buttons. At the bottom right of the window, there are links for "Terms of Use" and "Privacy & Cookies".

Microsoft account

Microsoft

Create account

someone@example.com

[Get a new email address](#)

[Add a user without a Microsoft account](#)

Back Next

[Terms of Use](#) [Privacy & Cookies](#)

Picture 1. 4: Account creation without Microsoft

- Type the details of the user in the window displayed then click the Next button

Microsoft account

Create a user for this PC

If you want to use a password, choose something that will be easy for you to remember but hard for others to guess.

Who's going to use this PC?

User name

Enter your user name.

Make it secure.

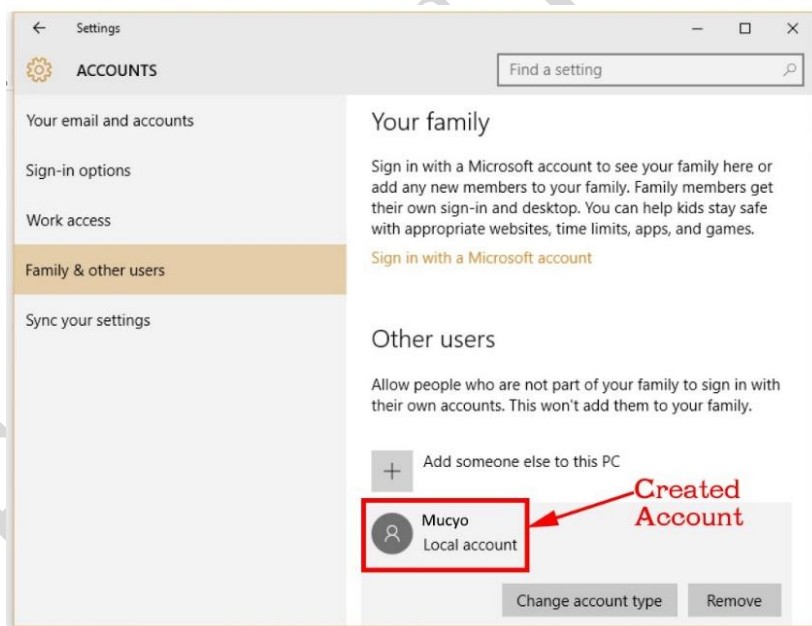
Enter password

Re-enter password

Next Back

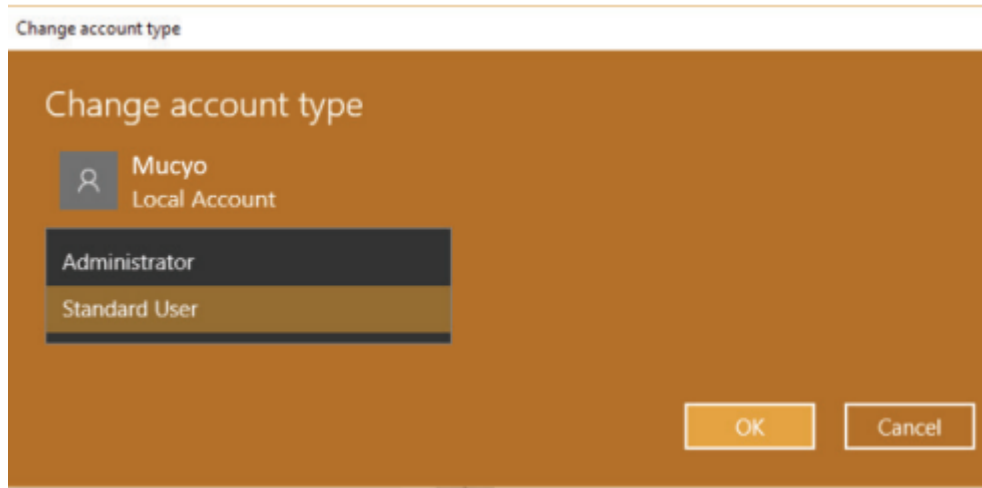
Picture 1. 5: User account window creation

- The account is automatically created with a default account type of Standard.



Picture 1. 6. The Manage Account Dialog Box

- Click on the box written Standard User. Select Administrator. Click OK



Picture 1. 7. The change account type dialog box

c) Switch between users without logging off

If a computer has more than one user account, it is possible to change from one user account to another without logging off the active user account. Here below are different methods used to switch from users:

Method 1: Using the Start menu

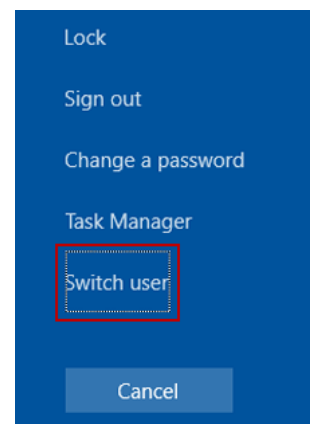
- Click the **Start Menu** and then click on **Switch User** command.
- Select the desired user account.

Method 2: Using the keyboard shortcut

- Press **Ctrl+Alt+Delete** and then click on **Switch User** command.
- Select the desired user account

Method 3: Using the Shut Down dialogue box

- Open the **Shut Down** windows dialog by pressing **Alt+F4**.
- Click the down arrow. Choose **Switch user** in the list and click **OK**.



Picture 1. 8. : Switching user dialog box

Method 4: Using the User Icon

- Click the **Start** button on the desktop.
- Click the **User** icon on the top-left corner in the **Start Menu**.
- Select the desired user account such as Guest from the menu displayed.



Picture 1. 9.Switching the user account icon

1.1.2 Password

A **password** is a secret code used to prevent unauthorised access to a computer hardware, software, as well as the data and information stored in the computer. A password provides the user with authority to access information in a computer.

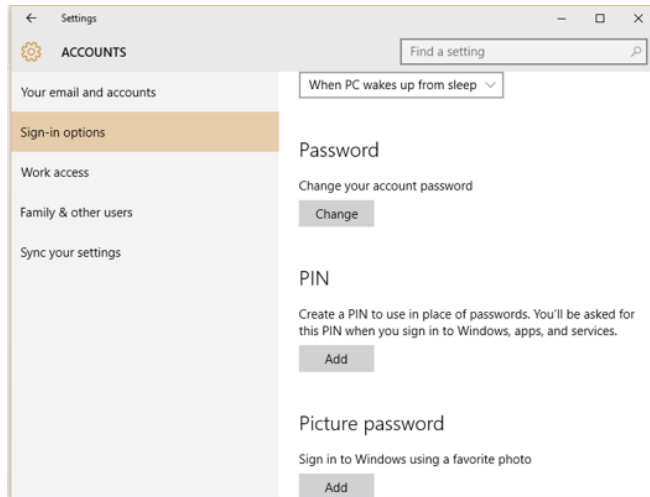
A good password should be strong and easy to remember; but it should be difficult to guess

The following are characteristics of a strong password:

- It should have at least eight (8) characters.
- It should contain a combination of uppercase and lowercase letters. It should also have numbers and symbols if they are allowed. An example of a strong password is: Nkosi%88#
- It should be changed frequently, for example after every one month.

a) Procedure for creating a password using windows

- Click the **Start** button. Select **All Apps** from the **Start menu**.
- Click on **Settings** then select **Accounts icon**. A dialog box appears.
- Click on Sign-in options in the left pane.
- Click on **Change** button in the right pane under **Password** as shown below.



Picture 1. 10: Managing sign-in options

b) Procedure for changing the password

- Type the new password and re-type it again in the window provided as shown below then click **Next**.

Picture 1. 11: Make changes to your Password

- Enter a new password in the **New Password** box. Re-type the password in the Reenter password box.
- Type a hint in the **Password hint** box. The hint is displayed when you cannot remember the password.
- Click the **Next command**. A final dialog box is displayed. Click on **Finish** as shown in the picture below.

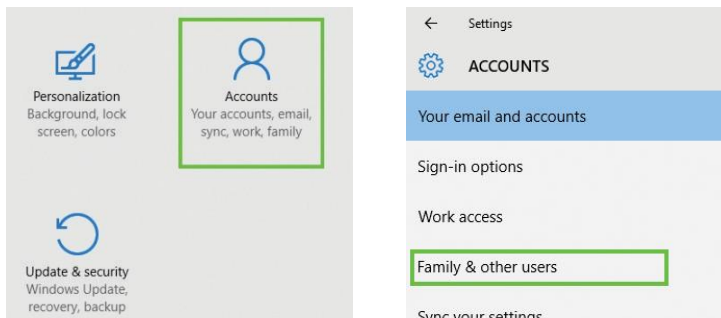
Picture 1. 12: The final step in changing the password

c) Why do you need to change your password?

- It is advisable to change one's password every few months for security reasons. Sometimes you may forget to log out of your account in a computer that is used by many people. In this case, someone may find your account open and gain access to your data.
- If someone gets to know your username and password, he or she could log into your computer and change, delete, or corrupt your files.
- If someone gets to know your username and password, he or she could log into your computer and share your files with other users.

d) Setting user privileges

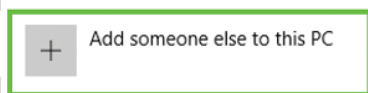
- Click the **Start Menu**. Select **Settings**. Click **Accounts**.
- Select **Family & other users**. Click “**Add someone else to this PC.**”
- Select “**I don't have this person's sign-in information.**” Select “**Add a user without a Microsoft account.**” Enter a username.
- Type the account's password twice, enter a clue, and select **Next**.



Picture 1. 13. Setting user privileges

Other users

Allow people who are not part of your family to sign in with their own accounts. This won't add them to your family.



Picture 1. 14. Add someone else to this PC" (Windows Other Users settings)

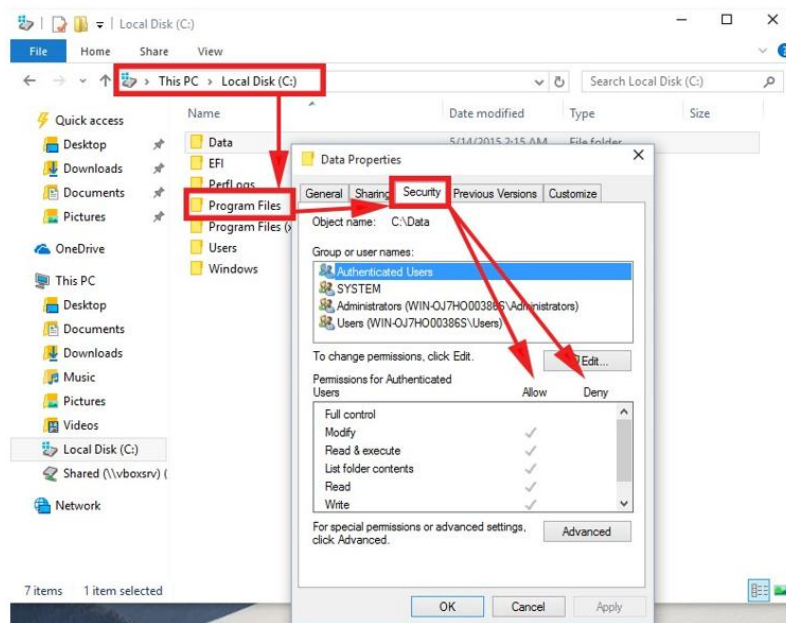
e) Allow and block specific programs

When an account is created in Windows 10, the user privileges are automatically defined with the type of account that has been created. It could either be a standard or an administrator's account.

The administrator account has more privileges than the standard account. The administrator is able to allow and block specific programs.

To do this, follow these steps:

- Double click on **This PC** icon on the desktop.
- Open the drive where the application files are located, for example, C:
- Open the **Program files** folder.
- Right-click on the specific application file and select **Properties**.
- Click on the **Security** tab of the window.
- Click on the **User profile** name to allow or deny access to the user under Group or User names.
- Click on the **Edit button** and give permissions to the user profile accordingly
- Click **Apply** then **OK**
- Close the window and check if the changes are effective.



Picture 1. 15. Allowing and blocking users from accessing specific programs

Application activity 1.1

Part A: Fill in the missing words with the correct answers.

- 1) _____ is the permission given to a computer user to perform an action.
- 2) is a secret code used to prevent unauthorized access to a computer, data, and software.
- 3) One who gains unauthorized access to a computer system is a _____.
- 4) A strong password should have at leastcharacters.

Part B: Answer these questions.

- 1) Define the term “data protection”.
- 2) Explain the meaning of these terms:

(a) User privileges

(b) Password

3) Identify the two types of password accounts.

4) Outline the procedure for setting up any one of the password accounts named in question 3 above.

1.2 Harmful programs

Activity 1.2

During ICT class, a student downloaded a program. After that, the computer started acting strangely unknown apps showed up, some files disappeared, and it became very slow. Ms. Uwase explained that this was caused by malware, a program that can damage a computer or access it without permission.

1) What signs show that the computer might have been infected by a harmful program?

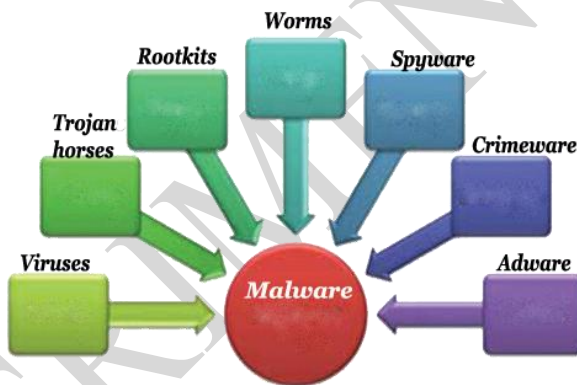
2) Explain in your own words what a virus.

1.2.1 Definition

The term **harmful program** refers to software that interferes or poses security problems to the computer hardware, software, data, and information. It is also known as **malware (malicious software)**, which refers to software specifically designed to gain access or damage a computer without the user's knowledge

1.2.2 Categories of harmful programs

Harmful programs are in different categories depending on their effect on a computer. Here below are in the image the categories of harmful programs:



Picture 1. 16. Categories of harmful programs

Malware (malicious software) is a program that interferes or poses security problems to the computer hardware, software, data, and information. It is also known as **malware (malicious software)**, which refers to software specifically designed to gain access or damage a computer without the user's knowledge. The categories of harmful programs include the following: viruses, Trojan horses, Rootkits, Worms, Spyware, Crimeware, and Adware.

a) Viruses

- A **virus** is a program that makes a copy of itself over and over again in a computer. This is called **replicating**.

- A virus is loaded into a computer without the user's knowledge. It can spread from one computer to another.
- It inserts copies of itself into other computer programs, data files, or the boot sector of the hard disk thereby erasing or altering computer files, filling computer memory, or making the computer fail to function.

Symptoms of a computer infected by viruses

- A computer system takes longer than usual to load programs and carry out other operations.
- The computer unexpectedly restarts without instruction.
- The file size in the computer becomes unusually large.
- The size of Random-Access Memory (RAM) may become smaller.

b) Worms

A worm is a malicious program that replicates itself and finally occupies the system memory of the storage media. Worms use the network to duplicate.

A worm does not alter files. It resides in the active memory and duplicates itself. They are noticed only when they occupy a lot of space in memory, thereby slowing or halting tasks during processing.

c) Trojan Horses

A Trojan horse is a harmful program that disguises itself as genuine software. An unsuspecting user is tricked into downloading and running it on his or her computer system. After it is activated, it can irritate the user by popping up windows or changing the desktop, deleting files, stealing data, or activating and spreading other malware.

Trojan horses do not replicate like worms and viruses; instead they spread through user interaction such as opening an e-mail attachment or downloading and running a file from the Internet.

d) Rootkit

A rootkit is a collection of malicious programs designed to a computer or areas of its software and hide there. It may also hide the existence of other malicious programs. A rootkit hides the fact that an operating system of a computer has been compromised. It does so sometimes by replacing important executable files in the system.

A rootkit is activated even before an operating system boots up therefore it is difficult to detect.

Rootkits themselves are not harmful. They are used to hide viruses, worms and other harmful programs by disguising them as necessary files that your anti-virus will overlook.

e) Spyware

Spyware is a program that monitors or tracks personal information or internet activity and sends that information to someone else.

Spyware can also gather information about e-mail addresses and even passwords or credit card numbers by recording the key strokes entered into a web form.

Spyware relays the collected information to the author who uses it for marketing or advertising purposes. Some malicious authors sell the gathered information to another party.

- f) **Crimeware:** This is a type of harmful software that is designed to carry out or facilitate illegal online activities.
- g) **Adware:** It is a software that automatically displays or downloads advertising material (often unwanted) when the user is browsing the internet.

Application activity 1.2

Part A. Complete the sentences using the words “**malware, replicate (copy), genuine, advertisements**”

- 1) Another name for harmful programs is _____.
- 2) A computer virus can _____ itself many times in a computer.
- 3) A Trojan horse pretends to be _____ software.
- 4) Adware displays unwanted _____ on a computer.

Part B: Answer the following questions.

- 5) What is a harmful program?
- 6) Mention two signs that show a computer has been infected by malware.
- 7) Name any two categories of harmful programs.

1.3 Source of viruses

Activity 1.3

The school computer lab was infected by a virus after a student downloaded a game from the internet and used a USB flash drive. Computers became slow, unknown programs ran in the background, and browser settings changed.

- 1) What user account privileges have allowed this virus to spread?
- 2) What type of harmful programs could have caused this?
- 3) Where did the virus come from?

There are many sources of computer viruses. They include the **internet** and **storage devices or media**.

1.3.1 Internet

The following are some sources of computer malware:

- **E-mail attachments:** A virus can come as a file attached to an e-mail message. When the file is opened, the virus is activated. This causes damage such as destroying files on the hard disk. It may also get forwarded to every contact listed in the address book.
- **File sharing:** Internet users are able to share or exchange files that are in their computers. This makes it possible for such computers to be vulnerable to virus infection risks because the shared files may be infected.
- **Malicious websites:** These are sites that trick the user into giving away information or downloading a virus. They may have a code that finds and exploits weak security points in a computer. If such a site finds a security weakness, it downloads a virus to the computer and installs it without the user knowing.

1.3.2 Storage devices or media

A common way of spreading viruses among computers is the use of removable storage media. When copying data, the data together with the malware are transferred to the storage medium. When the medium is connected to another computer, the malware is automatically copied to the next computer and this last one become attacked. Examples of such devices are flash disks, memory card, external hard drives.

Application Activity 1.3

- 1) Which of the following is an example of a virus spreading through the internet?
 - a) Using a printer
 - b) Opening an infected e-mail attachment
 - c) Turning off the computer
 - d) Installing a legal software using a CD
- 2) How can storage devices spread computer viruses?
 - a) By increasing computer speed
 - b) By automatically deleting files
 - c) By transferring infected files from one computer to another
 - d) By cleaning the hard disk
- 3) Explain how an e-mail attachment can spread a computer virus.
- 4) Describe how removable storage devices such as flash disks can transmit viruses from one computer to another.
- 5) What is the meaning of the term harmful program?

1.4 Measures to prevent harmful programs

Activity 1.4

David downloaded a free game from an unknown website. After installing it, the computer started showing pop-ups and unknown apps appeared.

- 1) What mistake did David make?
- 2) What kind of harmful program could this be?
- 3) How could he avoid this problem next time?

The following measures can be used to prevent harmful programs:

a) **Install an antivirus program**

An antivirus refers to computer software that is used to prevent, detect, and remove malicious software. It is also known as **anti-malware** software. In a computer installed with an antivirus program, always do the following:

- Keep the antivirus up-to-date at all times. This ensures that the antivirus is able to detect and remove any emerging viruses.
- Scan and detect viruses. To scan means to go through all the parts of the computer system carefully in order to detect the presence of viruses.
- **Quarantine** infected files and programs to prevent the virus from spreading.

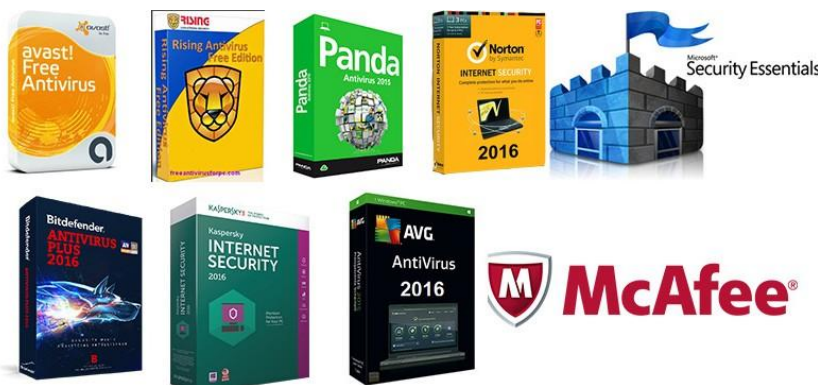
Quarantined files are no longer capable of infecting the computer system.

- Delete or remove the files infected with viruses.
- Notify the user in case a virus is detected.

Antivirus programs can be purchased from seller outlets directly or downloaded from the Internet.

It is advisable to install the latest antivirus software in the computers and regularly have it upgraded so that it is able to deal with any new viruses.

Examples of antivirus programs include Microsoft Essentials, Kaspersky, Norton, Bitdefender, Windows Defender, Avast, Panda, Macfee, and Rising among others.



Picture 1. 17. Examples of antivirus software

b) Scanning storage

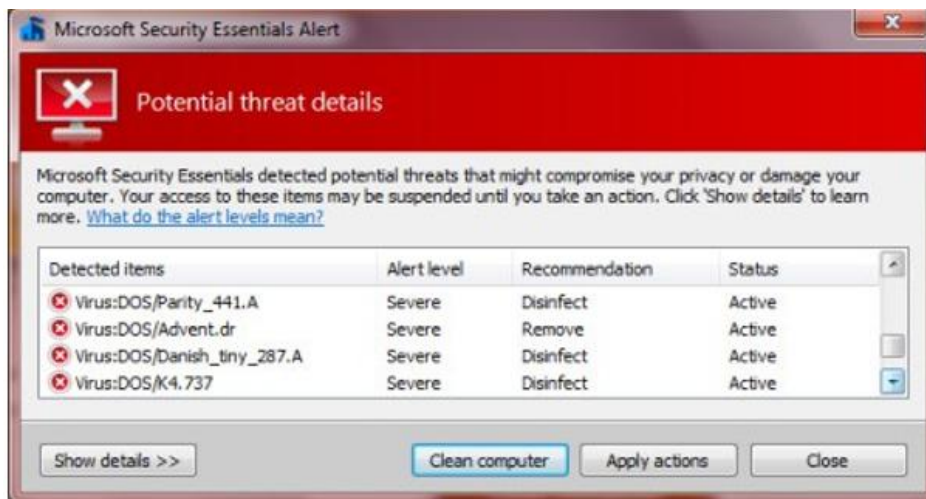
Scanning storage devices such as flash drives hard drives and memory cards is essential for keeping computers safe from harmful programs such as viruses, worms, trojans, and spyware.

Importance of scanning storages:

- It prevents the spread of malware from one computer to another.
- It protects important files from corruption or deletion.
- It helps keep the computer running smoothly and safely.
- It ensures data confidentiality by preventing unauthorized access.

Here below is the procedure to scan any removable device connected to the computer:

- Open **This PC** then right click on the device you want to scan
- Choose the antivirus you want to use during scanning
- Click on scan option from the drop-down list
- Choose the antivirus you want to use during scanning



Picture 1.18: Scanning the computer for malicious software

Note: The use of **cloud-based storage service** to store and share files and folders is a safe way to avoid the spread of computer viruses through secondary storage devices. Cloud-based services refer to data storage services where data are stored, maintained, managed, and backed up over the internet. Such services include Onedrive, Dropbox, Google drive.

c) **Keep the antivirus software updated**

Companies that sell antivirus programs keep updating them. It is important to always keep the antivirus program installed updated at all times. Microsoft always releases security updates that can help protect one's computer.

d) **Install anti-spyware software**

This is a type of program designed to prevent and detect unwanted spyware program installations. It also removes those programs that have been installed.

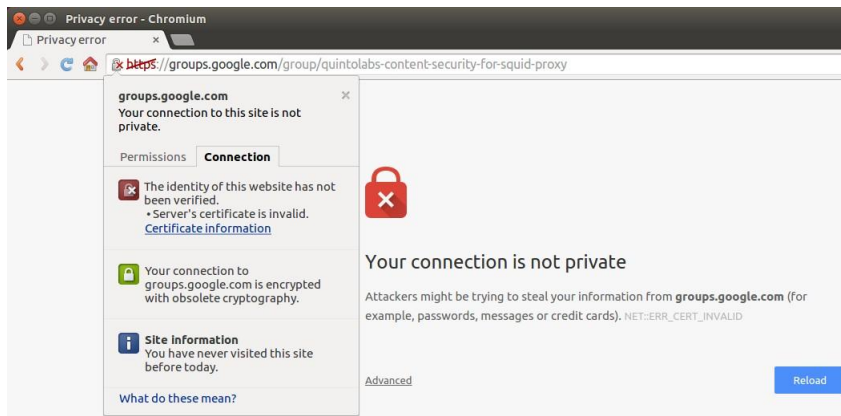
e) **Scan emails for viruses**

Do not open email attachments unless you are sure of the source. Since computer viruses are often contained in email attachments, scan the attachments for viruses before opening them. Most email services, for example, Google and Yahoo automatically scan emails for viruses before downloading any attachments.

f) **Avoiding non-secure sites**

The following are some control measures to be observed when dealing with non-secure websites:

- Block sites that you do not want the computer to connect to through the browser. Examples are sites that may contain indecent information and graphics.
- Google Chrome tells you if the site you're trying to visit is dangerous or deceptive.



Picture 1.19: Warning on dangerous sites

Application activity 1.4

Part A: Fill in the missing words to complete the sentences:

- 1) An antivirus software _____ infected files to prevent virus spread.
- 2) Computer viruses are mainly spread through secondary storage devices.
_____ and _____ are examples of secondary storage devices.
- 3) A _____ is a set of hardware and/or software equipment that is used to prevent hackers from invading private networks.
- 4) List three examples of antivirus programs
- 5) Why data protection is necessary in an organisation.
- 6) State three ways in which an antivirus software can protect a computer system from virus attacks.
- 7) Differentiate between spyware and adware.

End unit assessment.

PART A.

- 1) A strong password should:
 - a) Be easy to remember and only letters
 - b) Contain at least eight characters, uppercase and lowercase letters, numbers, and symbols
 - c) Only include numbers
 - d) Be the same for all users
- 2) Which of the following is a program that replicates itself to infect other files?
 - a) Trojan horse
 - b) Virus
 - c) Adware
 - d) Spyware
- 3) A Trojan horse:
 - a) Replicates automatically like a virus
 - b) Disguises itself as genuine software to trick users
 - c) Occupies memory without user interaction
 - d) Displays unwanted advertisements only

4) Which type of malware monitors a user's activity and sends information to someone else?

a) Worm b) Spyware c) Rootkit d) Adware

5) Adware is primarily:

a) Used to steal passwords b) Used to copy itself across systems
d) c) Used to display unwanted advertisements Used to delete files permanently

Part B

1) Differentiate between a Standard account and an Administrator account.

2) Identify and explain two types of harmful programs.

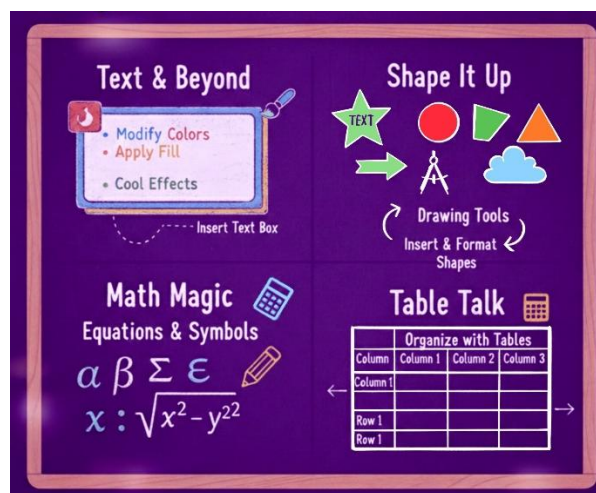
3) List four sources of computer viruses.

4) Why is data backup important?

5) What is the role of a firewall

6) Explain why students should avoid downloading software from unknown websites

UNIT 2: OBJECTS IN A DOCUMENT



Key Unit Competency:

Manipulate shapes, symbols, equations, pictures, tables, and charts into a document properly.

Learning objectives

At the end of this unit you should be able to:

Knowledge and Understanding:

- Understand what objects are in a document and how to insert them.
- Know the features used to modify objects, tables, and charts.
- Understand how to present data using different types of charts and how to format them properly.

Skills:

- Illustrate information using various objects such as shapes, tables, equations, and charts.
- Use drawing and formatting tools to improve the appearance of objects and charts.
- Create and format tables, equations, and charts within a document.

Attitudes and values:

- Appreciate the value of using objects to enhance document presentation.
- Develop creativity and professionalism when designing documents.
- Demonstrate responsibility and ethical use of software tools.

2.0. Introduction

This unit deals with insertion and manipulation of graphics in a Word document. The term graphics refers to anything in the form of images or pictures. The graphical objects could be in form of textboxes, tables, pictures, equations, symbols, Word Art and tables.

Introductory activity

Imagine a big technology company announces a competition at your school. Your challenge is to create the most attractive and convincing one-page digital profile

about a cause you care about such as protecting the environment, supporting a school ICT club, or promoting sports.

To win, your page must catch people's attention at first sight and clearly communicate your message.

- 1) Which objects in Word would you use to make your page more attractive?
- 2) How would you organize your information so that people understand your message quickly and easily?

2.1 Insert and format text boxes

Activity 2.1

The school is preparing for the "Annual Science Fair." The headteacher asks the ICT club to create a digital poster. One student, Keza, has a beautiful picture of a robot in the center of the page, but she wants to add a "Did You Know?" fun fact in the top-right corner and a "Date & Venue" label slanted across the bottom. Every time she types, the text moves her picture or stays stuck at the top of the page.

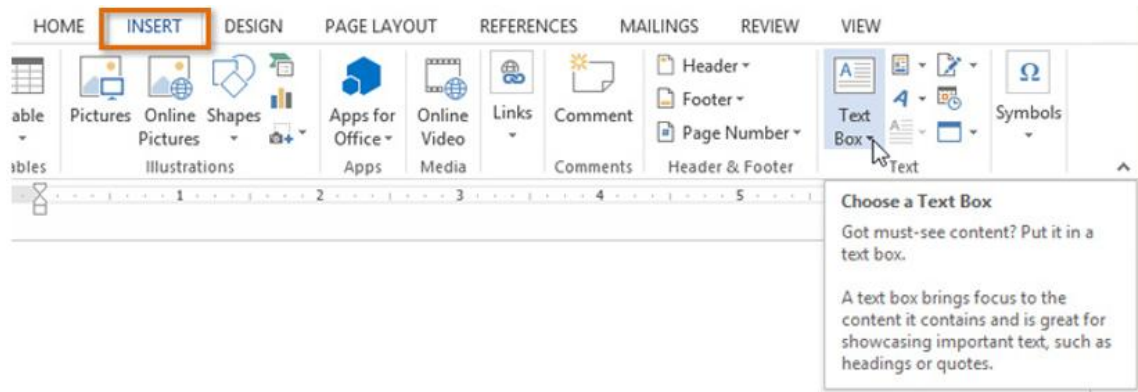
- 1) Why is Keza having trouble moving her text?
- 2) If you were Keza, how would you solve the problem of placing text anywhere on the page without disturbing the rest of the layout?

2.1.1. Inserting a text box in a document

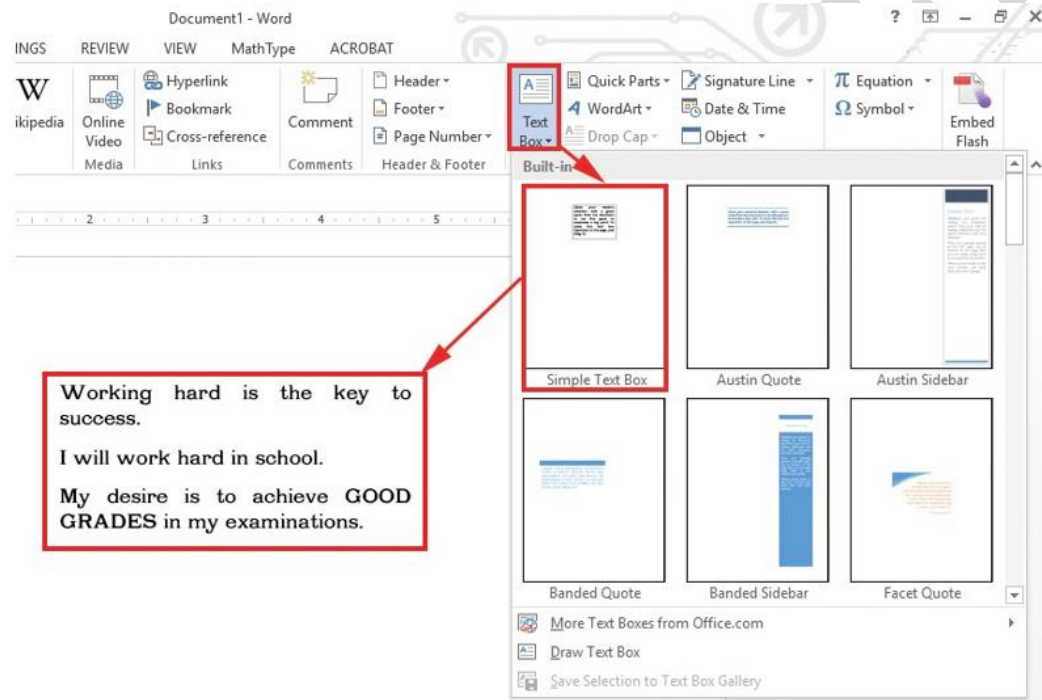
The Word Processing program allows the user to insert a text box in a document. A **text box** also known as a **text field** is a graphic that is intended to enable the user to input textual information to be used by the program.

Method 1: Using the Insert Tab to insert a text box in a document

- Click on **Insert tab** from the menu bar, select **Text box** from the **Text group**. A drop-down menu appears as shown in the picture.
- Click on the desired text box in the Built-In section or click on **Draw text box** to insert a customized one.
- If **Draw text box** option is selected, the cursor changes to a plus sign in the document. Click and drag it in the desired location. Otherwise, begin typing in the box and the content in it will automatically be deleted.
- Type the desired text.



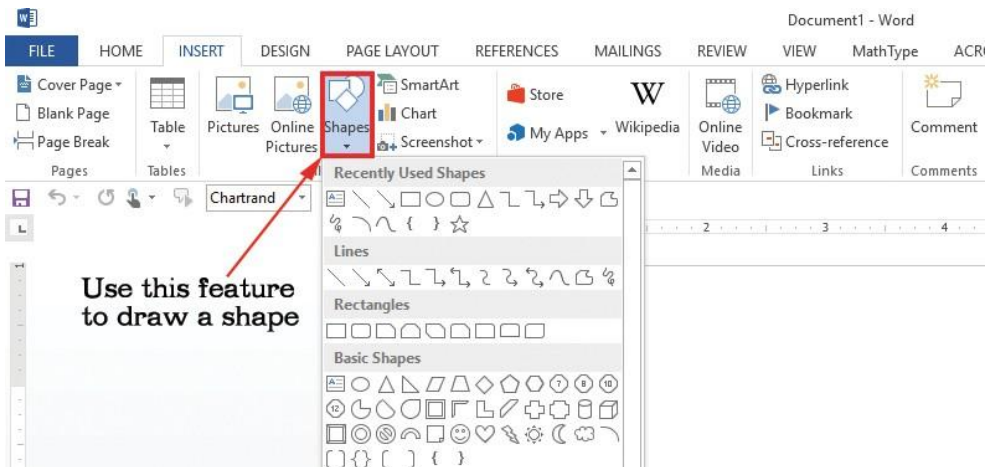
Picture 2. 1. Selecting Text box from the insert menu



Picture 2. 2. Inserting a simple text box

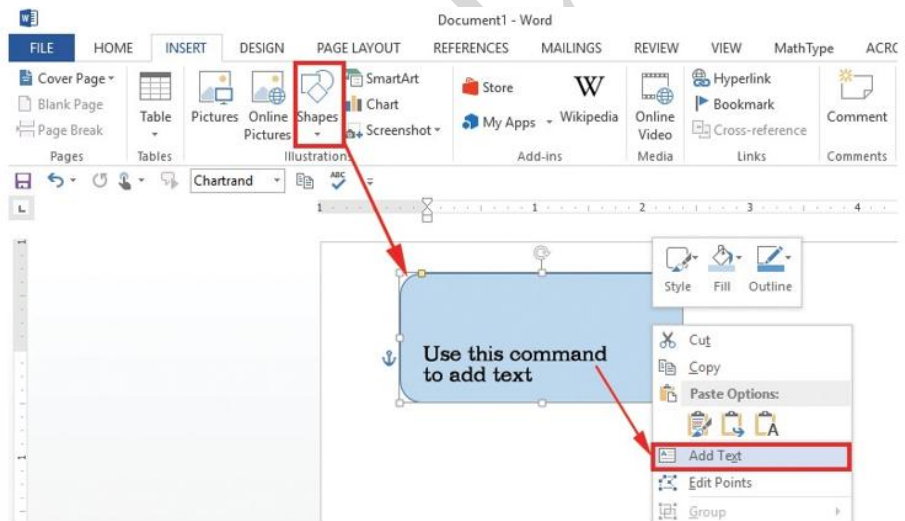
Method 2: Using the Shapes Feature to insert a text box in a document

- Click on **Insert** tab from the menu bar.
- Select **Shapes** from the Illustrations group.
- A drop down menu appears as shown in the picture below.

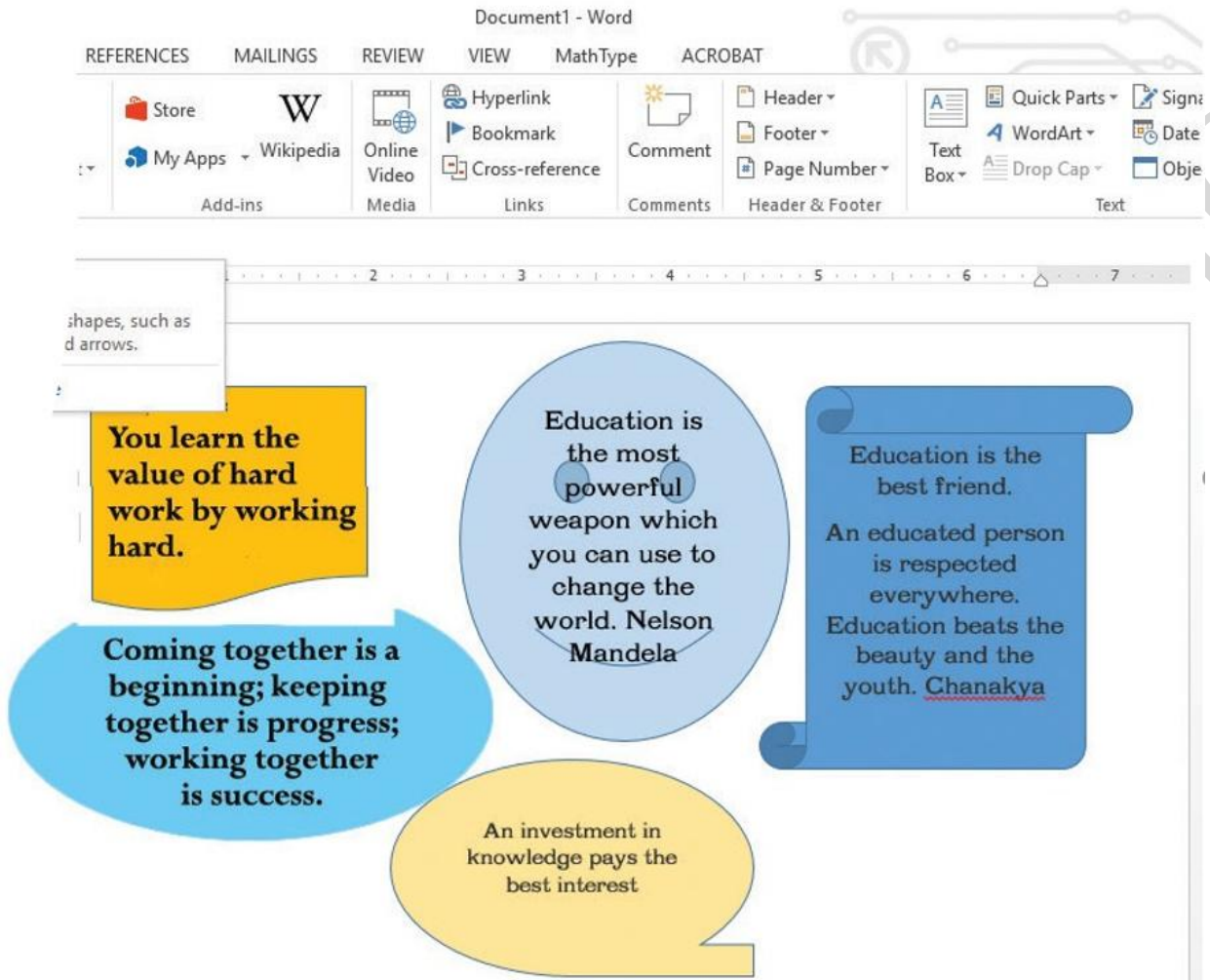


Picture 2. 3. Using the shapes feature to draw a text box

- Click on the desired text box in the Built-In section use it to draw the desired shape.
- Right-click the drawn shape and choose **Add Text** from the pop-up menu.



Picture 2. 4. Using the shapes feature to draw a text box



Picture 2. 5. Using the shapes feature to draw a text box

2.1.2. Formatting a text box

When a text box is highlighted in a document, the **Drawing Tools** tab which comes along with the **Format** tab in the menu bar is activated.

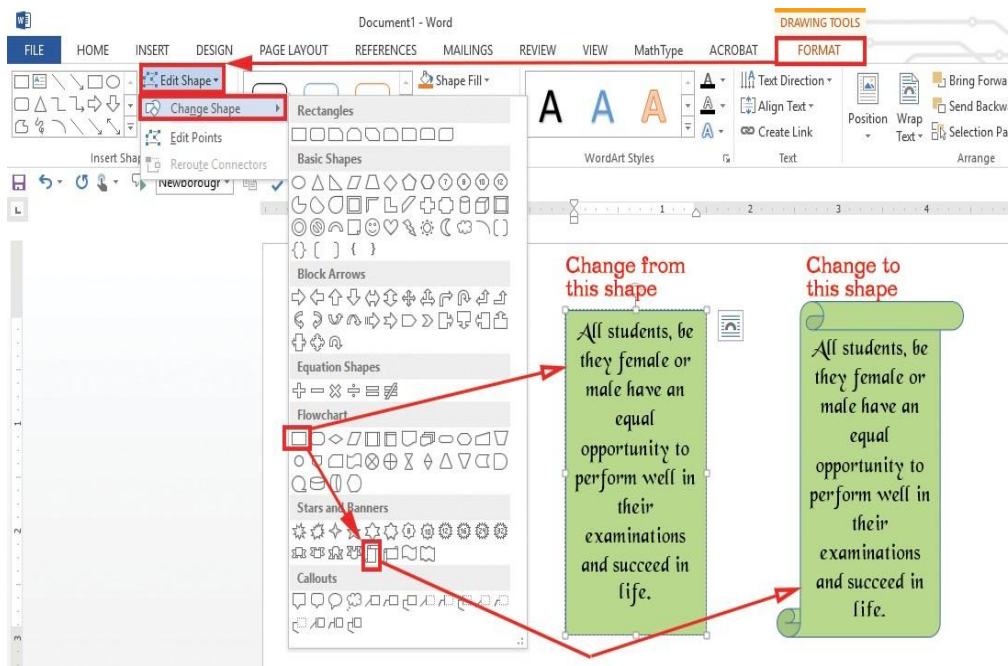
There are various formatting features available some of which include editing the text box shape, creating a link and changing text direction.

a) Editing the shape of a text box

The editing feature is used to change the shape of a drawing. To use this feature, do the following:

- Click on the text box to select it.
- Click on **Format** tab under Drawing Tools tab in the menu bar.
- Select **Edit Shape** under **Insert Shapes** group. A drop-down menu appears.
- To change the shape of the text box, select the desired shape from **Change Shape** option as shown in Picture below.
- To edit points of a text box, click on the **Edit Points** option. The points are selected

and the shape changes to red colour.



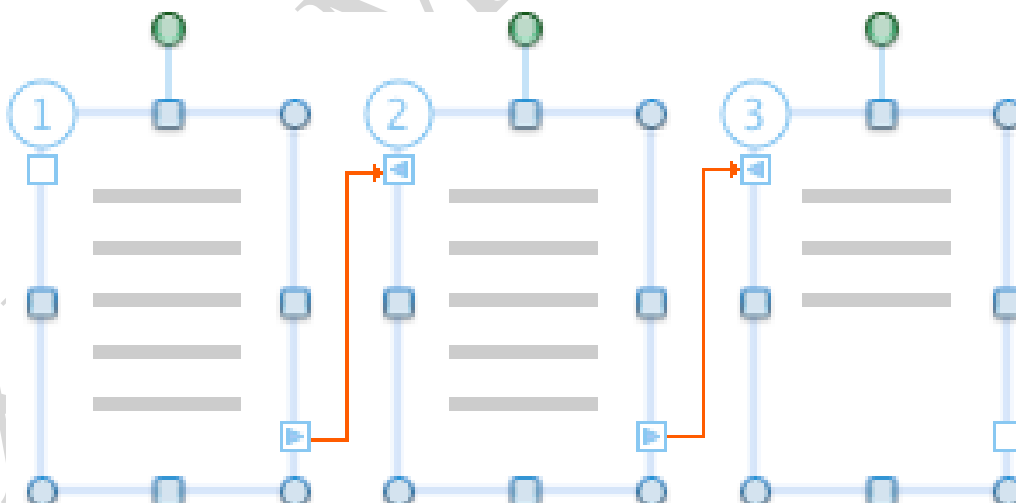
Picture 2. 6. Changing the shape of the a text box

- Once you select the points, you can drag them to the desired shape.

b) Creating link between text boxes

You may have text that you would like to run in a series of text boxes. You want the text to flow from one text box to the next and ensure that the reader will be able to follow the text.

If you add or delete text in one text box, **Create Link** will adjust the text in the rest of the text boxes accordingly.



Picture 2. 7. Text in linked boxes flows from one box to the next

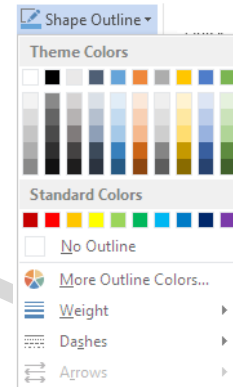
To create a link between two text boxes, do the following:

- Add two text boxes and ensure that they are empty.
- Click on the first text box.
- Click on **Format** tab under **Drawing Tools** tab in the menu bar.
- Select **Create Link** icon under **Text** group.

2.1.3 Modifying colours and lines for text boxes

The default color for lines is black. To change this color to a desired one, do the following:

- Click on the text box.
- Click on **Format** tab under **Drawing Tools** tab in the menu Bar.
- Select **Shape Outline** in the **Shape Styles** group. A drop down menu appears as shown in the picture .



Picture 2. 8. Filling a shape with color

Click on the desired colour under **Theme Colours** to change the border colours.

To remove the borders:

- Click on **NO OUTLINE**
- Click on **Weight** and select the desired line to change the line thickness or to change the line style.
- Click on **Dashes** and select an option in the side kick menu.
- Click on **No Outline**.

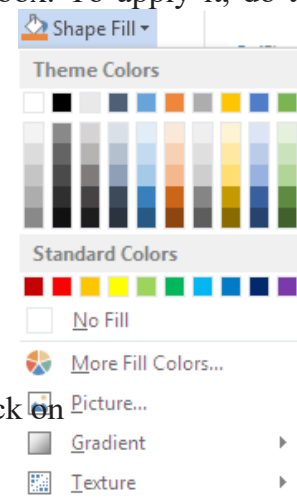
2.1.4 Applying Fill

The feature of **Fill Effects** adds colour inside the text box. To apply it, do the following:

- Click on the text box.
- Click on **Format** tab under **Drawing Tools** tab in the menu bar.
- Select **Shape Fill** in the **Shape Styles** group.

A drop-down menu appears as shown in the picture below.

- Click on the desired colour under **Theme Colours** to fill the shape.
- To remove fill added, click on **No Fill** box.
- Click on **Gradient** and select the desired fill style, click on



Picture 2. 9. Filling a shape with colour

Texture and select an option in the side kick menu.

Application Activity 2.1

Part A: Fill in the missing words to complete the sentences

- 1) The term graphics refers to anything in form of ___ or___.
- 2) A _____ is an object created to contain text.
- 3) The direction of text in a text box can be changed to the desired_____.

Part B: Answer the following questions.

- 4) Define the term text box.
- 5) State two ways in which a text box can be used.
- 6) List three ways in which a text box can be added in a document.

Part C: Do this practical activity

- 7) Create a new document and insert three text boxes in the document.
- 8) Create a link between the second and the third text box.

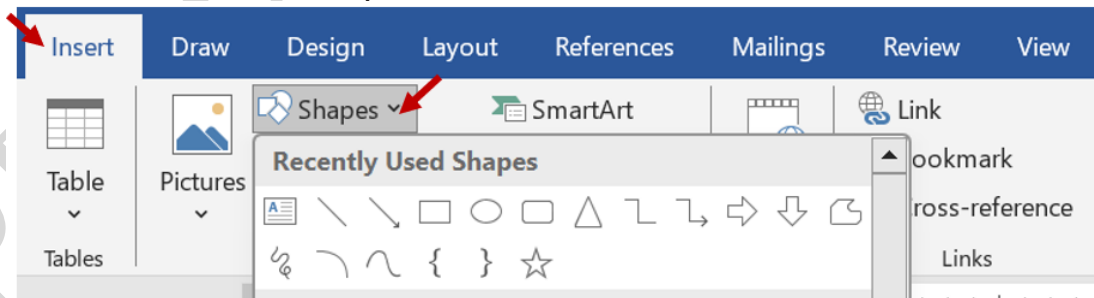
2.2 Adding shapes

Activity 2.2

- 1) Which tab is used to insert shapes in a document?
- 2) Mention one way you can change the appearance of a shape.
- 3) Why do we rotate a shape?
- 4) What is grouping used for when working with shapes?
- 5) What is WordArt used for in a document?

2.2.1 Inserting shapes

- Click on **Insert** tab from the menu bar, select **Shapes** icon from **Illustration** group. A drop down menu appears as shown in the picture.
- Click on the desired shape, the pointer changes to a plus sign.
- Position the pointer at the desired area of the document and drag it in the desired direction to draw the shape.



Picture 2. 10. Options to insert shapes

2.2.2 Formatting shapes

Once a shape has been inserted in a document, one can format it. To format in this case

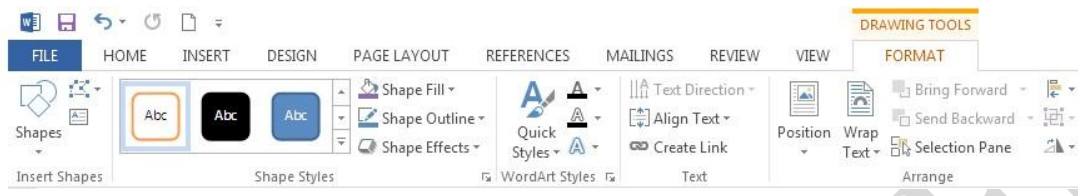
means

to arrange it in the desired way.

Some formatting styles that can be applied on a shape include changing the shape; rotating the shape to an angle; and grouping a number of shapes into one image.

a) Drawing tools

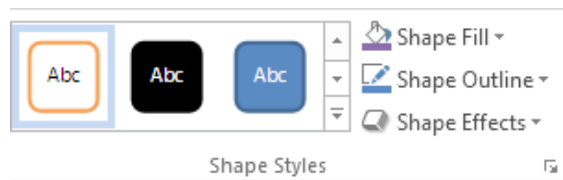
Drawing tools are normally used when another shape or line exists and is highlighted to activate the **Drawing tools** menu option in a document.



Picture 2. 11. The drawing tools tab

b) Shape Styles Group

Shape styles are used for formatting the shapes using fill colour, outline, or effects. Here is a picture of the Shape Styles group.



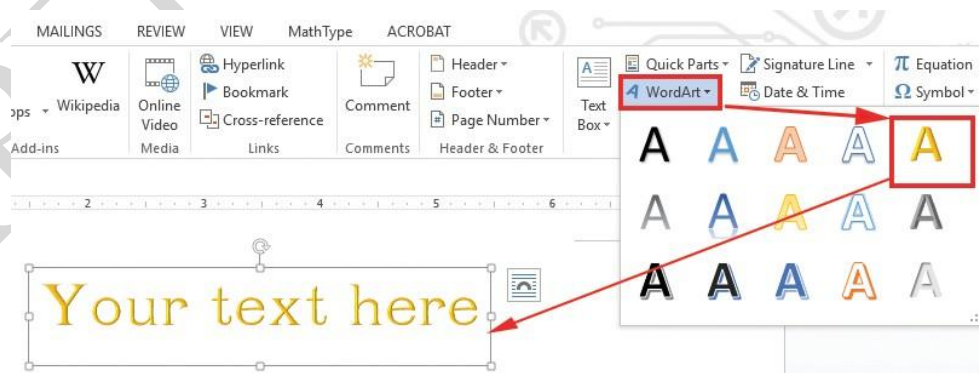
Picture 2. 12. Shape formatting styles

c) WordArt Styles Group

WordArt style is used for formatting text found in shapes using text fill, text outline, and text effects. WordArt enables you to create special text effects such as shadowed, rotated, stretched, and multicolored text.

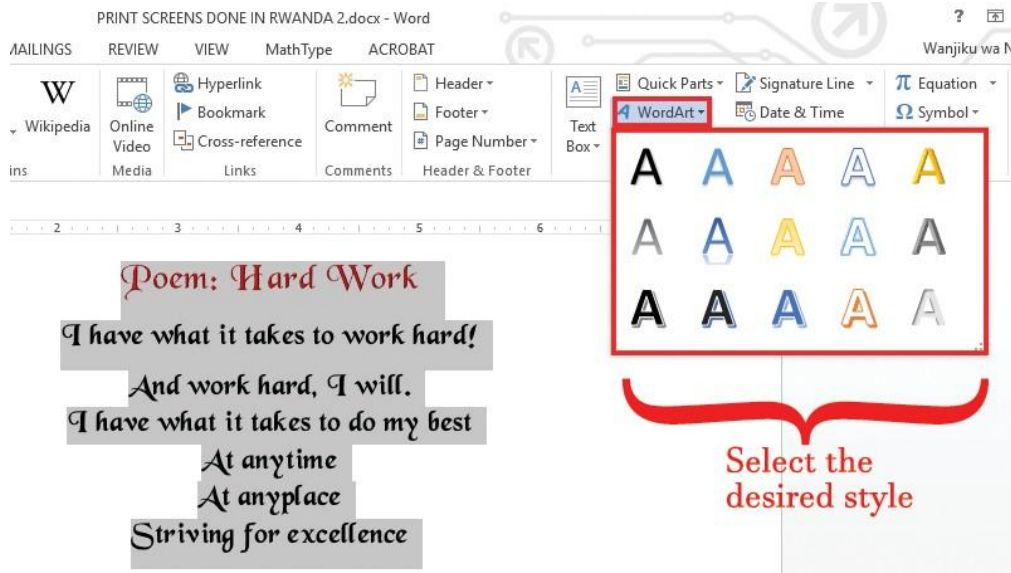
Method 1: Inserting WordArt

- On the **Insert** tab, click the **Insert WordArt** button.
- In the **WordArt** gallery, click the **WordArt** style that you desire.
- Replace the placeholder text with the text you want to format using **WordArt**.



Select *Picture 2. 13. Inserting WordArt styles*

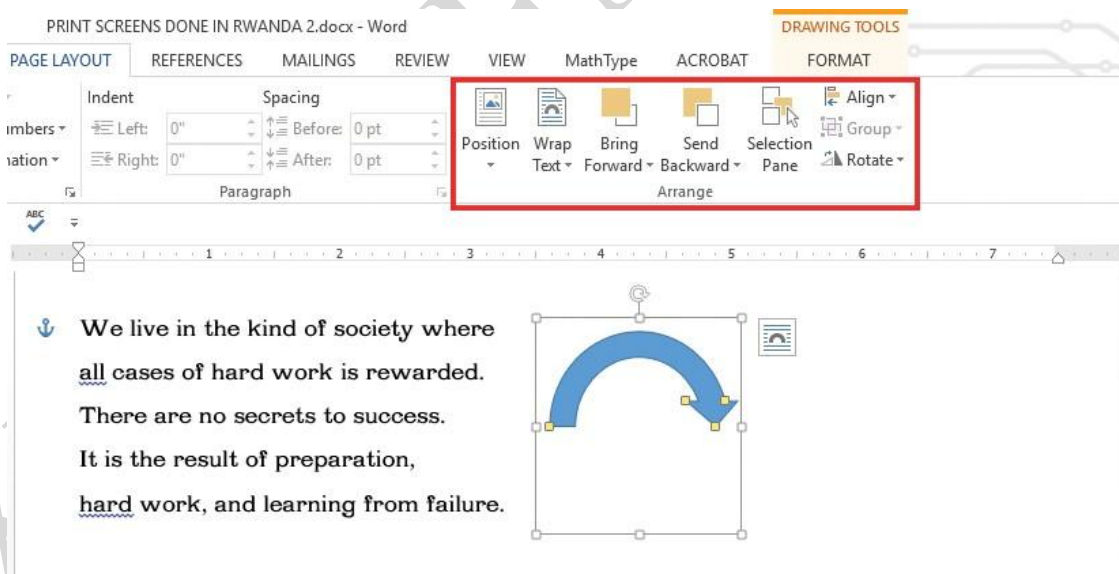
- Click the WordArt drop-down arrow in the **Text group**. Word will automatically create a text box for the text, and the text will appear in the selected style.



Picture 2. 14. converting text to a WordArt

d) Arranging group

It is used when arranging text and shapes in a document. It can also be used for changing position and rotating a shape.

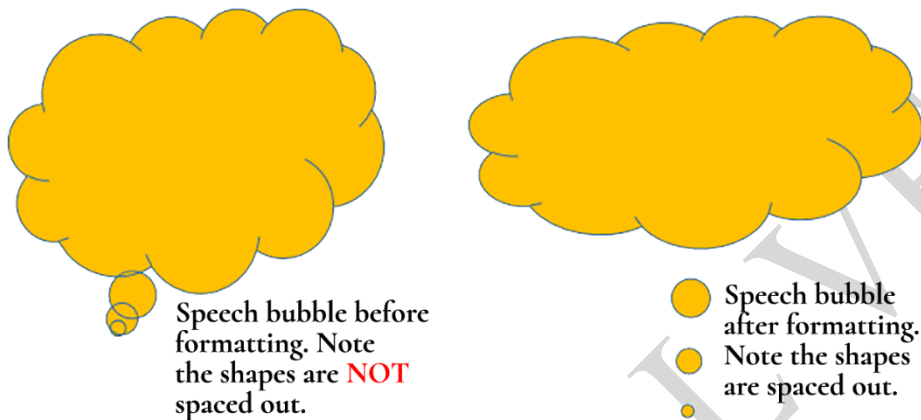


Picture 2. 15. Arranging text and shapes in a document

e) Changing shape

To change a shape that has already been inserted, do the following:

- Select the shape then click on the adjustment handle which is normally a yellow diamond and drag
- it either inward or outward.
- Release the mouse button once the desired shape is realized.

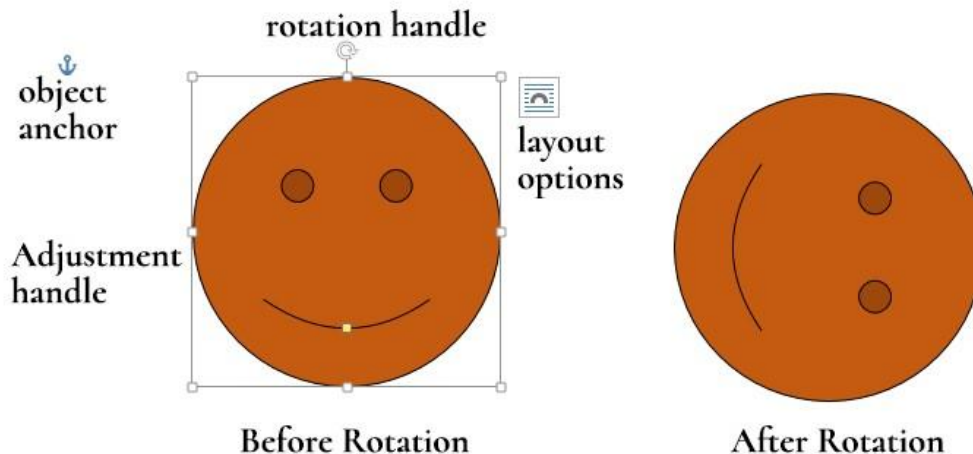


Picture 2. 16. Using the formatting features to change a shape

f) Rotating a shape

A shape can be rotated to a desired angle. Rotating is done to change the orientation of the shape. To rotate a shape, do the following:

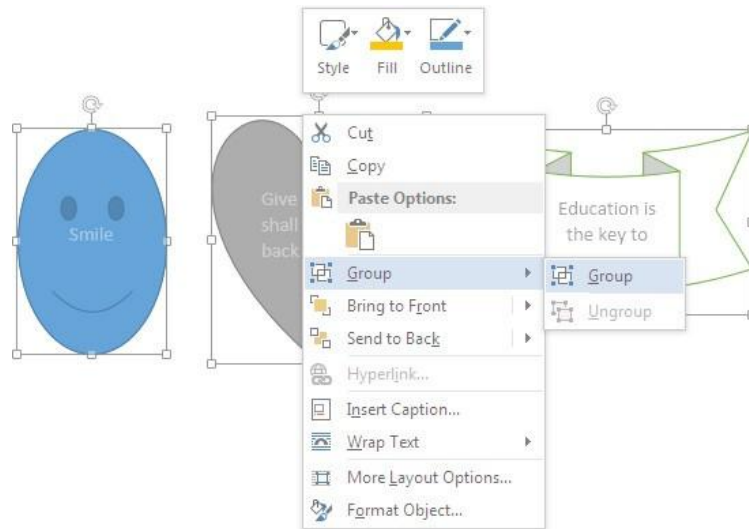
- (i) Click on the shape to be rotated.
- (ii) Click on the handle and drag the picture to the desired direction.



Picture 2. 17. Rotating is done to change the orientation of the shape

More than one shape can be grouped in one so that they are moved or formatted as a unit. To group shapes, do the following:

- Select the shapes by clicking on them as you hold the shift key down.
- Right-click on the selected shapes. A pop-up menu appears as shown in the picture below.
- Select **Group** then **Group** again from the next pop-up menu. A canvas is automatically inserted around all the shapes and hence making it easier to format all the shapes at once. Immediately the shapes you selected get grouped as one shape.

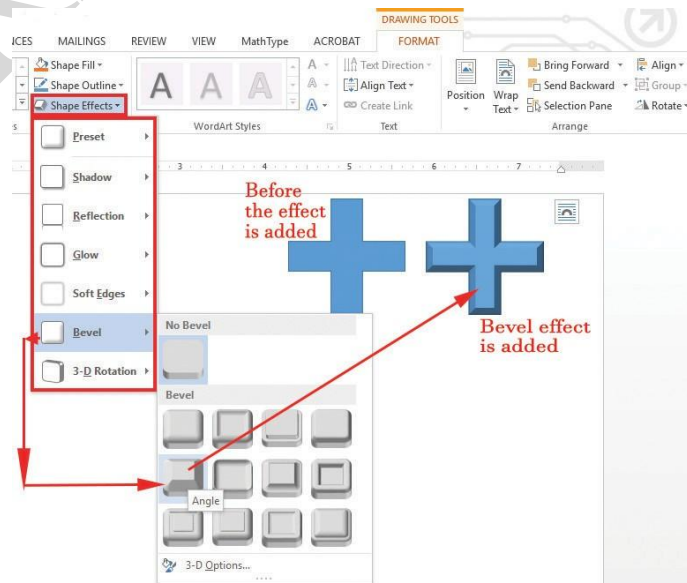


Picture 2. 18. Grouped objects

h) Shape effects

Some examples of shape effects include shadow, reflection, glow, soft edges, bevel and 3-D rotation. To add a shape effect, do the following:

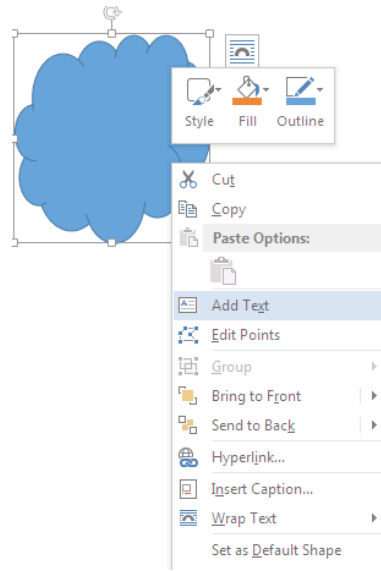
- Select the picture that click on **Shape Format** tab in the **Drawing Tools** tab from the menu bar
- Select the **Shapes Effects** icon from **Shape Styles** group. A drop down menu appears.
- Click on the desired effect to display other options in a pop-up menu.
- Select the desired effect to automatically apply it to the shape.



2.2.3 Adding text in shapes

Text can be added in a document using the following procedure:

- Right-click on the shape and select **Add Text** option from the pop-up menu as shown in the picture below.
- Type the desired text.
- Click away from the shape.



Picture 2. 20. Adding text to a shape

Application activity 2.2

Part A: Give the correct answers

- 1) More than one shape can be ____ into one so that they are moved or formatted as a unit.
- 2) _____ feature is used when arranging text and shapes in a document.
- 3) Shadow, reflection, glow, soft edges, bevel and 3-D rotation are examples of _____.

Part B: Answer the following questions

- 4) Outline the procedure of inserting a parallelogram in a document.
- 5) List three formatting features that can be applied to a shape.
- 6) State the purpose of grouping of objects.

Part C: Do this practical activity

- 7) Create a new document and insert a speech bubble, a round corner rectangle, and a 16-sided star.
 - a) Save the document as **Shapes**.
 - b) Rotate the speech bubble to 180° .

2.3 Inserting pictures and Word Art objects

Activity 2.3

The Community Event Flyer

Students are tasked with creating a document for advertising a restaurant business. Students start with a blank document. They must insert a WordArt title to make the event name "pop" with a curved effect. Then, they must search for and Insert a picture of a cupcake. To make the flyer look professional, they have to use Text Wrapping (Square) so the event details wrap neatly around the image, and apply a Picture Style to finish the design.

- 1) An inserted picture is hiding your text and you want the text to go around that

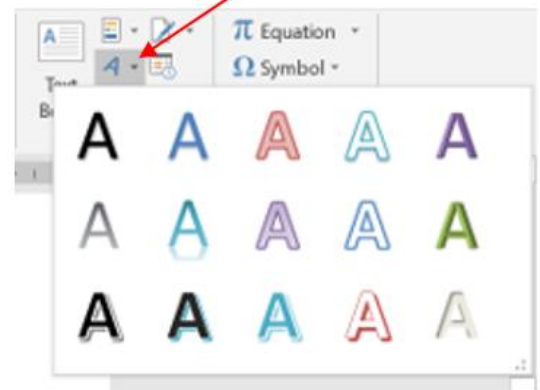
- picture. Which setting should you change?
- 2) What is the difference between regular text and WordArt?

In Microsoft Word, you can make your documents more attractive by adding pictures and **WordArt**.

WordArt allows users to create decorative and stylized text. Unlike normal text, WordArt can be shaped, colored, and given visual effects to make it stand out. It is commonly used for titles, headings, or important announcements in documents.

2.3.1 Insert WordArt

- Click the **Insert** tab on the menu bar ribbon.
- In the **Text group**, click **WordArt**.
- A gallery of WordArt styles appears.
- Click the style you like.
- In the textbox that appears, write the text you want.



Picture 2. 18: WordArt Style

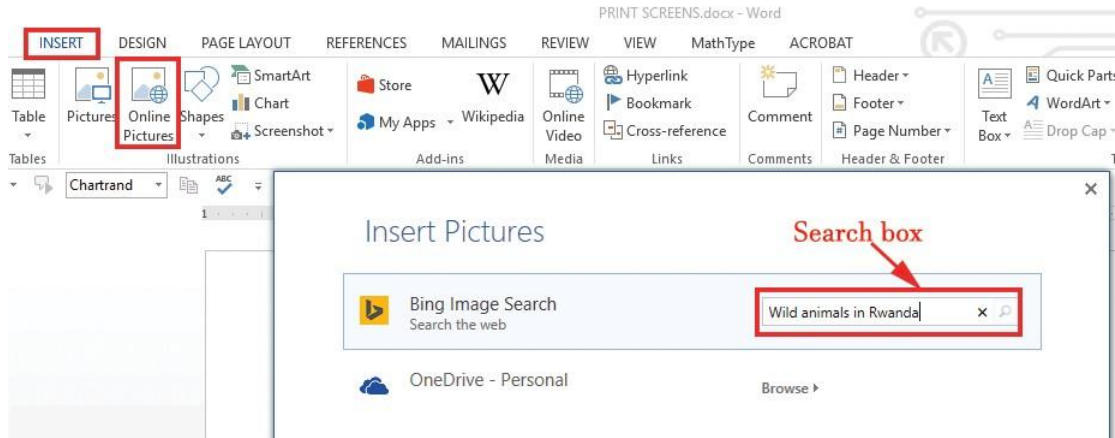
Modifying WordArt

- Change the font and size by selecting the WordArt text then click on the **Home** tab, then choose one tool from the **Font group** to change the font type, size, or make it bold/italic.
- Apply **Text Fill** and **Outline** by clicking the WordArt text then go to **Drawing Tools/Shape Format** tab then in the **WordArt Styles group** click on **Text Fill** or **Text Outline**.

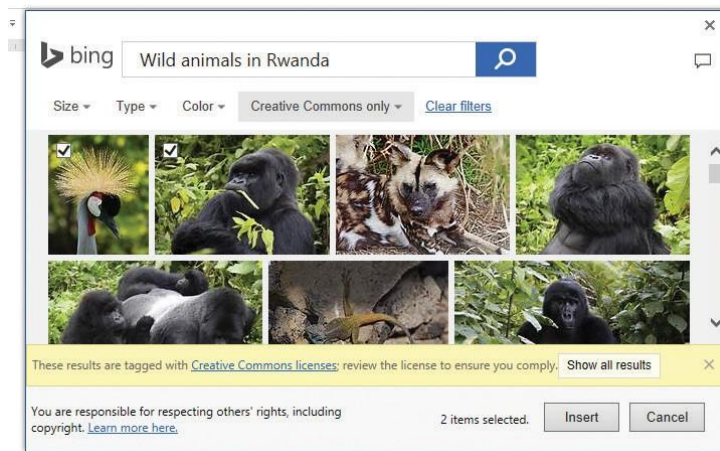
2.3.2 Inserting pictures from the internet

- Position the insertion point where the picture is to be inserted.
- Click **Insert** tab from the menu bar in the **Illustrations** group and click **Online Pictures** icon. If there is internet connection, a dialog box is displayed.
- Type the name of the image to be located in the box provided. Click on **Go** button to display the result.

- Click on the desired image then select **Insert** button.



Picture 2. 21. Inserting Picture dialog box

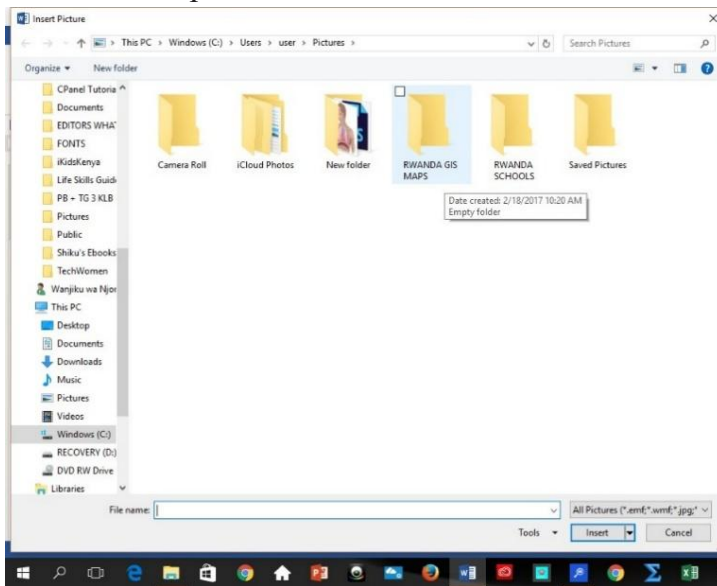


Picture 2. 22. Images displayed based on the search

2.3.3. Inserting pictures from an existing file

- Position the insertion point where the picture is to be inserted.
- Click **Insert** tab from the menu bar.
- Under **Illustrations** group, click **Pictures** icon. A dialog box appears.
- Select the location where the picture is stored.

- Select the picture then click **Insert** command.



Picture 2. 23. Inserting a picture from file

Application activity 2.3

Fill in the dashes with the correct words from these: Textbox, Insert Tab, Text, Illustrations, WordArt.

- 1) The tab in the menu bar where the online pictures icon is found is ...
- 2) Words written in artistic formats:...
- 3) A group in the ribbon where the WordArt icon is found is ...
- 4) A group in the ribbon where the online pictures icon is found: ...
- 5) A box in which a WordArt object is inserted is ...

2.4 Equations and symbols

Activity 2.4

Aline is preparing a Mathematics and ICT assignment using Microsoft Word. She types some content and wants to include a copyright sign after her name.

She needs to insert special characters such as quotation marks and dashes in her sentences.

She also wants to write mathematical equations for her assignment. However she notices that she cannot find some symbols on the keyboard. The mathematical equations do not look well when typed using normal text.

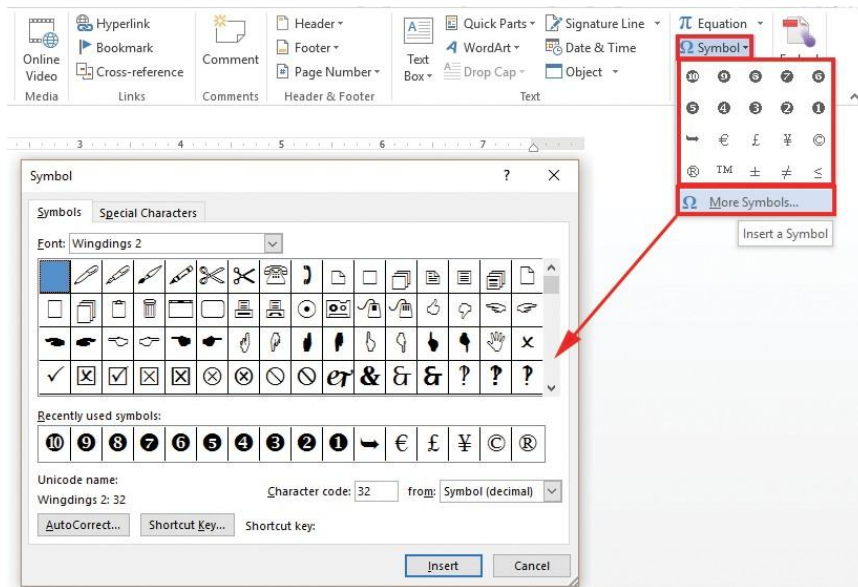
- 1) How can Aline write the equations in a proper mathematical form?
- 2) Which menu or tabs in Microsoft Word can help her solve each of these problems?

2.4.1 Inserting symbols and mathematic equations

Symbols are non-alphabetical signs that have specific meaning when used in a particular

context. An example is the © **symbol**, which is a copyright symbol. To insert a symbol, use the following procedure:

- Position the insertion cursor where the symbol or the special character is to be inserted.
- Click **Insert** tab from the menu bar. Under **Symbol** group, click **Symbol** icon. A drop-down menu appears containing a list of symbols.
- However, if the desired symbol is not in the list, click on **More Symbols** command to display a window for those symbols as shown in the picture.
- Select the desired symbol, then click the **Insert** button.

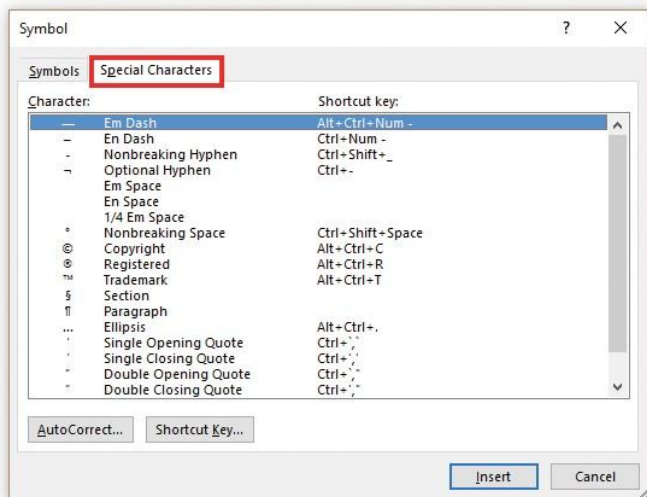


Picture 2. 24. Inserting symbols

2.4.2. Inserting special characters

To insert a special character, use the following procedure:

- Position the insertion cursor where the special character is to be inserted.
- Click **Insert** tab from the menu bar. Under **Symbol** group, click **Symbol** icon. A drop down menu appears containing a list of symbols and a **More Symbols** command.
- Click on **More Symbols** command, select **Special Characters** tab to display a dialog box as shown in the picture below.
- Choose the desired character then click the **Insert** button.

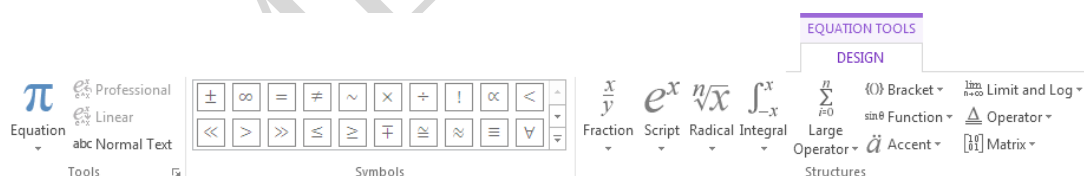


Picture 2. 25. Inserting special characters

2.4.3 Writing equations

An equation is a mathematical statement that has two expressions, usually separated by equal sign. Equations comprise numbers, letters, operators, and symbols. To insert an equation in a document, do the following:

- Position the cursor where the equation is to be inserted.
- Click **Insert** tab from the menu bar.
- Under **Symbol** group, click **Equation** icon. The **Equation Tools** menu is added along with **Design** in the menu bar.
- Click on **Design** to reveal the ribbon under it.



Picture 2. 26. Writing equations

- Select the desired equation from the **Symbols** group or **Structures** group to automatically insert it in the document. For example, to insert a quadratic equation, click on the equation in the **Radical** icon from the **Structures** group. The **picture below** shows a sample of a quadratic equation.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Picture 2. 27. A mathematical expression used for calculating the

Application activity 2.4

1) Create the document shown in the picture below as it appears.

1. The area of a circle: $A = \pi r^2$
2. The Binomial Theorem is given by: $(x + a)^n = \sum_{k=0}^n \binom{n}{k} x^k a^{n-k}$
3. The formula for expanding a sum is: $(1 + x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \dots$
4. The following are examples of 2×2 matrices: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix}$
5. Evaluate the following integrals: $\int_1^2 (4x^2 - 3^x) dx$

2) Save the document under the name MyEquations.

2.5. Tables

Activity 2.5

The school ICT teacher asks Jean to prepare a document showing students' names, subjects, and marks for his class. Jean types the information in a paragraph format like this:

John Mathematics 78 English 82

Alice Mathematics 85 English 90

Emmanuel Mathematics 90 English 89

Gad Mathematics 83 English 89

After typing, he notices that the information is not well organized, it is difficult to compare marks, some headings should be combined into one space and he needs to add more students and subjects.

- 1) Explain how tables in Word can help solve these problems.
- 2) Open Word then click on Insert tab then create a table.
- 3) Write the above information in the table you have created.

2.5.1 Inserting a table

A table is a feature that is used to present information in an organised layout. It consists of rows and columns intersecting to form cells where data is entered and manipulated. A **cell** is a box formed by the intersection of a row and a column in a table.

In the text below, two ways of creating a table are developed namely **using the Graphic Grid** and **using the Insert Table** feature.

a) Using the Graphic Grid

- Position the cursor where the table is to be inserted.

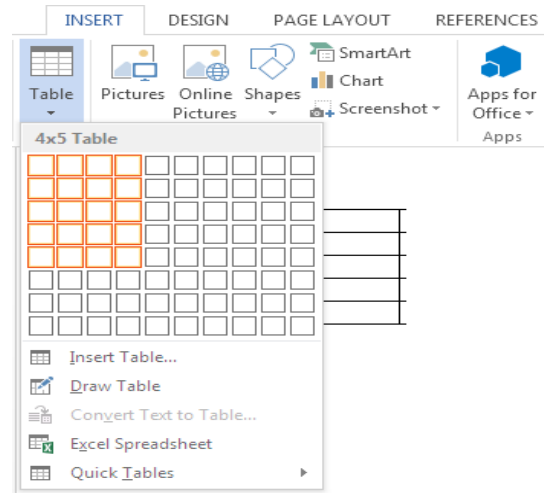
- Click on **Insert** tab from the menu bar and select **Table** command under the **Tables** group.

A drop-down menu appears as shown in the picture below.

- Select the desired number of rows and columns from the grid

b) Using the Insert Table

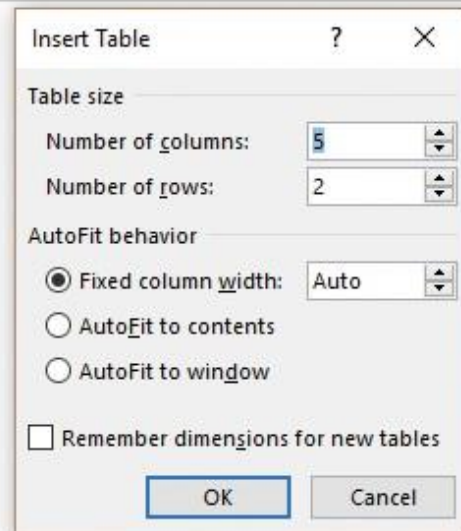
- Position the cursor where the table is to be inserted.
- Click on **Insert** tab from the menu bar and select **Table** command under the **Tables** group.



Picture 2. 28. Using the graphic grid to insert

- Select **Insert Table**. A dialog box appears as shown in the picture.

- Type the number of columns required in the **Number of columns** box. You can also use the down and up arrows on the window to decrease or increase the number of rows.
- Type the number of rows required in the **Number of rows** box.
- Select the desired option under **AutoFit behavior** and click **OK** to apply.



Picture 2. 29. Inserting a table dialog box

2.5.2. Inserting columns and rows in a table

To add a new column(s) in an already created table do the following:

- Place the cursor in any cell within the column to precede or succeed the new one.
- Click **Layout** tab from the menu bar and select **Insert Left** or **Insert Right** command under the **Rows & Columns** group.

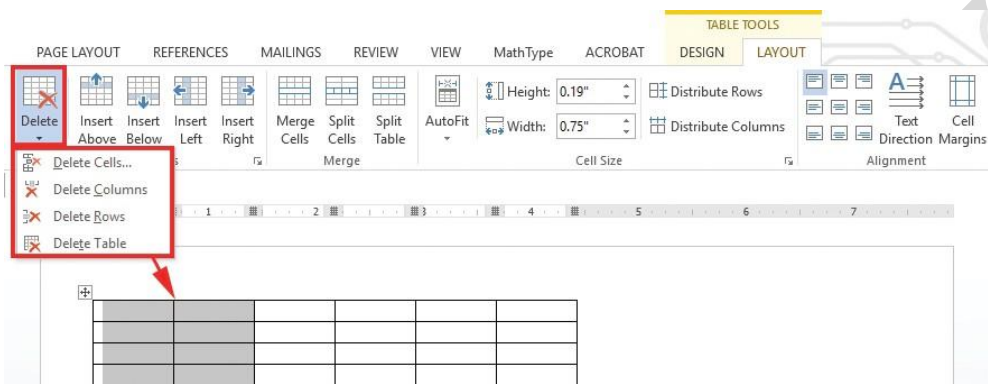
To add a new row(s) in an already created table do the following:

- Place the cursor in any cell within the row to precede or succeed the new one.
- Click **Layout** tab from the menu bar and select **Insert Below** or **Insert Above** command under the **Rows & Columns** group.

2.5.3 Deleting a row, table and column

To delete a row(s) in an already created table do the following:

- Select a cell in the respective table.
- Click **Layout** tab from the menu bar and select **Delete** command under the **Rows & Columns** group. In the drop-down menu that click on the desired option.



Picture 2. 30. Deleting cells, rows, tables, and columns

Options	Functions
Delete Columns	Erases an entire column.
Delete Rows	Erases an entire row.
Delete Table	Erases a table.
Delete Cells...	Erases the selected cells.

Table 2. 1: Options available in delete command

2.5.4. Merging cells and splitting cells

Merge cells command is used to combine more than one cells to appear as one large cell while split cells command is used to divide a cell to more than one section hence appearing as many different cells.

a) Merging cells

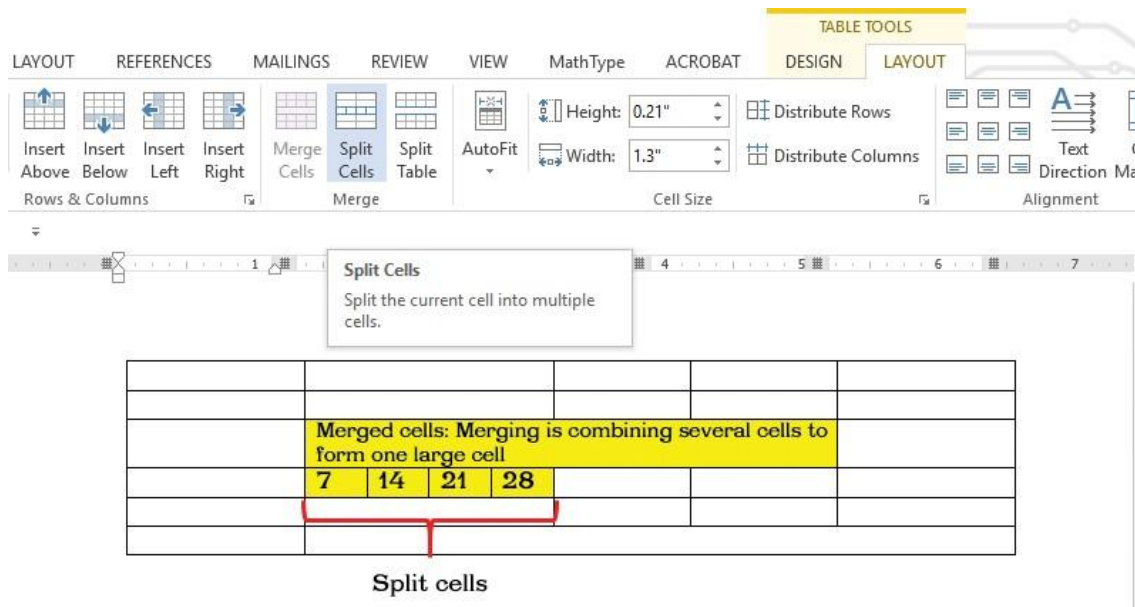
The following is the procedure for merging cells:

- Select the cells to be merged.
- Click **Layout** tab from the menu bar and select **Merge Cells** icon under the **Merge** group. The cells are automatically combined.

b) Splitting cells

- To split a table, select the cell from where the table is to be separated

- Under the **Table Tools** click on **Table Layout** then click on **Split cells**.



Picture 2. 31. Merging and splitting cells

Application Activity 2.5

Part A: Fill in the dashes with the correct answers

- 1) A table is a feature that consist of ____ and ____.
- 2) A box formed by the intersection of a row and column in a table is known as ____.
- 3) _____ command is used to combine more than one cells to appear as one large cell.

Part B: practical activity

- 4) Create the following table and save it as Table1.

Name	Class	House
Habimana	2a	Muhanga
Muteteli	2b	Huye
Sebahire	2a	Rubavu
Nyamahoro	2b	Gatsibo

- a) Insert three columns to the right of the column containing house.
- b) Add two rows after the row containing Nyamahoro details.
- c) Add another row on top and include the title "Examination Analysis"
- d) Save it as Table 2.
- e) Delete Nyamahoro's details.
- f) Add the following details in the three columns added:
Column 1(History,67, 89, 98),
Column 2 (Business, 77, 56, 34) and column 3 (English, 74, 87, 65).
- g) Merge the cells containing the title.

- h) Split the cell containing the name Habimana into two.
- i) Save it as **Table 3**.

END UNIT ASSESSMENT

Part 1. Section A:

- 1) Complete the following statements:
 - a) The term graphics refers to -----
 - b) A -----is an object created to contain text.
 - c) More than one shape can be -----into one so that they are moved or formatted as a single unit.
 - d) Shadow, reflection, and glow are examples of -----

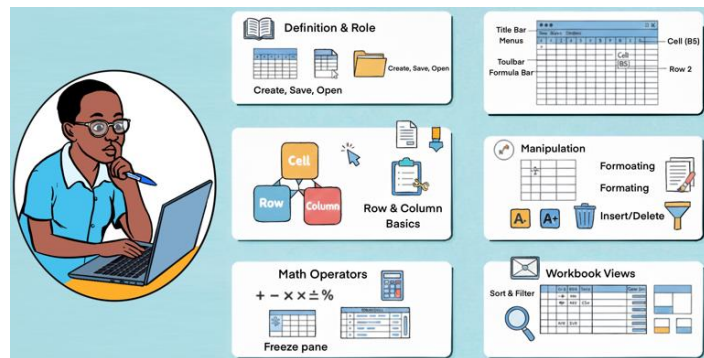
Section B: Answer the following questions

- 2) Explain the use of a textbox in a Word document.
- 3) Explain how a text box can be added in a Word document.
- 4) What is the purpose of the adjustment handle?
- 5) How do you remove a shape's border?

Part 2. Practice Activity

- 6) Create a linked flowchart using shapes, text boxes, and special symbols
- 7) Insert one Rounded Rectangle and one Parallelogram. Change the color of the rectangle using a Gradient fill and the parallelogram using a Texture fill.
- 8) Inside the rectangle, type your name followed by the Copyright symbol (©).
- 9) Insert two empty text boxes. Create a link between them.
- 10) In the first linked box, type: "Success is not final, failure is not fatal: it is the courage to continue that counts." Ensure the text flows into the second box.
- 11) Select all objects and group them so that they move as one unit.

UNIT 3: SPREADSHEET BASICS



Key unit competency:

By the end of this unit, you should be able to:

Apply basic manipulation of cell content using arithmetic operations.

Knowledge and understanding:

- Identify the basic features of a spreadsheet environment.
- Explain how to perform basic mathematical operations
- Explain the importance of sorting and filtering

Skills:

- Apply various methods to move, delete, and enter data from and into cells.
- Adjust row height and column width
- Use arithmetic operations to manipulate cell data.
- Analyze spreadsheet data using sorting and filtering tools effectively.

Attitudes and values

- Show accuracy and attention to detail when entering and formatting data.
- Appreciate the importance of organization and neat presentation in digital work.
- Demonstrate responsibility in saving and managing files properly.

3.0. Introduction

A spreadsheet program is an interactive program that enables users to layout and organise numerical data in a tabular form. Spreadsheets contain rows and columns that intersect to form cells into which data are entered. The spreadsheet program in this unit is Microsoft Excel 2019.

Introductory activity

Your school administration has asked the ICT Club to help the academic office improve how students' exam results are processed and presented. Previously, teachers calculated totals and averages manually, which often led to errors and wasted time.

- 1) Why do schools and organizations prefer using Excel instead of manual calculations?
- 2) What problems might occur if exam results are calculated manually?

3.1 Starting with Spreadsheet

Activity 3.1

- 1) Click on Start and write in the Search box “Excel” then click on the Excel icon to open Excel.
- 2) Save your Spreadsheet document under your name.
- 3) a) From your observation explain how to open Excel
b) Explain how to save an Excel document

3.1.1 Features of Spreadsheet

A Spreadsheet has the following characteristics:

- They consist of a workbook and worksheets and every worksheet consists of cells where data is entered.
- Every individual cell has a unique cell address used for identification.
- It contains inbuilt formulas known as functions that enable the user to perform calculations quickly.
- It allows presentation of information using charts.

3.1.2 Create, save, and open a workbook

A workbook refers to a file that can contain one or more worksheets used to organize different kinds of related data.

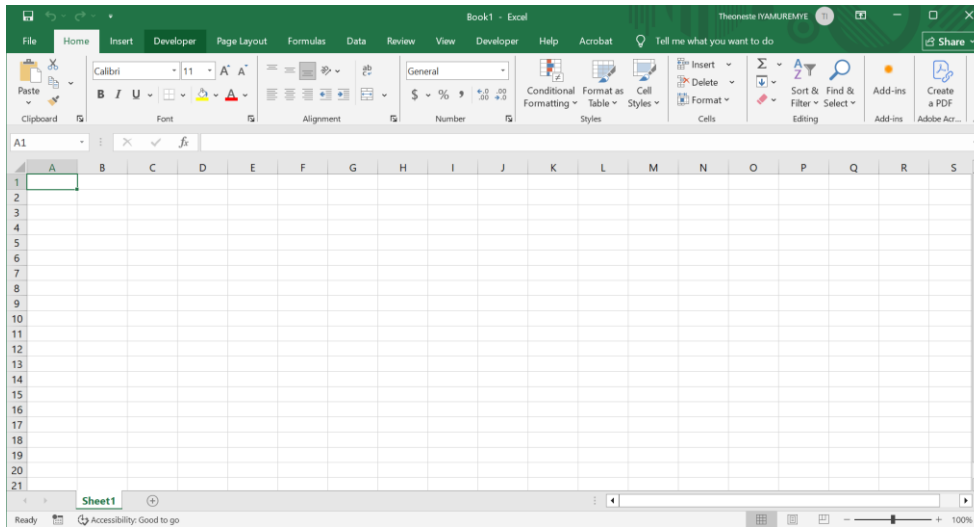
A worksheet is a form containing cells that are organised in rows and columns. A worksheet in a computer screen resembles a sheet of paper with multiple columns that are used to analyse numerical data.

a) Opening Microsoft Excel program

Open Microsoft Excel program by using one of these three methods:

- Click **START** button, select **All Apps** then click on **Microsoft Office** to display a list of programs contained in the office suite. Click on **Excel 2019**.
- Click **START** button then select **Microsoft Excel 2019** if it was pinned on the start menu.
- Click **Microsoft Excel 2019** icon if it was pinned on the task bar.

Any of the options above will open Excel in a window like the one below:



Picture 3. 1. Excel blank workbook

b) Creating a workbook

When Excel is launched, it automatically opens a workbook containing a worksheet with a default name of **Sheet1**.

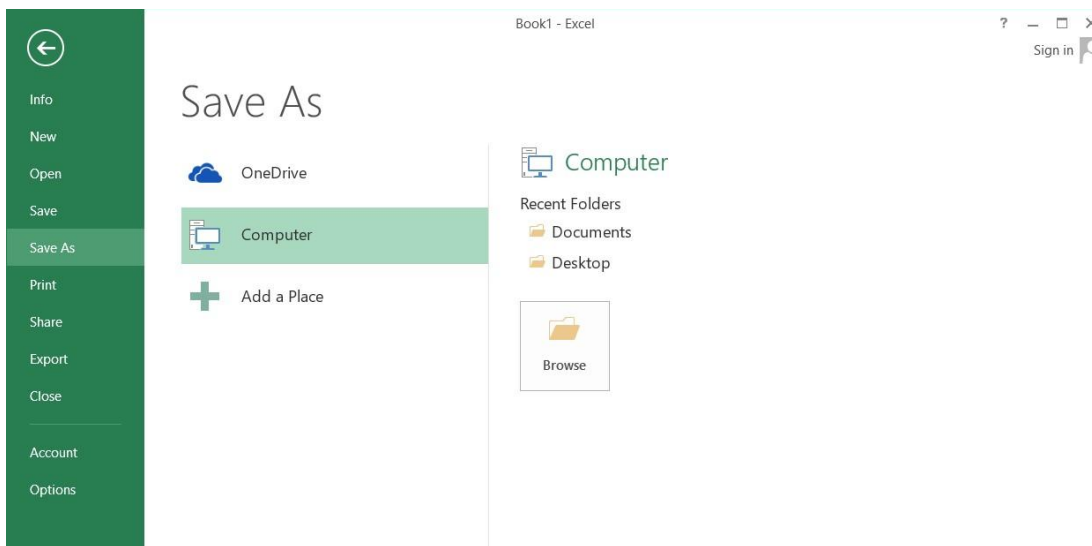
After the program is opened, the following steps can be used to create a new workbook:

- Click on the **File** tab from the menu bar.
- Select **New** from the pull down menu.
- Click on the desired template from the right pane. For example, to create a new blank workbook, click on the **Blank workbook** option.

a) Saving a workbook

To save a workbook for the first time, do the following:

- Click on the **File** Tab in the menu bar.
- Click the **Save As** option from the pull down menu that appears. The Save As window appears as shown below



Picture 3. 2. Saving a workbook

- Click on the location where the workbook is to be stored if it displayed on the Save As window. Otherwise click on **Browse** or Computer icon to display the **Save As** dialog box.
- Type the name of the workbook in the **File name** box. To select another storage location click on it on the left pane of the window.
- Click **Save** button.

Once a workbook has been saved, a user can make changes to it and save it under the same name or a different name. To save changes on an existing workbook, do one of the following:

- Click the **File** tab, then select **Save** command.
- Click the **Save** icon on the **Quick Access Tool bar**.

b) Closing a workbook

To close an active workbook click **Close** button located on the top right corner of the title bar of the active worksheet. Or press **ALT + F4** on the keyboard or click the **File** Tab then select **Close** command.

c) Opening an existing workbook

To open an already saved workbook, do one of the following:

- Click the **File** tab then select **Open** command.
- Select the location where the workbook was stored such as Desktop.
- Click on the name of the workbook from the list displayed or click **Browse** to locate the file. Finally click **Open**.

Application activity 3.1

- Which of the following best defines a Spreadsheet?
 - A Word Processing tool for writing long essays.
 - An interactive program that organizes numerical data in a tabular form.
 - A system used only for browsing the internet.

- d) Hardware used to print documents.
- 2) What is the intersection of a row and a column called?
- a) A Label b) A block
c) A Cell d) A Workbook
- 3) Which of the following is an example of Spreadsheet software?
- a) Microsoft Word b) Lotus 1-2-3
c) Adobe Photoshop d) Windows 10
- 4) Why is a Spreadsheet considered dynamic compared to a paper ledger?
- a) Because it uses different colors.
b) Because it automatically recalculates totals when data is changed.
c) Because it can be printed on large paper.
d) Because it only stores text
- 5) In a Spreadsheet, rows are identified by while columns are identified by
- a) Letters / Numbers b) Symbols / Colors
c) Numbers / Letters d) Names / Addresses
- 6) List three common examples of spreadsheet software.
- 7) Differentiate between a Workbook and a Worksheet.

3.2 Spreadsheet environment

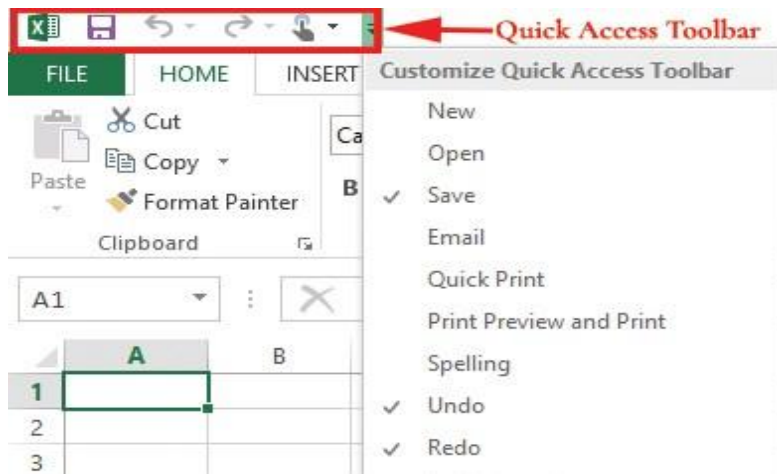
Activity 3.2

- 1) List the main part of Spreadsheet interface
- 2) What do the letters at the top and the numbers along the side of the spreadsheet represent?
- 3) Where do you type data in a spreadsheet?

An Excel window has different elements that are used to do different tasks as outlined below:

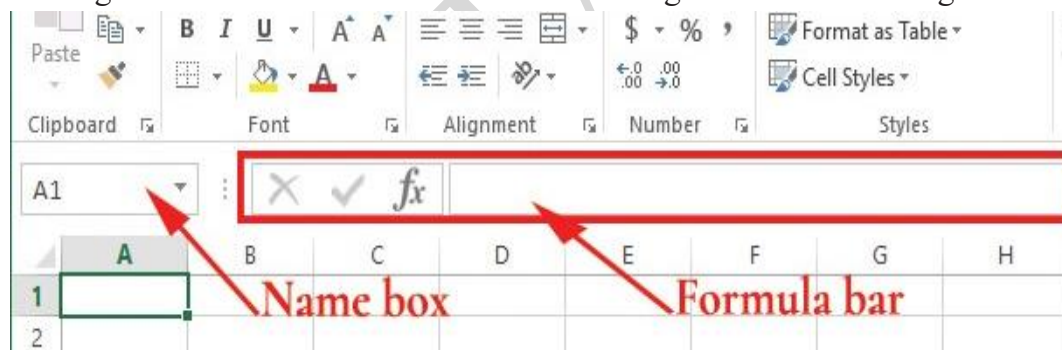
- **Title bar:** It displays the name of the application currently in use as well as the name of the active workbook. When a new spreadsheet is opened, Ms Excel allocates default names such as Book1, Book2 and so on.
- **Buttons:** The title bar also contains three buttons which are: Minimise, Restore Down/Maximise and Close buttons at the top right corner.
- **Menu bar:** This bar displays the following tabs: File, Home, Insert, Page Layout, Formulas, Data, Review and View.
- **Ribbon:** The ribbon consists of icons of commands organised and classified into groups. Clicking on any tab on the menu bar displays a ribbon unique to it. For example, clicking on **Page Layout** tab displays a ribbon with 5 groups namely Themes, Page Setup, Scale to Fit, Sheet Options and Arrange.
- **The Quick Access Toolbar:** It is a small customisable toolbar of the most commonly used commands located at the top left corner of the workbook window. By default, this bar contains Save, Undo and Redo commands.

It can be made to appear above or below the ribbon or commands can be added or removed from it.



Picture 3. 3. Customizing the Quick Access Toolbar

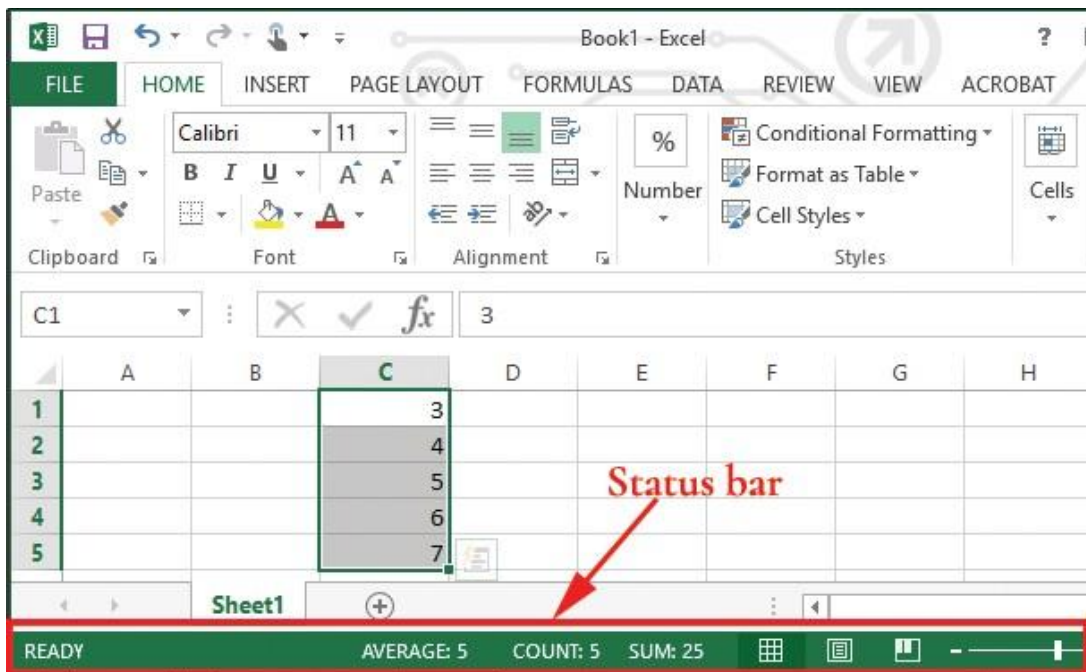
- **Scroll bar:** The scroll bar allows one to move or navigate through a large worksheet. There are two types of scroll bars the vertical scroll bar and horizontal scroll bar. Vertical scroll bar allows upward or downward movement on the worksheet while horizontal scroll bar allows movement from left to right and vice versa.
- **Formula bar:** It provides the user with a box for entering or editing data or formula in a cell. It is located below the ribbon by default. It contains the cell naming box and the formula box used for entering a formula or editing data.



Picture 3. 4. formula bar

The formula bar contains the name box and the formula box

Status bar: It is located at the bottom of the window and it displays status on options that are selected to appear on the status bar. For example, to display average, count, or sum of selected cells is looking at the **status bar**. The picture below shows a display of the status bar.

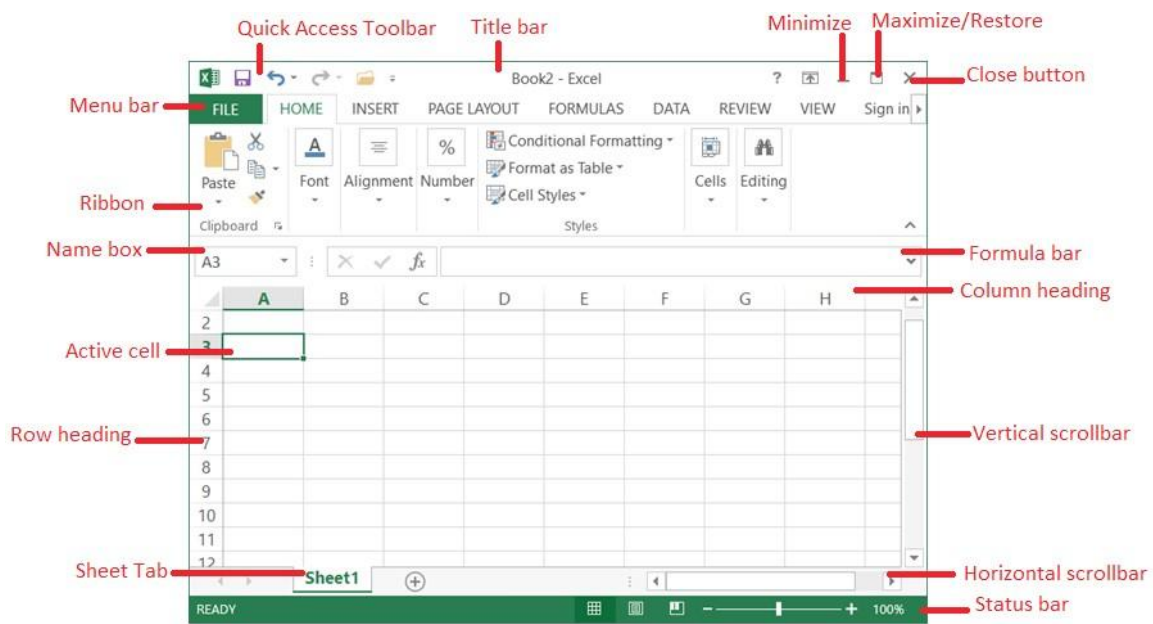


Picture 3. 5: The status Bar

- **Cell:** It is an intersection of a row and a column where data is entered.
- **Active cell:** It is also referred to as **cell pointer** or **selected cell**. It is a rectangular box highlighting the cell being worked with and where data is being entered. By default A1 is usually the active cell. To make a cell active simply click on it.
- **Name box:** It displays the name of the active cell.
- **Column Title:** It is also known as **column heading** and is labelled in letters. This is a row containing the column labels.
- **Sheet Tab:** It contains the name of the worksheets in the workbook.
- **Row Title:** It is also known as **row heading** and is labelled in numbers. This is a column containing the row labels.

Parts of the Microsoft Excel Window

Below are the parts of Microsoft Excel window:



Picture 3. 5. The Microsoft Excel application window

Application activity 3.2

Part A: Fill in the dashes with the correct answers

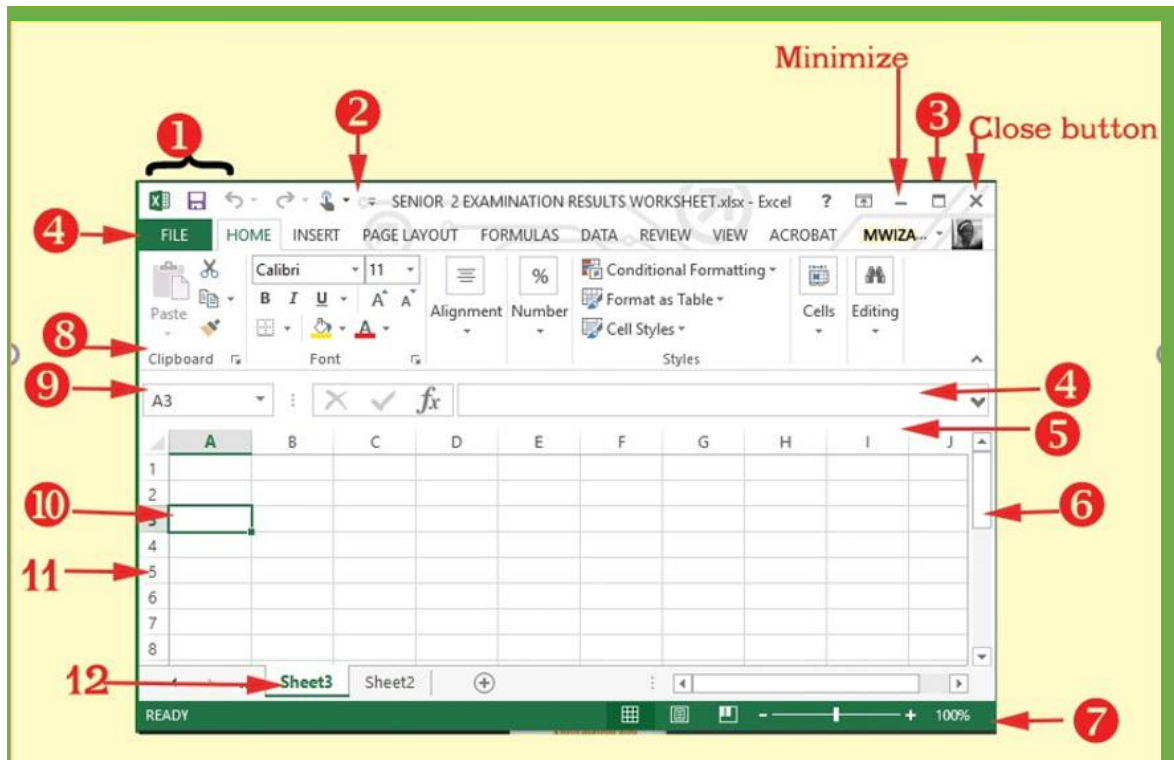
- 1) The default name of the first sheet in a workbook is _____.
- 2) A ribbon consist of __ of commands organised in groups.
- 3) The intersection of a row and a column where data is entered is known as_____.

Part B. Answer these questions

- 4) State two roles of a spreadsheet program.
- 5) Outline two possible ways of starting the Microsoft Excel program.

Part C: Label the diagram shown below:

- 6) Identify and label the parts numbered 1 to 12.



3.3 Cell, row, and column basics

Activity 3.3

- 1) What is the difference between a Word and Excel environments?
- 2) Explain why rows and columns are identified with letters and numbers.
- 3) What is the importance of rows and columns in a Spreadsheet?

A row is a range of horizontal cells having a unique number. They run vertically on the left side of the worksheet while a column is a range of vertical cells labelled using a unique letter. They run horizontally at the top of the worksheet.

3.3.1. Cell content

There are different types of data in a cell. These include Labels, Values, Functions and Formulas.

- **Labels:** They are text or alphanumeric characters that are entered in a cell which cannot be manipulated mathematically. By default, all labels are aligned to the left of a cell.

Entering a label

- (i) Click on the cell where the text is to be entered then type the text.
- (ii) Press the **Enter** key or click on a different cell.

- **Values:** They are numeric data that can be manipulated mathematically. They include numbers, dates, time and currency among others. A value can be entered as a label if it is preceded by an apostrophe. By default, all values are aligned to the right of a cell.

Entering values

- (i) Click on the cell where the value is to be entered then type the value
- (ii) Press the **Enter** key or click on a different cell.

- **Formulas**

They are mathematical expressions created by the user. Every formula must have the following components:

- (i) Begin with an equal sign (=).
- (ii) Contain mathematical operators such as +, -, / and *.
- (iii) Cell addresses that contain values.

- **Function**

They are in-built formulas in Excel. Every function must begin with an equals sign (=). Formulae can be typed in a cell in a worksheet but when the enter key is pressed, the actual cell displays the result of the calculation.

- a) **Entering data in a cell**

When data is typed in a cell, it is displayed in the selected cell as well as in the formula bar. The data enters the cell when the **Enter** key or **Arrow** key is pressed. Text can be wrapped on a cell by pressing **ALT+ Enter** while typing.

- b) **Selecting data**

This refers to highlighting a cell containing data that is supposed to be formatted. The table below summarizes the different ways of selecting data in a worksheet.

Selecting data in	Procedure
A single cell	Click on the cell; or use either the arrow or tab key on the keyboard.
Block cells	Do one of the following: • Click on the first cell of the block, hold down the shift key, and press the arrow key to extend the selection in the desired direction. • Click on the first cell of the block, press and hold down the left mouse button and drag the cursor until the desired number of cells are selected. • Click on the first cell of the block, hold down the shift key, and click on the last cell of the block.
Entire row/ column	Go to the column title or row title and click on the row number / column letter.
Non-adjacent cells	Hold down the control (CTRL) key as you click on the desired cells.
Entire worksheet	Click on the Select All button or press CTRL + A .

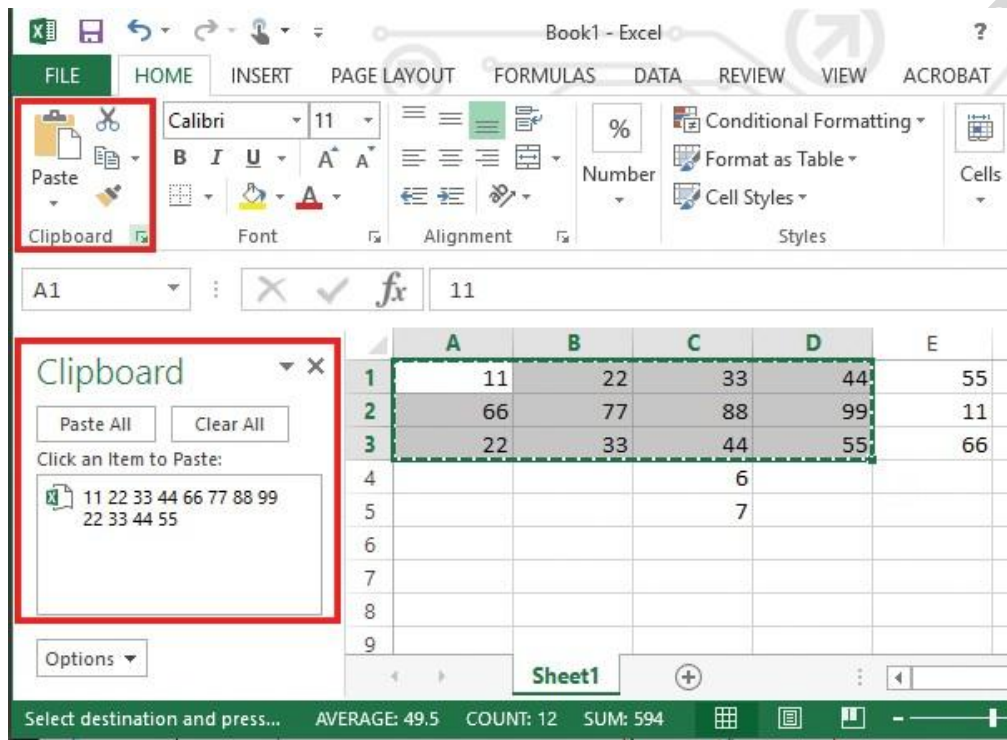
Table 3. 1: Different steps for selecting data

- c) **Copy and paste**

Copying is the process of creating a duplicate of data. The content remains in its original location as a copy of it is stored on the **clipboard**. A clipboard is a temporary storage location for items to be moved or copied. **Pasting** is the process of inserting the current clipboard content to a new location. To copy and paste, do the following

- (i) Select or highlight the data.

- (ii) Right-click on the highlighted data then select **Copy** from the pop-up menu. Press **Ctrl + C** on the keyboard. From **Home** tab, click **Copy** icon under the Clipboard group.
- (iii) Position the cursor where the duplicate is to be placed.
- (iv) Paste it by doing one of the following:
 - From **Home** tab, click **Paste** icon under the **Clipboard** group.
 - Or right-click at the cursor position and select **Paste** from the pop-up menu that appears.
 - Press **Ctrl+ V** on the keyboard.



Picture 3. 6. The Paste option in Excel

d) Cut and paste

Cutting is also known as moving. It is the process of changing the position of data to a new location. The data is deleted from its original location.

To move data, do the following:

- (i) Select or highlight the text or object.
- (ii) Use one of the following options:
 - From **Home** tab, click **Cut** icon under the **Clipboard** group.
 - Or right-click on the highlighted text or object then select **Copy** from the pop-up menu.
 - Press **Ctrl + X** on the keyboard.
- (iii) Position the cursor where the data is to be transferred.
- (iv) Paste it by using the procedure described under the section “copy and paste” specifically the content related on “pasting”.

3.3.2 Cell, row and column manipulation

a) Changing row height

The row height can be made smaller or larger than the default by following this procedure:

- (i) Place the mouse pointer at the boundary between two rows on the row title.
- (ii) When the mouse pointer changes to a two headed arrow, click and drag to the desired direction.
- (iii) Release the mouse button to apply the new height.

b) Changing column width

The column width can be made narrow or wider than the default by following this procedure:

- (i) Place the mouse pointer at the boundary between two columns in the column title.
- (ii) When the mouse pointer changes to a two headed arrow, click and drag to the desired column width.
- (iii) Release the mouse button to apply the new width.

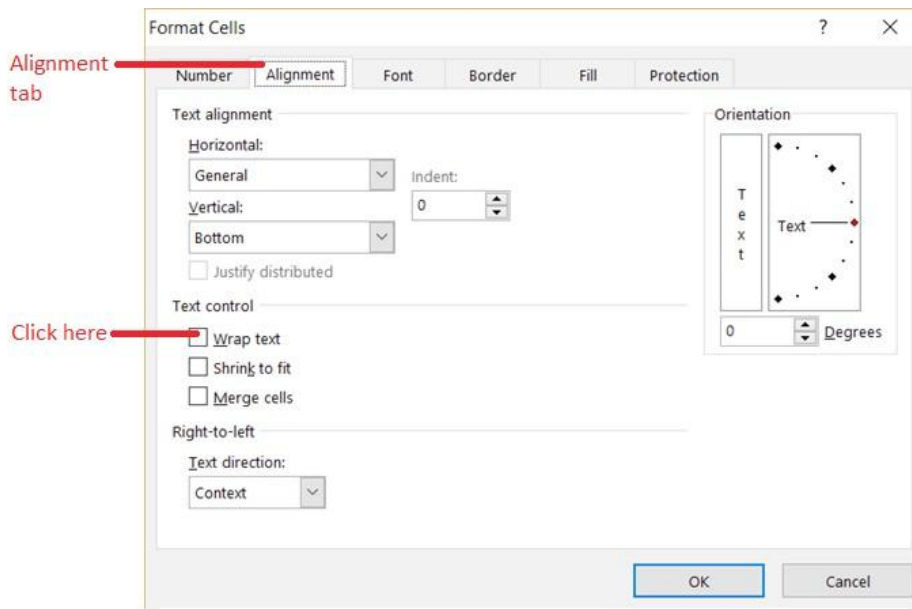
c) Wrap text

Wrap text feature enables a cell to contain content that is larger than its size and all the content is displayed in multiple lines. The cell is automatically enlarged as data is entered. To wrap text, do the following:

- (i) Select the cell(s).
- (ii) Click on the **Home** tab then select **Wrap text** icon in the **Alignment** group.

Alternatively:

- (i) Select the cell(s).
- (ii) Click on the **Home** tab then select **Format** icon under the **cells** group.
- (iii) Click on **Format Cells** from the drop down menu.



Picture 3. 7. Wrapping text

- (iv) Click on the **Wrap text** under the **Text control** section.
- (v) Click **OK** to apply and close the dialog box.

d) Merge cells

Merge cells refers to the combination of two or more horizontally or vertically adjacent cells to become one large cell that is displayed across multiple columns or rows. The cell reference for a merged cell is the upper-left cell in the original selected range. To merge cells, do the following:

- (i) Select the cells to be merged.
- (ii) Click on the **Home** tab then select **Merge & Center** icon in the **Alignment** group. The cells are automatically merged.

C2

X

✓

fx

Rwanda's Population in the Year 2002

	A	B	C	D	E	F
1						
2			Rwanda's Population in the Year 2002			
3			Age Group	Male Population	Female Population	
4			0-4	655,090	665,141	
5			5-9	563,851	577,688	
6			10-14	536,876	558,349	
7						

Sheet1

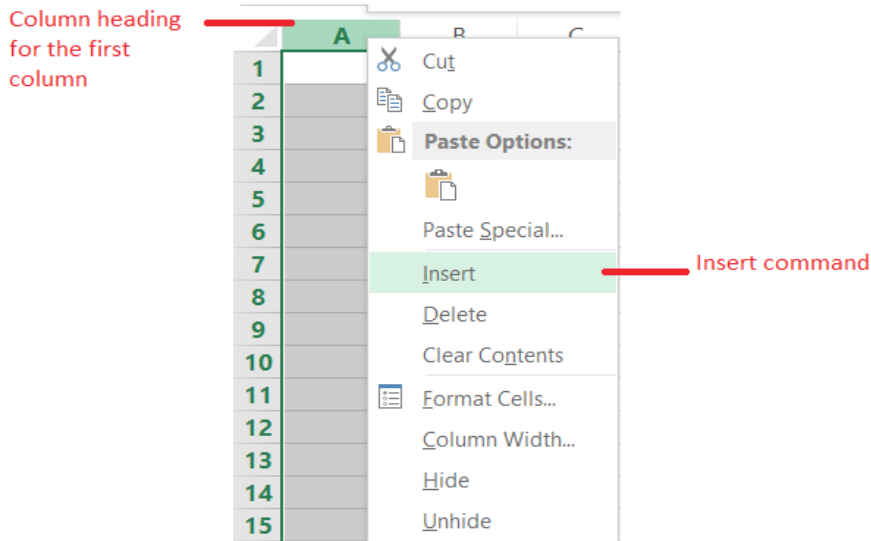
+

Picture 3. 8. Cells that have been merged

e) Inserting a column or a row

It is possible to insert a row or a column between existing rows or columns respectively. To insert a row or a column, do the following:

- Select the cell next to which the row or column is to be inserted.
- Do a right click and choose Insert



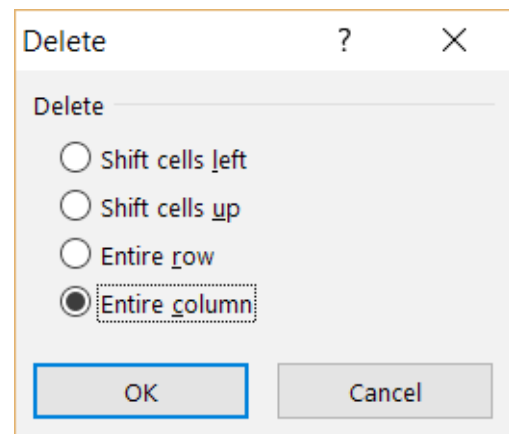
Picture 3. 9. Click insert to add a column to the left

- In the new dialog box select **Entire Row** or **Entire Column** depending on whether you want to insert a row or a column. Immediately the row or column get inserted.

f) Deleting rows and columns

To delete rows or columns in a worksheet do the following:

- Highlight a cell or cells to be deleted.
- Right-click on the selected cell(s) then select **Delete** option from the pop-up. A dialog box appears.
- Select **Entire row** option to delete the selected row(s) or **Entire column** option to delete the selected column(s) then click **OK**.



Picture 3. 10. Delete an Entire row

g) Moving rows and columns

To move a row or column in a worksheet, do the following:

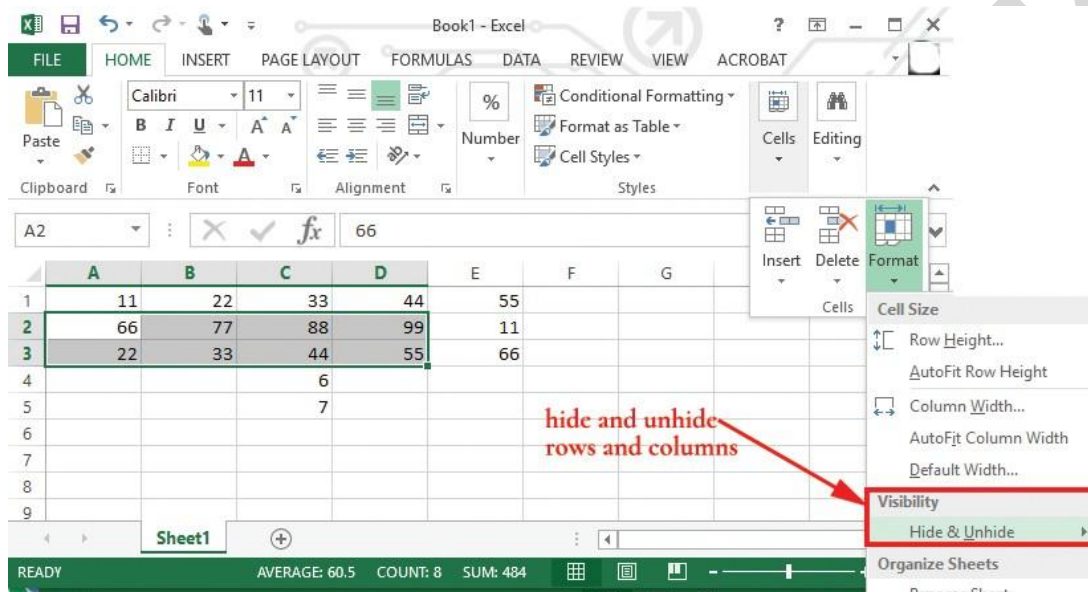
- (i) Select the row or column to be moved.
- (ii) Move the mouse pointer to the edge of the selection until it changes from a regular cross to a four-sided arrow cursor.

- (iii) Then drag the column or row to a new location.
- (iv) Release the mouse button to place the row or column to its new location.

h) Hide and unhide row or columns

This feature is used to temporarily hide rows and columns from view in order to prevent certain data from being changed or deleted. It is also used to simplify the appearance of a spreadsheet. To hide or unhide a row or a column; do the following:

- (i) From **Home** tab, on the **Cells** group click on **Format**.
- (ii) From the resulting drop down menu, point to **Hide & Unhide** then select the desired option from the side kick menu.



Picture 3. 12: Hide and unhide rows and Columns

Application activity 3.3

Part A: Answer the following questions

- 1) Differentiate between a row and a column.
- 2) Differentiate between a function and a formula.

Part B: Answer the following questions

3) The following are data that can be entered in a cell. Categorise each one of them as either formula, label, or value.

- (i) 45 (ii) Obey (iii) 34+56 (iv) =56+50

4) Create a workbook with one worksheet and enter the following details:

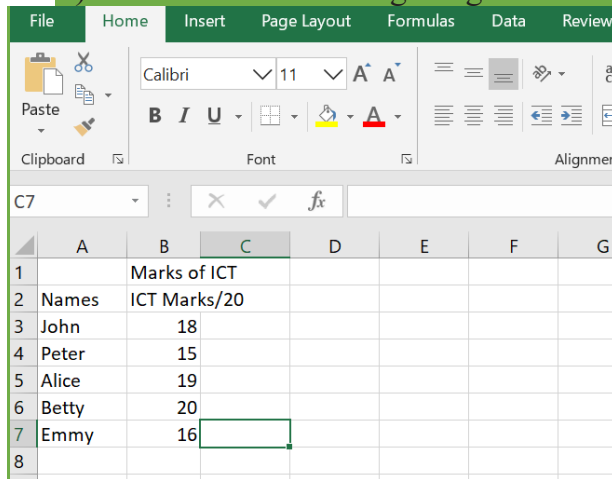
No.	Name	Maths	Science	English
1	Umutoni	56	64	93
2	Rwema	78	76	45
3	Kamali	90	88	90
4	Mulisa	45	34	46
5	Mwiza	67	78	77

- i) Insert two rows on top and type the title Final Analysis on the second row.
- ii) Merge the cells containing the title.
- iii) Hide the column labelled “No”
- iv) Save the document as “Analysis”.

3.4 Formatting a cell

Activity 3.4

1) Observe the following image and answer the questions



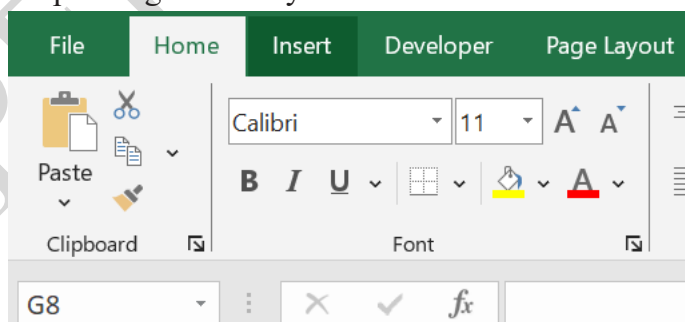
- a) Explain what to do for allowing "Marks of ICT" fit in its row.
- b) How can you change the background color of "ICT Marks/20"?

Formatting is the process of improving the appearance of a data in a worksheet. Some of the formatting options in Microsoft Excel are font, text alignment and orientation, cell borders and fill colours, and formatting numbers and text.

3.4.1 Font

To format font in cells using this option, do the following:

- Select the cells.
- Click on the **Home** tab then click on any one of the tools shown in the picture below depending on what you want to do.



Picture 3. 11. Font tools in Excel

In the table below are the font tools shown in the preceding image and their uses:

Tool	Use
B	For making the text bold
I	Used to make the text italic
U	Used for making the text underlined
Calibri	Used for changing the font type. The one in the box is Calibri .
11	Used for changing the size of the characters
A	Used for changing the color of the text. The color now available in the tool is red.
	It is used to fill color in the text

3.4.2 Text alignment and orientation

To format text alignment, do the following:

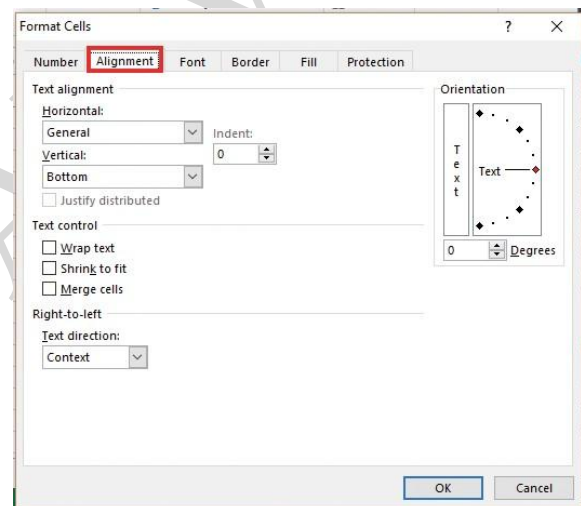
- Select the cells on which to apply the format.
- From **Home** tab, under the **Alignment** group, click on the dialog box launcher.
- Select the desired alignment option from the available options in the resulting dialog box.
- Click **OK** to apply.

The available options include:

- **Text Alignment:** It has three options, namely horizontal, vertical and indent.
- **Orientation:** Used to change the angle at which the text is inclined in the selected cells by using degrees.
- **Text control:** The options under **Text** control are used to adjust how the text should appear in a cell. This option include **Wrap text**, **Shrink to fit**, **Merge cells**.
- **Right-to-left:** It is used to specify reading order and alignment.

3.4.3 Cell borders

The worksheet gridlines do not appear when a worksheet page is printed. The cell border helps in ensuring that the cell outline appears in the print out. They are also used for enhancing the appearance of a document. To apply cell borders, do the following:

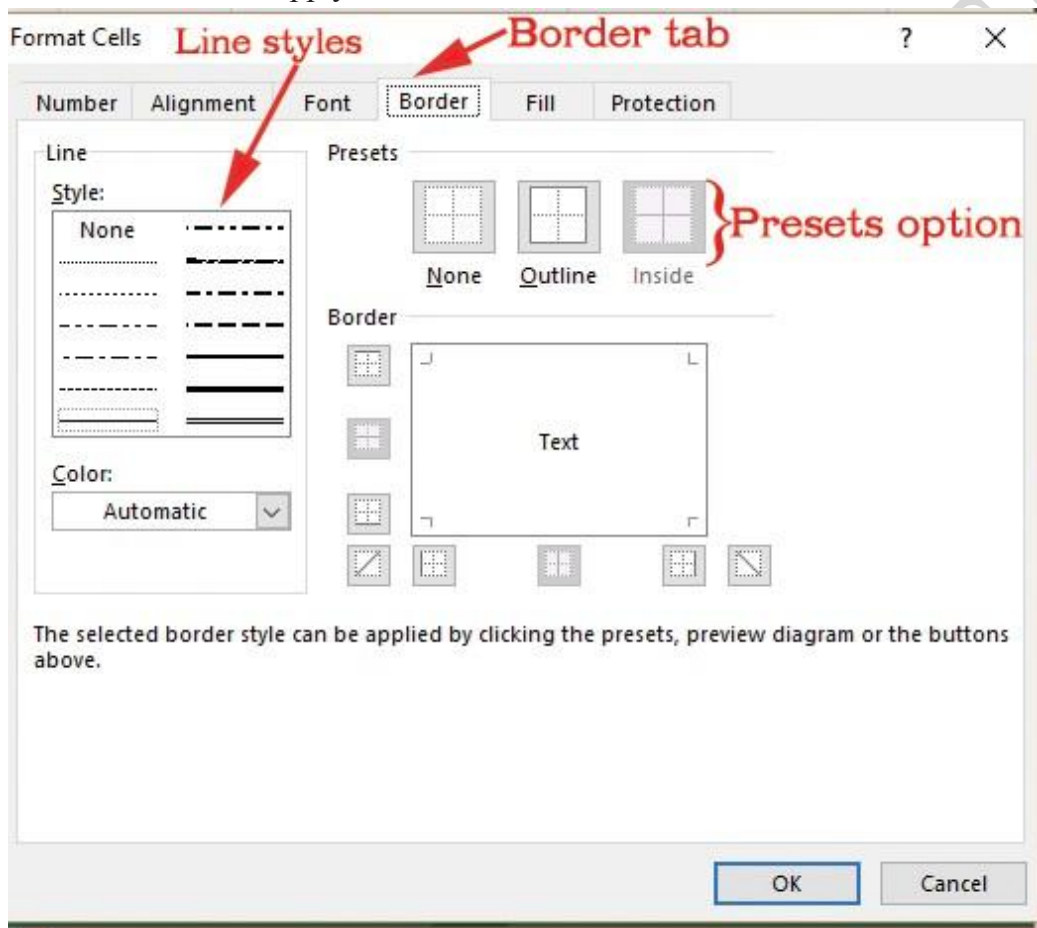


Picture 3. 12. Controlling and aligning text

- Select the cells to apply the format.
- From **Home** tab, under the **Font** group click on dialog box launcher.
- Click on **Border** tab. A dialog box will appear.
- Select the desired border style under **Style** option.
 - Click the buttons under **Presets** or **Border** to apply borders to the selected cells.

To remove all borders, click the **None** button. It is also possible to click areas in the text box to add or remove borders. To change the colour of the borders, click on the **Colour** list box and select the desired colour from the colour palette.

- Click **OK** to apply.



Picture 3. 13. Formatting cell borders

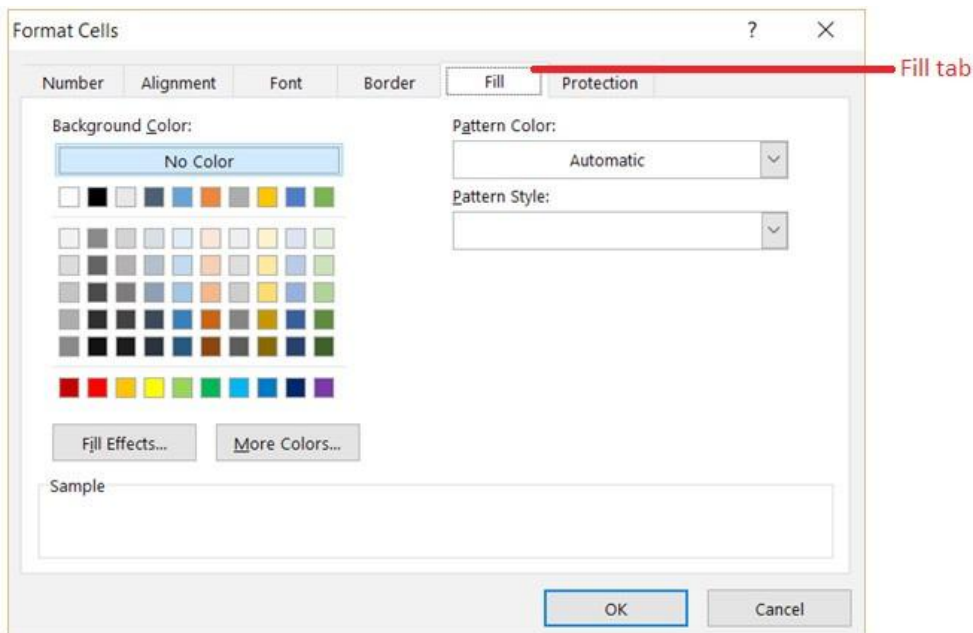
3.4.4 Fill color

To apply the fill colour command, do the following:

- Select the cells to apply the format.
- From **Home** tab, under the **Font** group click on dialog box launcher.
- Click on **Fill** tab. A dialog box will appear as shown in the picture.
- Select the desired fill style from the available options, namely background colour,

pattern colour, and pattern style and fill effects.

- Click **OK** to apply.

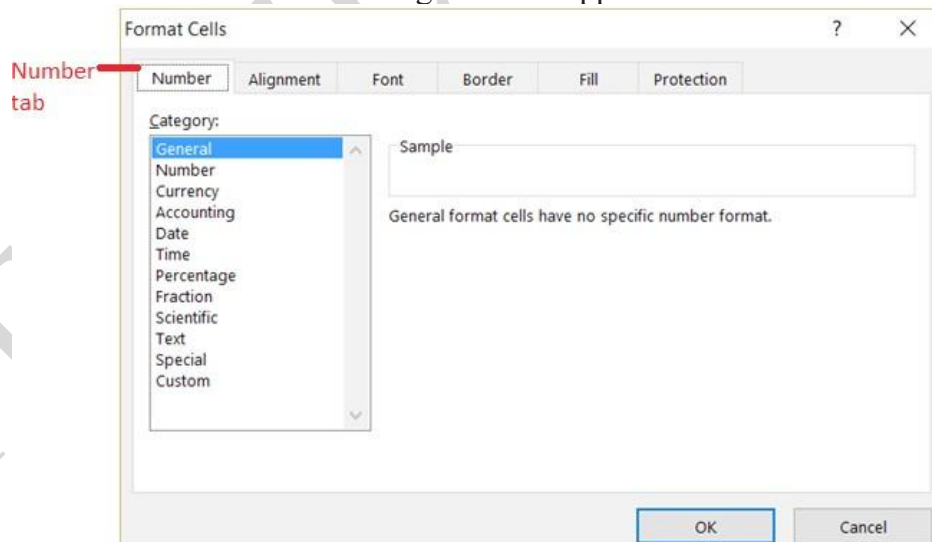


Picture 3. 14. Formatting colours

3.4.5 Formatting number

To change the format of numbers in Excel, select the cells to apply the format and do the following:

- From **Home** tab, under the **Font** group click on dialog box launcher.
- Click on **Number** tab. A dialog box will appear.



Picture 3. 17: Formatting number

- Under Category section, select an option and then select the desired options to specify a number format. The chosen format is shown as it will appear in the Sample box.
- Click **OK** to apply.

Application activity 3.4

Part 1: Choose the correct answer

- 1) Which of the following identifies a vertical line of cells in a spreadsheet?
a) Numbers b) Letters c) Symbols d) Dots
- 2) What is the correct way to write a cell address?
a) 1A b) A1 c) A-1 d) 1-A
- 3) If you are working in a horizontal line of cells, you are working in a:
a) Column b) Cell c) Row d) Workbook
- 4) What happens to the border of a cell when it becomes the "Active Cell"?
a) It disappears. b) It becomes thick and colored (usually green).
c) It starts blinking. d) It turns red.

Part 2. Answer the following questions

- 5) How can you tell which cell is active without looking at the grid?
- 6) Explain why a Spreadsheet uses both letters and numbers for its headers.

3.5 Worksheet basics

Activity 3.5

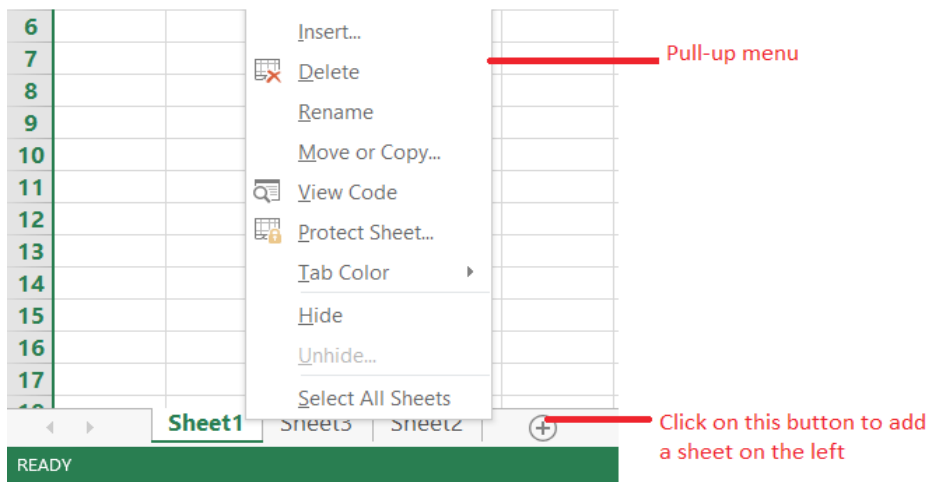
- 1) How can you insert a new worksheet in a workbook?
- 2) How do you delete a worksheet?
- 3) How can you rename a worksheet?
- 4) How do you copy or move a worksheet?

When excel is activated, a workbook is launched with a default worksheet. The user can insert, delete, rename, copy and move a worksheet. The following commands can be used when manipulating a worksheet.

3.5.1 Inserting a worksheet

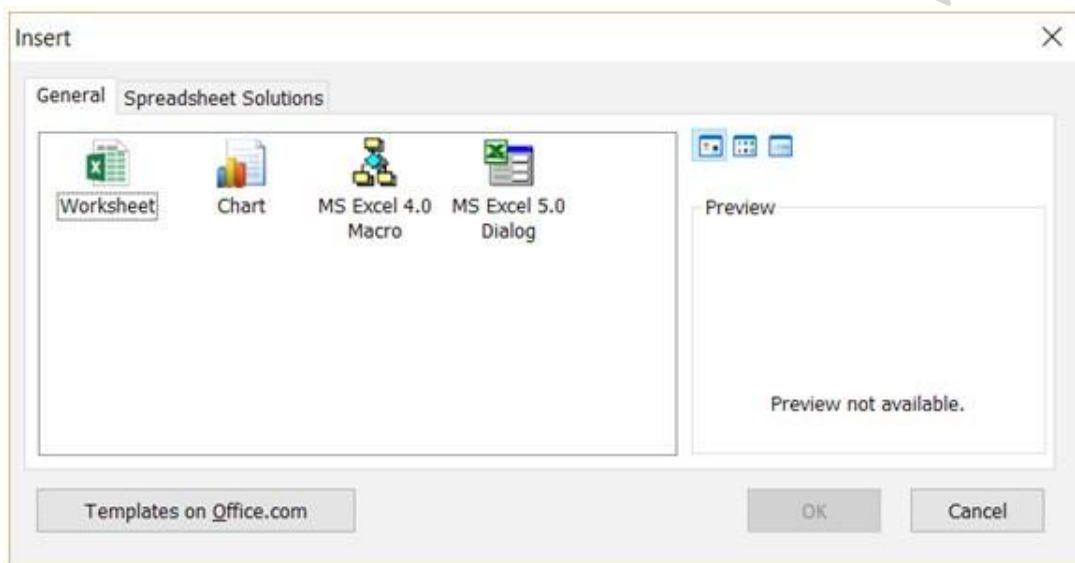
This feature enables the user to add a worksheet in a workbook. By default, the workbook in Microsoft Excel 2016 only comes along with one worksheet. To add another worksheet, do the following:

- Right-click on the name of the sheet. A pull-up menu appears as in the following picture.
- Select **Insert** option. A dialog box will appears.



Picture 3. 15: Adding a worksheet

- Click **OK** to apply. The new worksheet name appears before the current worksheet.



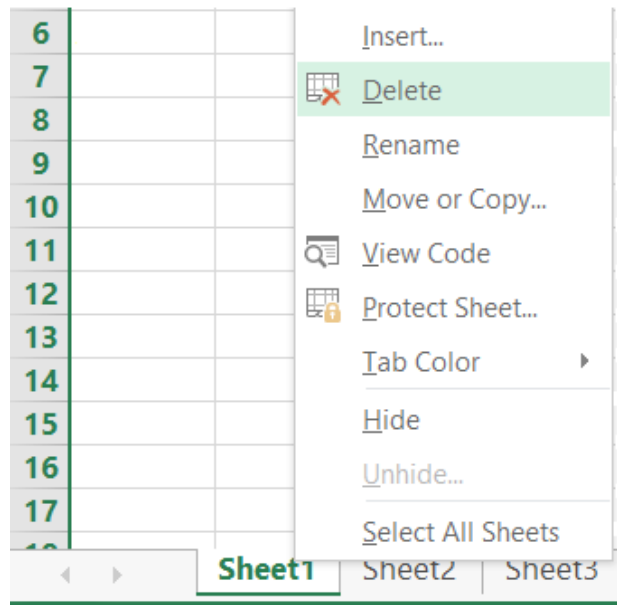
Picture 3. 16. Insert a worksheet

Note that you can insert a worksheet by simply clicking on the plus sign icon which is near the last worksheet name.

3.5.2 Deleting a worksheet

This option is used when the entire worksheet is to be deleted. Do the following:

- Right-click on the name of the sheet. A pull-up menu will appear
- Select **Delete** option. The sheet is automatically deleted

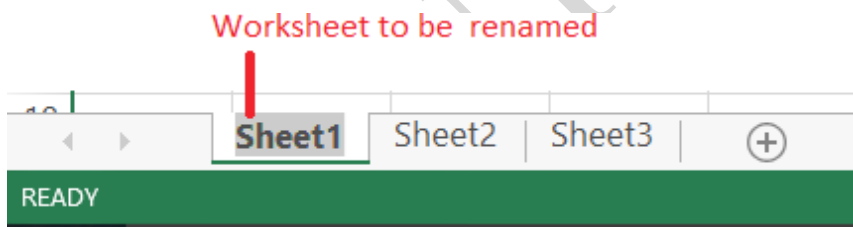


Picture 3. 17. Click on Worksheet name then select Delete to apply

3.5.3 Renaming a worksheet

Renaming is the process of assigning a worksheet a different name. To rename a worksheet, do the following:

- Double click on the name of the worksheet. The worksheet gets highlighted.

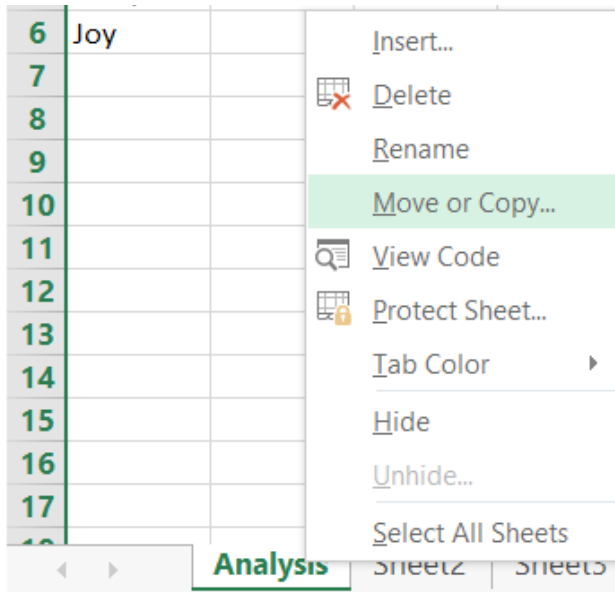


Picture 3. 18. Renaming a worksheet

- Type the desired name and then press **Enter**.

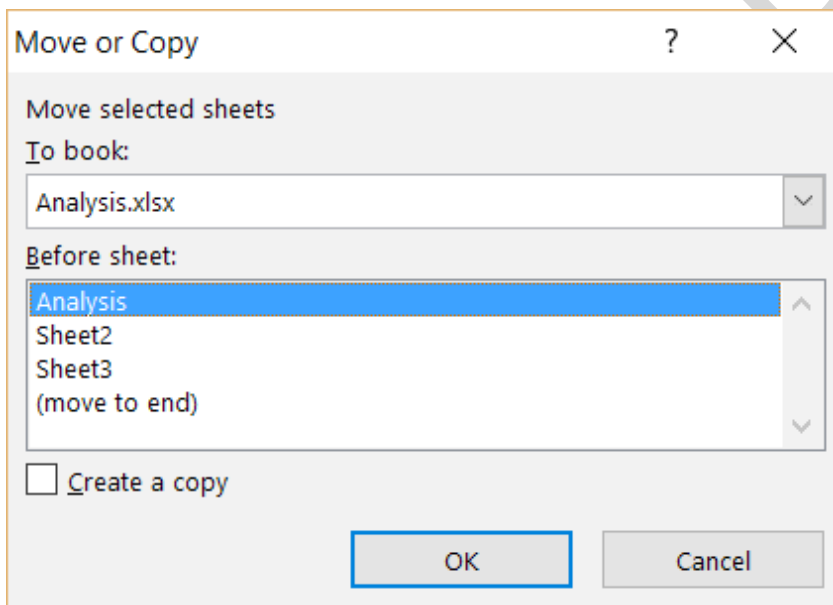
3.5.4 Copying and moving a worksheet

- Right-click on the name of the sheet. A pull-up menu will immediately appear.



Picture 3. 19. Moving or copying a worksheet

(ii) Select **Move or Copy** option. A dialog box appears as shown in the next picture.



Picture 3. 20. Moving a selected worksheet

iii) Select the book name in the **To book:** list and the worksheet to appear before it in the **Before sheet:** box.

iv) Click **OK** to apply.

When Copying click on the **Create a copy** check box to enable it duplicate the worksheet else it will be moved.

3.5.5 Grouping and ungrouping worksheets

Grouping worksheets allows people to treat multiple sheets as a single unit. When sheets are grouped, the tab colors lightens, and the word “**Group**” appears in the title bar at the top of the application.

Follow these steps to group worksheets:

- Select the first worksheet to be included in the group.
- Press and hold the CTRL key on the keyboard.
- Select the next worksheet needed in the group until you are done.
- Do whatever you want to do in the worksheet that you selected first and realize that the changes you make appear in the other worksheets.

Ungrouping the worksheets just click in any other worksheet immediately the grouping gets broken

Application activity 3.5

1) Open Excel and enter the following data.

No.	Name	Maths	Science	English
1	Umutoni	56	64	93
2	Rwema	78	76	45
3	Kamali	90	88	90
4	Mulisa	45	34	46
5	Mwiza	67	78	77

- a) Add two columns to the right of the column labelled English. Label the first column Total.
- b) Copy the content of this worksheet to another worksheet.
- c) Hide the column labelled No.
- d) Name the second worksheet as Final.
- f) Rename the first worksheet to Class.

3.6. Mathematical operations

Activity 3.6

- 1) What is an operator?
- 2) Give two examples of arithmetic operators and their uses
- 3) What are the advantages of using formulas in Excel?

In mathematics, the term **operator** refers to a sign or symbol used to evaluate mathematical operation. The most basic mathematical signs are the arithmetic operators which include addition (+), subtraction (-), multiplication (×), and division (÷).

$$Y = 15 + 3 \div 3 \times (12 + 5)$$

Brackets
operators operands

Picture 3. 21. A mathematical equation containing arithmetic operators

In mathematical computing, the same operators are used but multiplication and division operators are replaced with **asterisk (*)** and **forward slash (/)** respectively.

The table below gives a summary of the four mathematical operators supported in Excel and their functions:

Operator	Name	Description	Example (A=10, B=20)
+	Addition	Adds two operands	A+B returns 30
-	Subtraction	Subtract right operand from left	A-B returns -10
*	Multiplication	Multiplies binary operand	A*B returns 200
/	Division	Divides numerator by denominator	B/A returns 2

Table 3.2: Summary of the four mathematical operators

3.6.1 Addition operator

The addition operator (+) evaluates mathematical addition operation.

Example: Consider data of students 'performance in three subjects. The total marks for every student can be calculated in this way:

- Click on the first cell of the total column.
- Drag the mouse on the formula bar and type =B2+C2+D2
- Click on the bottom-right corner of that block and drag to the last cell of the total column.
- And release the mouse, the result will appear on all the specified cells of the column as it is shown below:

	A	B	C	D	E	F	G
1	Student's Name	Mathematics	Chemistry	ICT	Total		
2	Kundwa	46	50	68	164		
3	Kamali	67	52	90	209		
4	Sentwali	81	76	75	232		
5	Mukamana	38	49	64	151		
6	Habimana	70	92	86	248		
7	Mukeshimana	29	35	75	139		

Picture 3. 22. The result from addition Operation

3.6.2 Subtraction operator

The subtraction operator (-) evaluates mathematical subtraction operation.

Example: Using data stated in example above, Calculate the difference between Mathematics marks for Kundwa and Kamali. Perform the following steps:

- Click on the cell where the difference will appear.
- Write in cell the Subtraction formulas **=B3-B2**
- Click on enter button, the result will come in cell as it is shown bellow:

	A	B	C
1	Names	Mathematics	Chemistry
2	KUNDWA	46	50
3	KAMALI	67	52
4	SENTWALI	81	76
5	MUKAMANA	38	49
6	HABIMANA	70	92
7	MUKESHIMANA	29	35
9	Result	=B3-B2	

	A	B	C
1	Names	Mathematics	Chemistry
2	KUNDWA	46	50
3	KAMALI	67	52
4	SENTWALI	81	76
5	MUKAMANA	38	49
6	HABIMANA	70	92
7	MUKESHIMANA	29	35
9	Result	21	

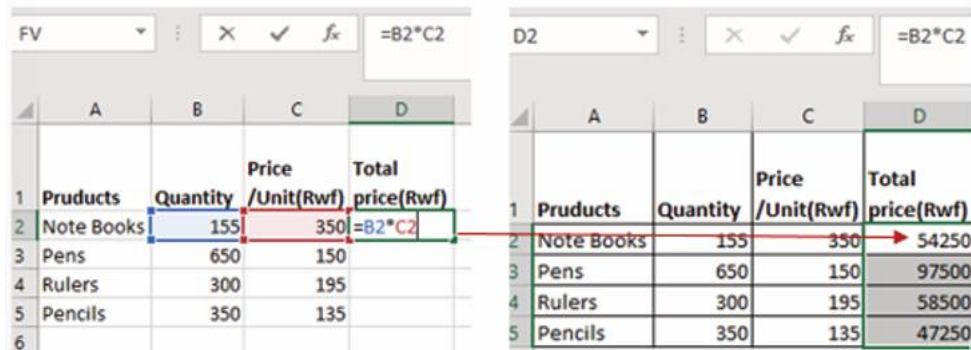
Picture 3. 23. The result from subtraction operation

3.6.3 Multiplication operator

The product of numeric values is performed by using a multiplication operator in formulas. Let's use the example of purchasing different products, where the total price as product will be obtained by multiplying quantity with price per unit. Here below, are the steps to be followed:

- Click on the first cell of the Product column.
- Drag the mouse on the formula bar and type **=B2*C2**

- Click on the bottom-right corner of that block and drag to the last cell of the Product column. Release the mouse button. You get the result as shown in the image below:



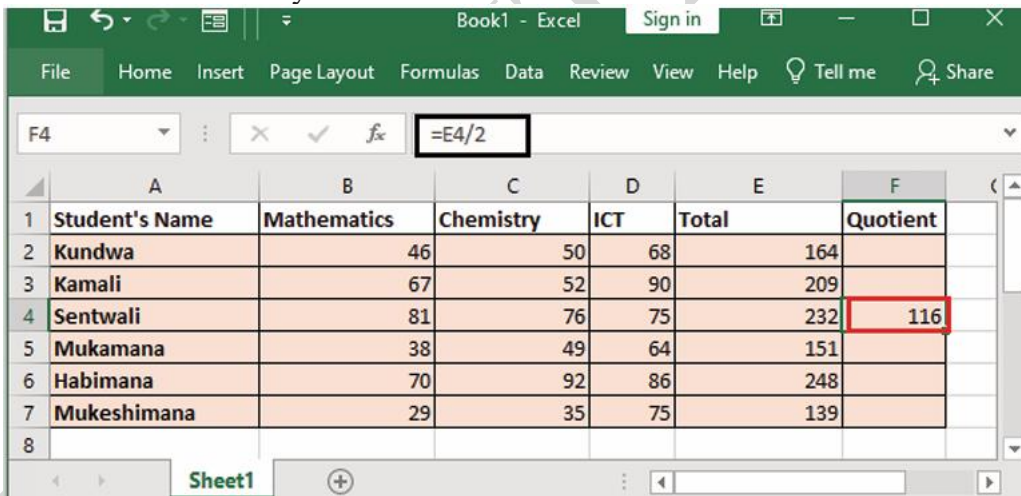
	A	B	C	D
1	Products	Quantity	Price /Unit(Rwf)	Total price(Rwf)
2	Note Books	155	350	=B2*C2
3	Pens	650	150	
4	Rulers	300	195	
5	Pencils	350	135	
6				

Picture 3. 24. The result from the multiplication operation

3.6.4 Division operator

The division operator is used in calculation of quotient of two numeric values. Use the same data of additional operator in order to understand how division operator works. **For example**, let divide total marks for Sentwali by 2. Proceed as follows:

- Select F4 cell of the Quotient column.
- Drag the mouse on the formula bar and type `=E4/2`
- Press Enter key to find the result



	A	B	C	D	E	F
1	Student's Name	Mathematics	Chemistry	ICT	Total	Quotient
2	Kundwa	46	50	68	164	
3	Kamali	67	52	90	209	
4	Sentwali	81	76	75	232	116
5	Mukamana	38	49	64	151	
6	Habimana	70	92	86	248	
7	Mukeshimana	29	35	75	139	
8						

Picture 3. 27: The result from division operation

Application activity 3.6

- Enter the marks below and perform different activities as indicated.

	A	B	C	D	E
1	Student's Name	Math	ICT Studies	Entrepreneurship	Total
2	Gahigi	60	50	50	
3	Mukamutara	38	74	76	
4	Uwimana	86	48	48	
5	Sentwali	40	80	90	
6	Habimana	61	40	39	

- Calculate the total mark for Gahigi.
- Calculate the difference between Maths and ICT marks for Gahigi.
- Multiply the Maths mark by 5 for Gahigi.
- Divide the total mark for Gahigi by 5.

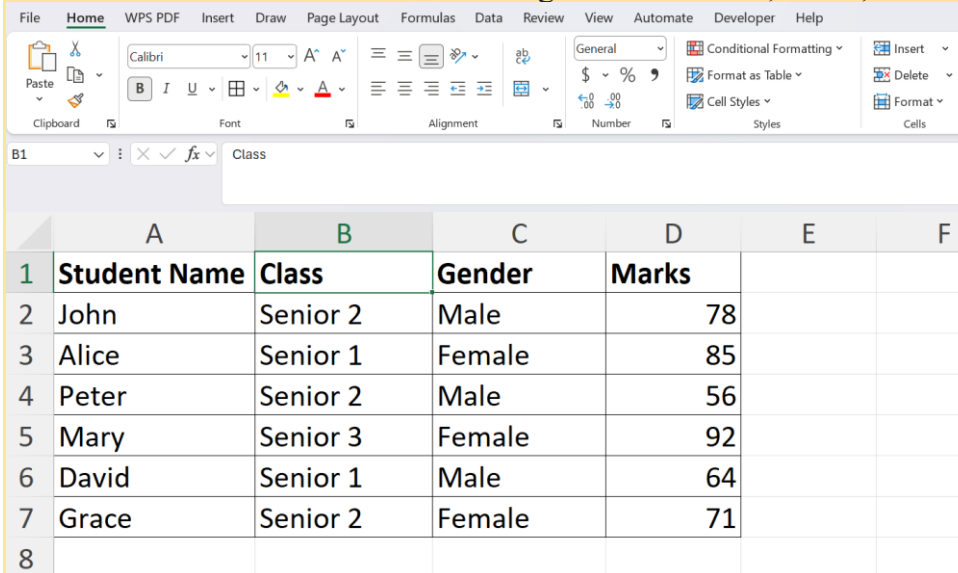
Part B: answer the following questions

- List three types of mathematical operators that can be used in calculations.
- State three components that must be contained in a formula.

3.7. Freezing panes

Activity 3.7

Observe the worksheet below containing **Student Name, Class, Gender, and Marks**.



	A	B	C	D	E	F
1	Student Name	Class	Gender	Marks		
2	John	Senior 2	Male	78		
3	Alice	Senior 1	Female	85		
4	Peter	Senior 2	Male	56		
5	Mary	Senior 3	Female	92		
6	David	Senior 1	Male	64		
7	Grace	Senior 2	Female	71		
8						

- When you scroll down and the column headings disappear, which Excel feature would you use to keep the headings visible?
- When you scroll to the right and student names are no longer visible, which feature helps keep the names displayed?

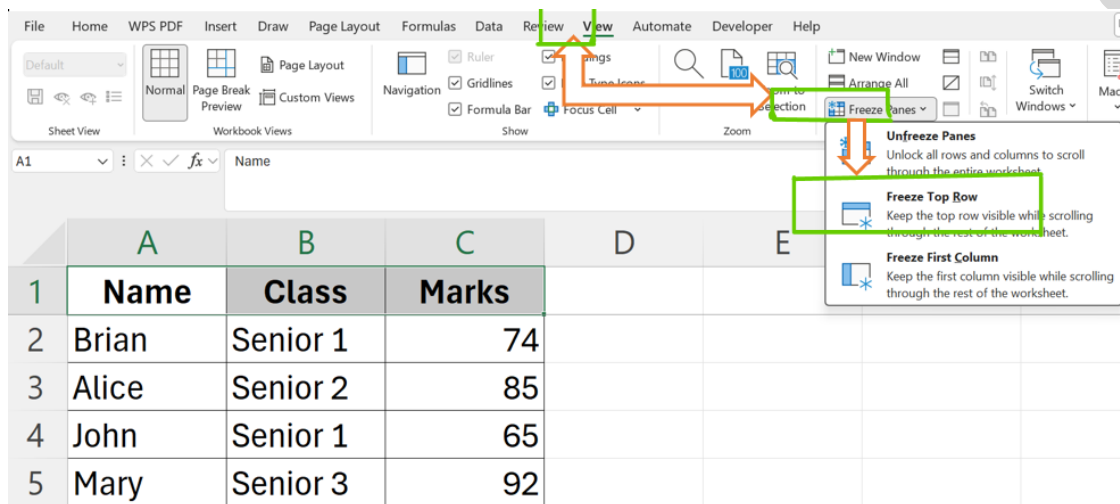
Freezing panes is a feature in Microsoft Excel that keeps selected rows or columns visible while you scroll through the worksheet. This is very useful when working with large datasets so that headings or key identifiers are always visible.

3.7.1. Freeze top row

This feature keeps the first row of the worksheet visible while scrolling down. It helps users avoid losing track of which column contains specific data as they move through the worksheet.

Steps to freeze top row

- Go to the **View** tab and click **Freeze Panes**.
- Select **Freeze Top Row**. Now, when you scroll down, the top row remains visible.



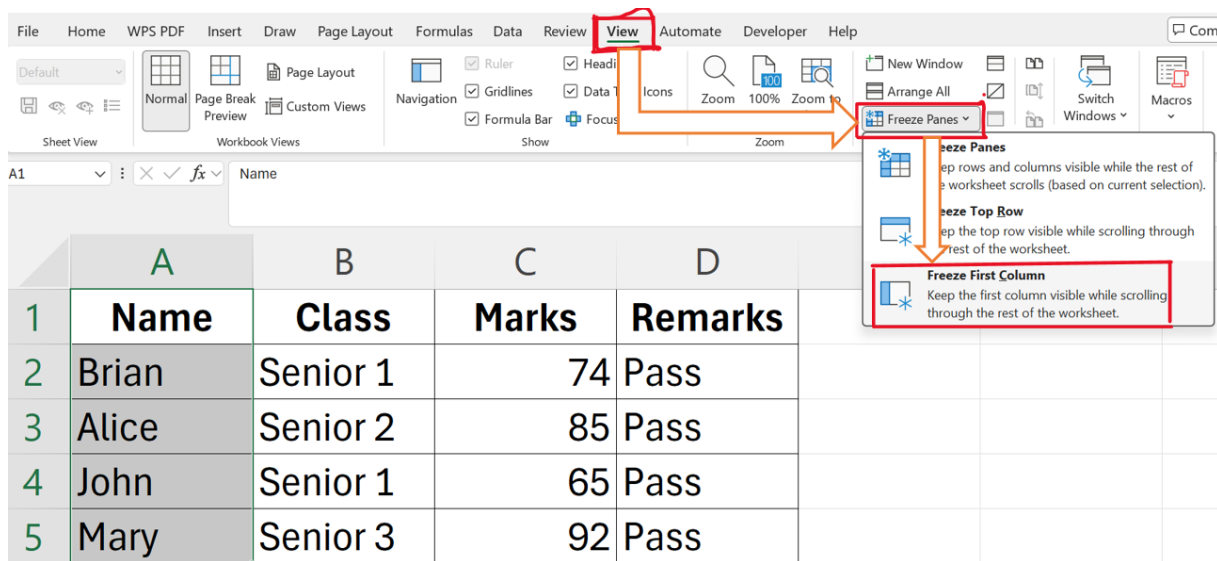
Picture 3. 25. Freeze Top Row

3.7.2. Freeze first column

This keeps the first column visible while scrolling to the right. It is ideal for **key identifiers** like student names, registration numbers, or IDs.

Steps to Freeze First Column

- Open your worksheet in Excel then go to the **View** tab and click **Freeze Panes**.
- Select **Freeze First Column**. Now, when you scroll horizontally, the first column remains visible.



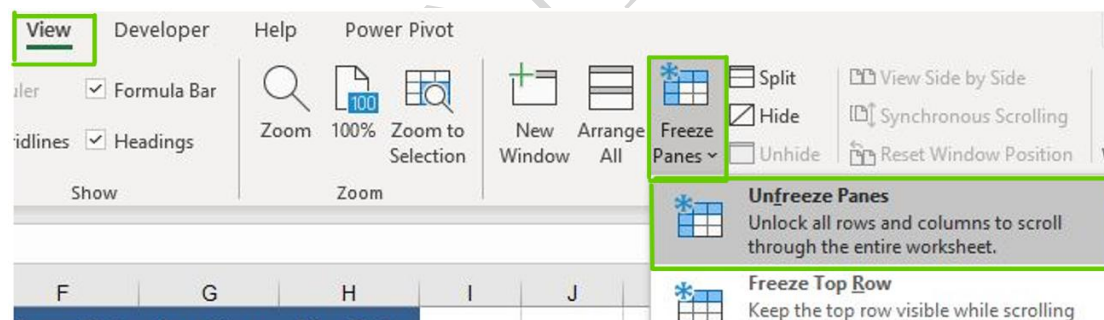
Picture 3. 26. Freeze first column

Note: Scrolling right will keep **Name** always visible, making it easier to track data per student.

3.7.3. Unfreeze Panes

Unfreezing panes will cancel all the freezing that was applied. Follow the steps mentioned below:

Go to **View** then click on **Freeze Panes** and click on **Unfreeze Panes**.



Picture 3. 27. Unfreeze panel

Application activity 3.7

Part A: Multiple choice questions

1) When working with a large worksheet and you want to keep column headings visible while scrolling down, which Excel option should you use?

- a) Split worksheet
- b) Freeze first column
- c) Freeze Top Row
- d) Page Layout view

2) Which Excel feature keeps student names visible while scrolling horizontally?

- a) Freeze First Column
- b) Split Worksheet
- c) Custom View
- d) Zoom

3) If you want to keep both the top row and first column frozen at the same time, which option should you choose?

- | | |
|------------------------|--------------------|
| a) Freeze Panes | b) Freeze Top Row |
| c) Freeze First Column | d) Split Worksheet |

Part B: Short answer questions

- 1) Why is Freeze Top Row useful in a worksheet containing many student records?
- 2) Give one example of when you would use Freeze First Column.
- 3) What happens to the frozen row or column when you scroll the worksheet?

3.8. Worksheet views

Activity 3.8

A worksheet contains a long list of students with their Names, Classes, and Marks. The teacher wants to work with the data normally on the screen and also view different parts of the worksheet at the same time.

- 1) Which worksheet view is best used for normal data entry and everyday work on the worksheet?
- 2) Which worksheet view feature allows a user to see two different parts of the same worksheet at once?

The Excel View options allow you to view or see the spreadsheet differently. You can adjust the Excel window to suit what you are currently working on by changing the view to match your current task.

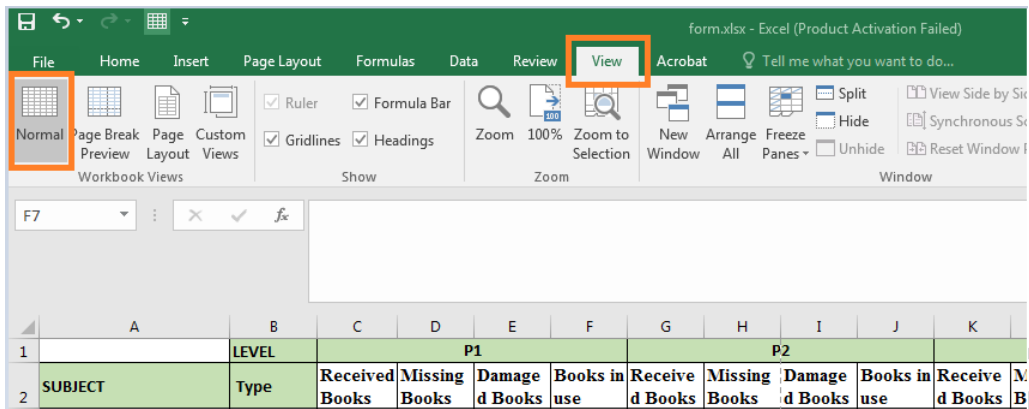
Excel offers three different views namely normal view, page layout view and page break preview.

3.8.1. Normal View

Normal View is the default view that opens automatically when a worksheet is opened. It is mainly used for entering data, editing cells, and creating formulas. It shows rows and columns clearly without showing printing details.

Here are the steps to be followed:

- ✓ Click the **View tab** on the Ribbon
- ✓ In the Workbook Views group, click **Normal icon**. Immediately Excel switches to **Normal view**.



Picture 3. 28. To switch to Normal View via the Ribbon

You can also switch to Normal view by clicking on its icon located at the bottom right of the Excel worksheet.



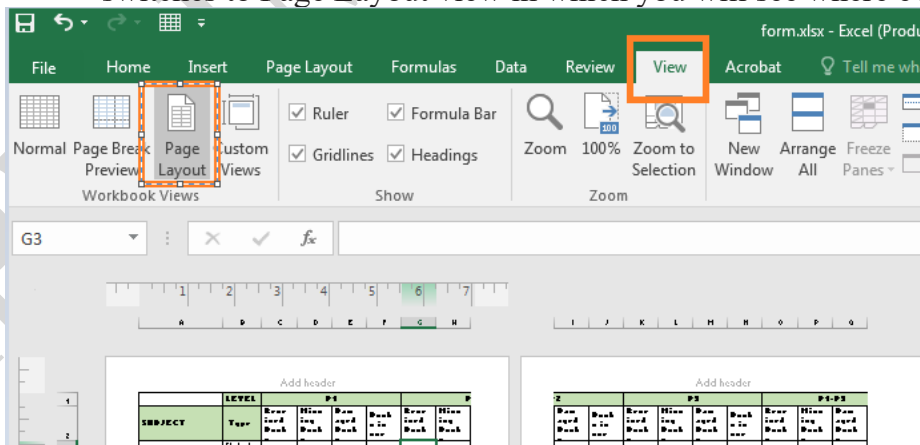
Picture 3. 29. Normal View via Shortcut Icon

3.8.2. Page Layout View

Page Layout View shows the worksheet exactly as it will appear when printed. It displays **page margins, headers, and footers**. Therefore it helps users arrange data properly before printing.

Here are the steps to be followed:

- ✓ Click the **View** tab.
- ✓ In the Workbook Views group, click the **Page Layout** icon. Immediately Excel switches to Page Layout view in which you will see where every page ends.



Picture 3. 30. Page Layout View via the Ribbon

You can also switch to **Page Layout View** by clicking the **Page Layout** button located at the bottom right of the Status bar.



Picture 3. 31. Page Layout View icon

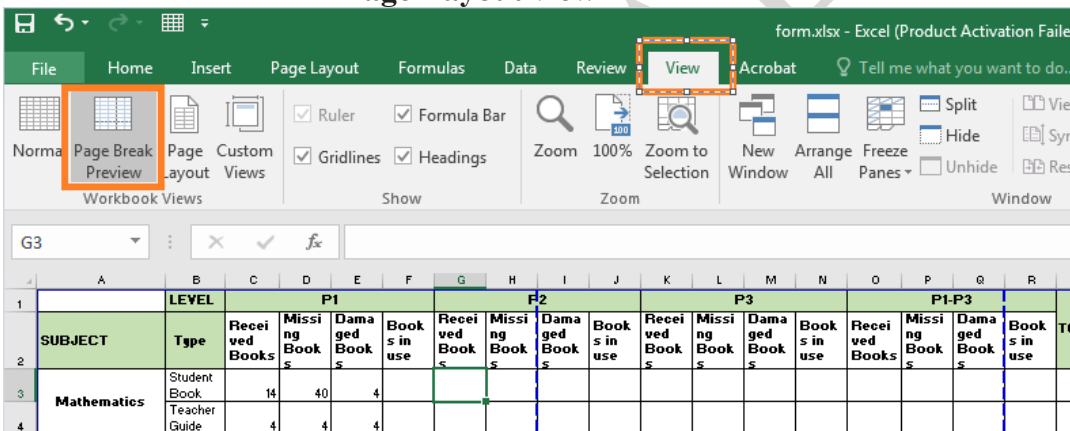
3.8.3. Page Break preview

This view displays worksheets as if you were going to print them out. This view is used to fine-tune a work sheet before printing.

- It displays **page boundaries** clearly.
- Users can move page breaks to fit data on fewer pages.

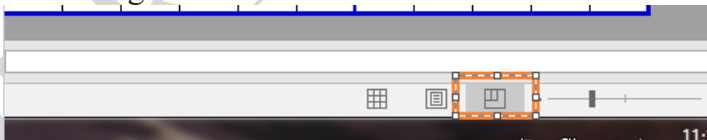
Here are the steps to be followed:

- ✓ Click the **View** tab.
- ✓ In the Workbook **Views** group, click the **Page Break Preview** icon. Immediately Excel switches to **Page Layout** view.



Picture 3. 32. Page Layout View via the Ribbon

You can also switch to the Page Break preview by clicking on its icon located at the bottom right.



Picture 3. 33. Page Layout View icon

Application Activity 3.8

Part A: Choose the correct answer.

1) Which worksheet view feature keeps the first row visible when scrolling down?

- | | |
|--------------------|-------------------|
| a) Split Worksheet | b) Freeze Top Row |
| c) Custom View | d) Zoom |

2) To switch views using the Ribbon, which tab must be clicked first?

- a) Home b) Insert c) Page Layout d) View

Part B: Answer the following questions.

3) State one use of Freeze First Column in a worksheet.

4) Explain the primary difference between Normal View and Page Layout View.

3.9. Sorting and filtering

Activity 3.9

A worksheet contains students' Names, Class, Gender, and Marks and the teacher wants to organize and view specific information easily.

- 1) How can you arrange the students' marks so that the highest marks appear first in the worksheet?
- 2) How can you display only students who scored more than 60 marks without deleting other records?

3.9.1 Filtering data

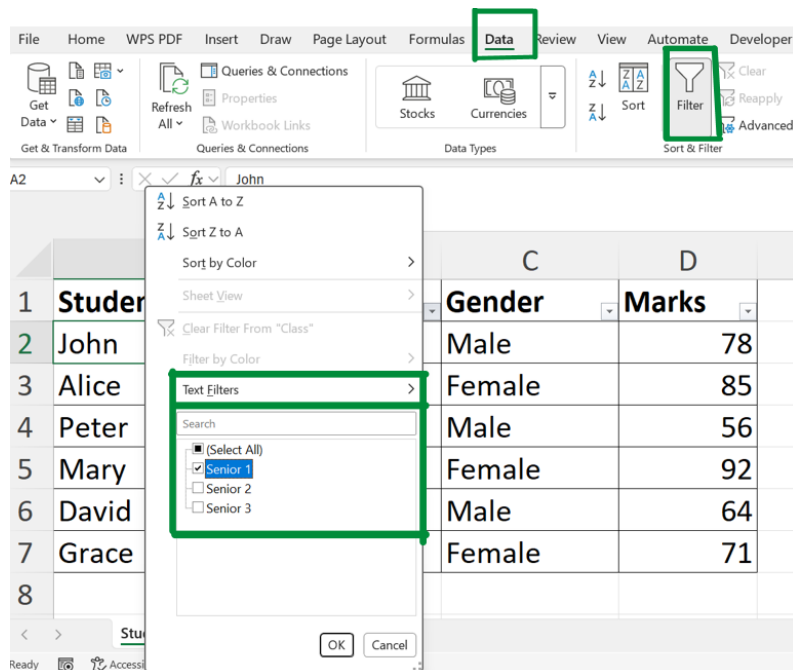
After entering data in Excel, it is also possible to filter, or hide some parts of the data, based on user-indicated categories. When using the Filter option, no data is lost; it is just hidden from view. Filtering displays only the data that meets specific conditions while hiding the rest of the data.

a) Filter by Text

This option shows only rows that contain specific text. An example is when we want to display only students that are in "Senior 1".

Follow these steps:

- Click on **Data** and on **Filter** then click the drop-down arrow in the text column.
- Choose **Text Filters** and select a condition.



Picture 3. 34. Filter by Text

b) Filter by Number

This option filters data based on numerical values. In this case the filter options include Greater than, Less than, Equal to and Between. An example is when we want to show only students who scored above 60 marks.

Steps to apply:

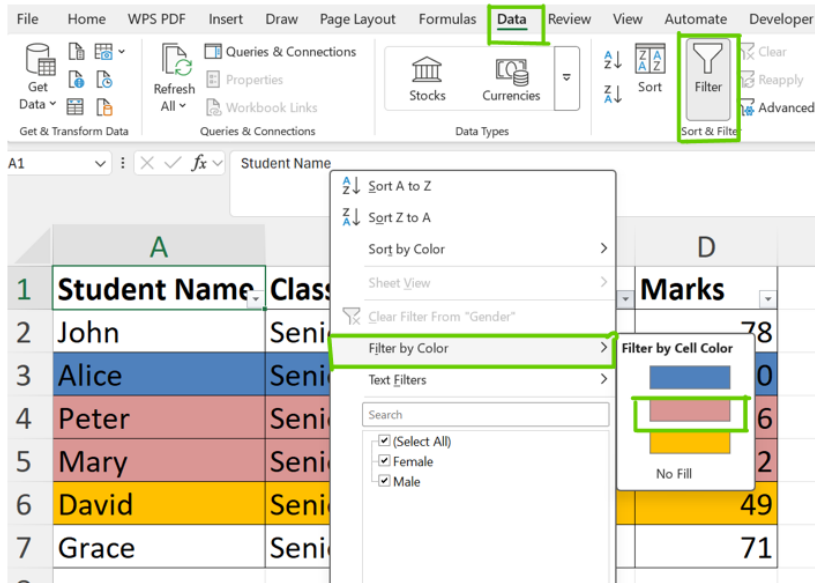
- Click on **Data** then click on the **Filter** tool.
- On the title row under the specific column on which you want to apply the filter Click the drop-down arrow.
- Choose **Number Filters** and set the condition.

c) Filter by Color

This option displays rows with cells that have a specific **cell color**, **font color**, or **icon**. An example is when you want to show only students whose marks are highlighted in red for low performance.

Steps to apply:

- Click on **Data** then on **Filter**
- Under the title on which the filter will be applied click the filter arrow
- Choose **Filter by Color** and select the color.



Picture 3. 35. Filter by Color

3.9.2. Sorting

Sorting is the process of arranging data in a specific order based on selected criteria such as names, numbers, or dates.

a) A to Z or Smallest to Largest

This option arranges data from the **lowest to the highest** or from **A to Z** making it easy to locate information and helping compare values quickly.

Example: Sorting student names from A to Z or arranging marks from the lowest score to the highest score.

Follow these steps to sort data:

- Select the column to sort then click on **Data**
- **Sort A to Z** (or Smallest to Largest).

Sorting Student Names from A to Z or Arranging Marks from Ascending Order

1. Click 'Sort A to Z'.

2. Sorted from A to Z and Smallest to Largest

	A	B
1	Names	Marks
2	Mary	85
3	Brian	74
4	John	65
5	Mary	92

	A	B	C	D	E
1	Class	Names	Marks		
2		Brian	74		
3	Senior 1	Alice	85		
4	Senior 2	John	65		
5		Mary	92		

2. Sorted from A to Z and Smallest to Largest

Picture 3. 36. Sort A to Z or Smallest to Largest

b) Z to A or largest to smallest

This option arranges data from the **highest to the lowest** or from **Z to A**. It Helps identify top or bottom values easily.

An example is sorting exam results to show the highest marks first.

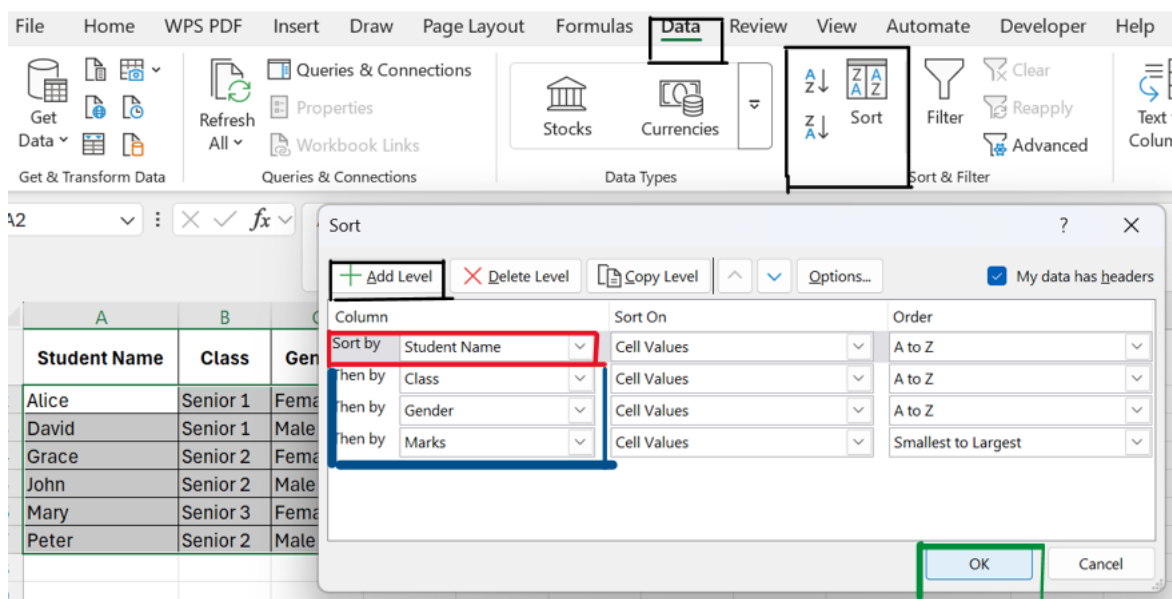
It is done by clicking on the **Data** tab then clicking on the **Z A** tool. The steps are the same as sorting from A to Z.

c) Custom sort

Custom Sort allows users to sort data using more than one column. Organize data in a more detailed and meaningful way. An example is when sorting students by Class, by Marks, by gender then by marks.

Follow these steps:

- Click on **Data** then click on the **Sort** icon
- In the new dialog box, choose the first column and order.
- Click **Add Level** to add another column to use as a second sorting rule.
- Add other columns you want then click **OK**.



Picture 3. 37. Custom Sort

Application activity 3.9

A. Multiple choice questions

- 1) Which Excel feature arranges data in alphabetical or numerical order?
a) Filter b) Sort c) Freeze Panes d) Split
- 2) What does filtering data do?
a) Deletes unwanted data b) Changes the values of data
c) Displays only selected data d) Rearranges data permanently
- 3) Which filter should be used to show students who scored above 70?
a) Text Filter b) Color Filter c) Number Filter d) Custom Sort
- 4) Which filter option is used to show only cells with a specific color?
a) Number Filter b) Text Filter c) Color Filter d) Custom Filter

B. Practical

- 5) Record the marks that at least 10 classmates got in Mathematics, Chemistry and Biology then do the following:
a) Sort the names from A to Z
b) Arrange your data so that the names are ordered starting from those who got more marks in Mathematics
c) Apply multiple sort so that the list gets ordered by Marks. Start with Mathematics, then Biology and finally Chemistry.

END UNIT ASSESSMENT

Part 1: Choose the correct answer:

- 1) Which of the following is the intersection of a row and a column?
a) Label b) Workbook c) Cell d) Block
- 2) In Excel, rows are identified by _____ while columns are identified by _____.
a) Letters / Numbers b) Symbols / Colors
c) Names / Addresses d) Numbers / Letters
- 3) What is the default name given to the first worksheet when Excel is launched?
a) Sheet1 b) Book1 c) Page1 d) Document1
- 4) Which of the following is an example of an in-built formula in Excel?
a) Label b) Value c) Function d) Active Cell
- 5) Which feature combines two or more adjacent cells into one large cell?
a) Wrap Text b) Shrink to Fit c) Merge Cells d) Format Painter
- 5) What is the correct cell address for a cell in the third column and fifth row?
a) 3C b) 5C c) C5 d) 3-5
- 6) Which bar displays the content or formula of the active cell?
a) Title Bar b) Formula Bar c) Status Bar d) Scroll Bar
- 7) Which command is used to change the position of data from one location to another by deleting it from the original spot?
a) Copy and Paste b) Cut and Paste c) Wrap Text d) Save As

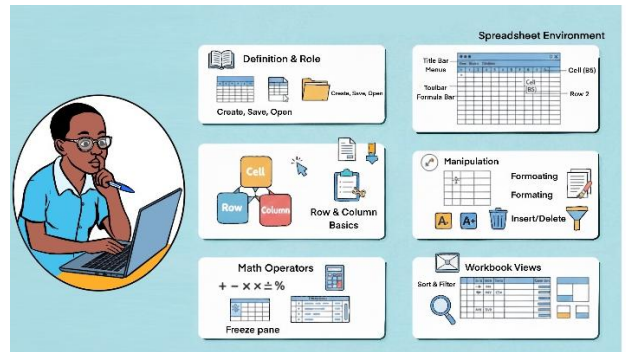
Part 2: Answer the following questions.

- 1) If you have a long student name that doesn't fit in a narrow column, which formatting feature should you use to see the whole name without widening the column?
- 2) Why is it important to use a "Cell Address" (like A1) in a formula instead of just typing the numbers (like 10 + 20)?

Part 3: Do the following questions

- 1) Create a workbook with one worksheet. Enter the headers "Item Name" in A1 and "Price" in B1. In A2, type "Bread" and in B2, type "800". Save the document as "Canteen_Inventory".
- 2) In a new row above your data, merge cells A1 and B1 and type the title "Daily Sales". Then, calculate the total of two items in cell B4 using a formula.

UNIT 4 COMPLEX FORMULA AND FUNCTION



Key Unit competency:

Apply conditional formatting, complex formulas, functions, absolute and relative referencing to the contents of a worksheet accurately and correctly.

Learning objectives

At the end of this unit you will be able to:

Knowledge and understanding:

- Explain how formulas and functions work in Excel and the role of different operators in calculations.
- Discuss the importance of cell referencing when working with formulas.
- Explain how conditional formatting is used to visually analyze and interpret data in worksheets.

Skills:

- Apply formulas and functions effectively to manipulate and analyze data using mathematical, logical, and text functions.
- Construct and manage complex formulas, using multiple operators, correct cell references, and links across worksheets.
- Use Excel tools accurately, including cell referencing and conditional formatting, to improve data organization and analysis.

Attitudes and values:

- Develop accuracy, responsibility, and discipline when creating and managing formulas and worksheet data.
- Appreciate the importance of well-organized and well-formatted worksheets for productivity, collaboration, and effective communication.

4.0. Introduction

Spreadsheets are programmed in such a way that they can accept different formulas and functions. Formulas may comprise a combination of actual values, cell references, functions and mathematical operators. When the correct formula is typed in the formula bar, the result of the formula appears in the active cell. In this unit you will learn how to automate different operations by using Excel formula and functions.

Introductory activity

Your school administration has noticed a difference in how exam results are processed. Some teachers still calculate totals and averages manually on paper or using calculators while others use computers with Excel spreadsheets.



One teacher uses Excel, where results are calculated automatically and displayed clearly using tables and charts while another teacher uses a calculator, entering marks one by one. This method takes more time and requires repeated checking. The third teacher calculates results manually on paper, surrounded by piles of papers, feeling stressed.

- 1) What differences do you notice between using a computer, a calculator and manual paper calculations?
- 2) Which method is the fastest and most accurate? Why?
- 3) What kinds of errors may occur when results are calculated manually or with a calculator?
- 4) How can Excel help reduce mistakes when calculating totals and averages?

4.1 Predefined operators and symbols in formula

Activity 4.1

Mutoni is the accountant of the school's Entrepreneurship Club. She has a list of items bought for the school canteen, including the quantity and the unit price. She needs to calculate the total cost for each item then apply a 5% tax to the final total.

Mutoni tries to type the calculations like she does in her math notebook: Total = 10 x 500. However, Excel does not show any result; it just displays the text she typed. She also realizes she isn't sure which symbol to use for "divide" or "multiply" because the keyboard doesn't have a $\times$ or a $\div$ key.

- 1) a) What is the first symbol Mutoni must type in a cell to tell Excel she is writing a formula?
- b) What happens if she forgets it?
- 2) Which keyboard symbols should Mutoni use for Multiplication and Division?
- 3) If Mutoni wants to add two prices first and then multiply by a quantity, which symbol must she use to ensure the addition happens first?

4.1.1. Predefined operators and symbols

Operators and symbols are the building blocks of any calculation. They tell the program exactly how to handle the data you've entered. Below are the operators and symbols that are found in an Excel formula:

Category	Symbol	Name	Description	Example
Arithmetic	+	Plus Sign	Addition	=A1 + 10
	-	Minus Sign	Subtraction	=B5 - A1
	*	Asterisk	Multiplication	=C2 * 5
	/	Forward Slash	Division	=D10 / 2
	%	Percent Sign	Percentage	=A1 * 15%
	^	Caret	Exponentiation	=A1 ^ 2
Comparison	=	Equal Sign	Equal to	=A1 = B1
	<>	Not Equal To	Not equal to	=A1 <> B1
	>	Greater Than	Greater than	=A1 > 50
	<	Less Than	Less than	=A1 < 10
	>=	Greater or Equal	Greater than or equal to	=A1 >= 20
	+<=	Less or Equal	Less than or equal to	=A1 <= 5
Reference	:	Colon	Range of cells	=SUM(A1:A10)
	,	Comma	Union of ranges	=SUM(A1, C1)
		Space	Intersection of ranges	=(A1:C1 B1:B10)
Text	&	Ampersand	Joins text (Concatenate)	="Hello " & A1
	"	Double Quotes	Identifies text strings	"Total Sales"
Grouping	()	Parentheses	Controls calculation order	=(A1+B1)*5

4.1.2 Complex formulas

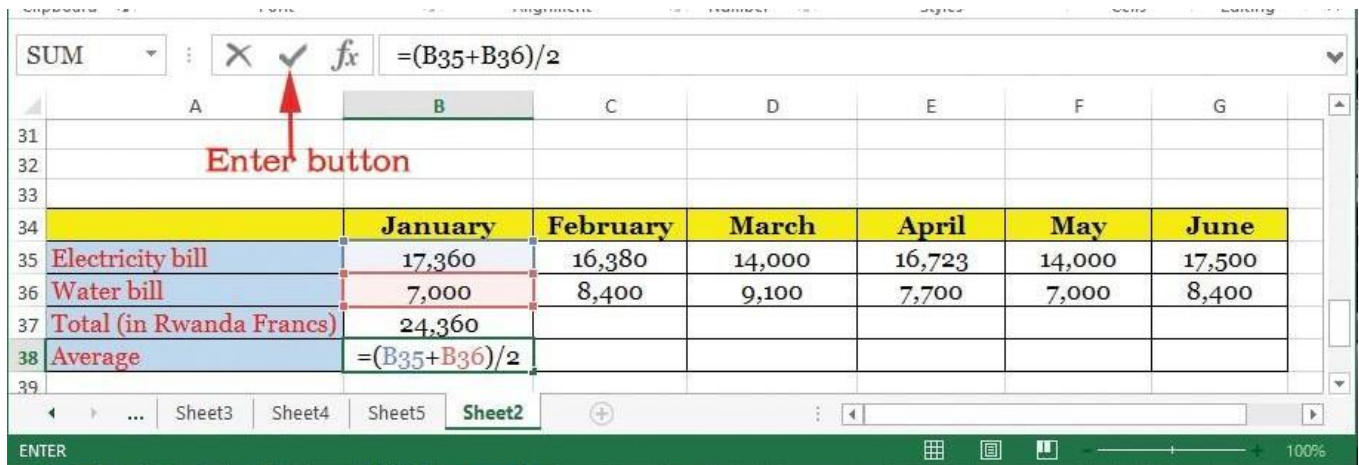
Excel has a provision of using complex formula. They are written using more than one operator. Example of a complex formula is $= (B2 + B3)/2$ written in cell B5. In this case, the operators used are addition and division.

a) Creating complex formulas

The following are the steps followed when creating complex formulas.

- Click the cell where the result of the formula is to be displayed such as B5 and type the equal sign.
- Type an opening parenthesis.
- Click on the first cell to be included in the formula or type the cell address such as B2.
- Type the mathematical operator such as (+).
- Click on the second cell in the formula or type the cell address such as B3.
- Type a closing parenthesis.
- Type the next mathematical operator such as (/).
- Type the next cell address or value such as 2.
- Press Enter, or click the Enter button on the Formula bar to end the formula.

For example, the picture below shows the formula $= (B35+B36)/2$ entered in cell B38.



Picture 4. 1. The formula for computing the average bill for January

Application activity 4.1

- Which of the following operators would you use if you wanted to calculate the square of a value in cell A1?
 - %
 - /
 - *
 - ^
- In the complex formula $= (B2 + B3) / 2$, why are the parentheses used?
 - To multiply the sum by the divisor
 - To ensure addition happens before division
 - To indicate the start of a formula
 - To identify the cells as numbers
- In the formula $= "Hello" \& A1$, what is the purpose of the double quotes?
 - To join the text together
 - To calculate the value of cell A1
 - To start the formula
 - To identify Hello as a text string
- If you change the formula $= (10 + 10) / 2$ to $= 10 + 10 / 2$, the answer changes from 10 to 15. Why do you think the parentheses are important?

4.2. Cell references

Activity 4.2

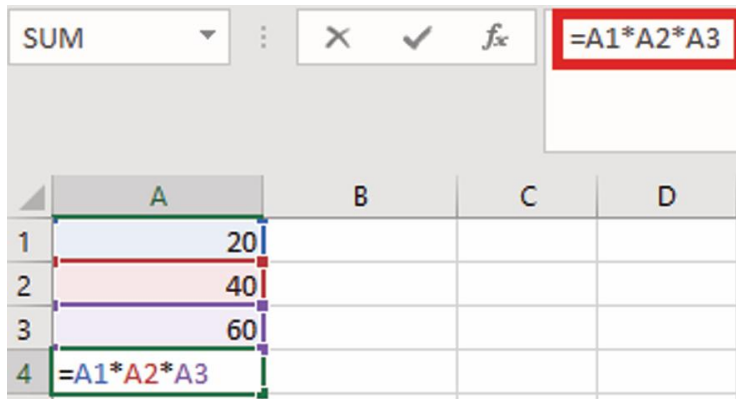
- In Excel, what does this cell address mean B5?
- If you have a formula in one cell and you "drag" it down to the next row, what does Excel do by default?
 - It deletes the formula.
 - It keeps the numbers exactly the same.
 - It automatically changes the row numbers to match the new spot.
 - It turns the text into a different color.
- If a "Relative cell reference" means the cell address moves when copied and an "Absolute cell reference" means the cell address stays the same, what does a "Mixed cell reference" mean?
 - One part of the cell address moves while the other part stays the same.
 - It is a mistake in the formula that needs to be deleted.
 - It uses cell addresses from two different spreadsheets at the same time.

Cell reference also known as **cell addresses**, consists of a column letter and row number and is used in formulas. There are three types of references in Excel namely relative cell

references, absolute cell references and mixed cell references in Excel.

4.2.1. Relative cell references

By default, all cell references are relative references. Relative references are changed when copied across different cells based on the relative positions of rows and columns. For example, suppose this formula $=B1*C1$ is to be copied from row 1 to row 2, the formula will become $=B2*C2$. Relative reference is useful when it requires to repeat a calculation across numerous rows or columns.

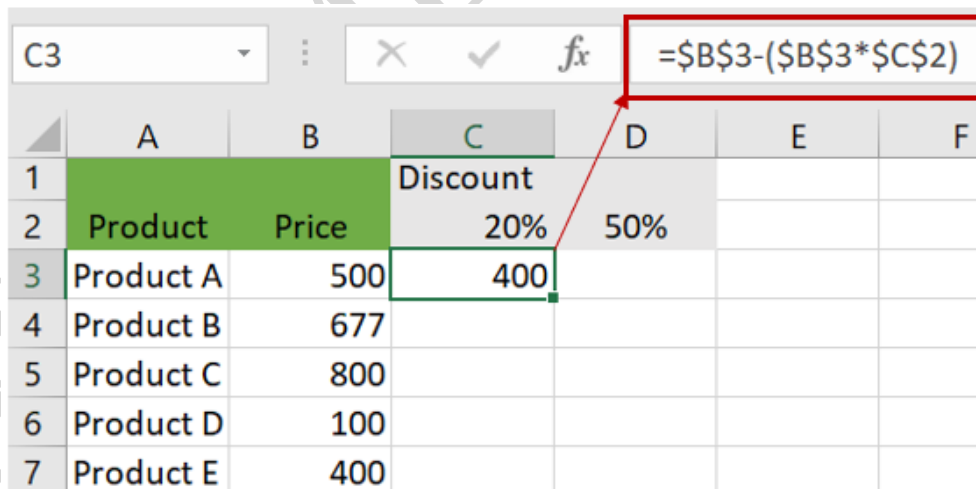


	A	B	C	D
1	20			
2	40			
3	60			
4	$=A1*A2*A3$			

Picture 4. 2. Using Relative cell references

4.2.2. Absolute cell references

Absolute cell reference is one in which the cell references being referred to do not change like they did in a relative reference. To set an absolute cell reference, use the \$ sign by pressing **f4** to create a formula for absolute referencing. The \$ sign means lock, and it locks the cell reference for all of the formulas, ensuring that the same cell is referred to all of them. To use a formula with absolute cell reference use the \$ (dollar sign) in front of every cell.



	A	B	C	D	E	F
1			Discount			
2	Product	Price	20%	50%		
3	Product A	500	400			
4	Product B	677				
5	Product C	800				
6	Product D	100				
7	Product E	400				

Picture 4. 3. Absolute cell reference

4.2.3. Mixed cell reference

In Excel a mixed cell reference is a mixture of both relative and absolute cell reference. In mixed cell reference, dollar signs are attached to either the letter or the number. For example, **\$B2** or **B\$4**. It's a mix of relative as well as absolute reference.

There are two types of mixed cell references:

- The row is locked while the column changes when the formula is copied.
- The column is locked while the row changes when the formula is copied.

	A	B	C	D
1	Product	Price	Discount	
2			20%	50%
3	Product A	500	400	
4	Product B	677	541.6	
5	Product C	800	640	
6	Product D	1000	800	
7	Product E	400	320	

Picture 4. 4. Applying a mixed cell reference

Application activity 4.2

- 1) Answer by True or False:
 - a) If you copy a relative formula like $=A1*B1$ down to the next row, it will automatically change to $=A2*B2$.
 - b) The \$ sign in a formula acts like a "lock" to keep a cell reference from moving.
 - c) To make a cell reference absolute, you must put a dollar sign in front of both the column letter and the row number (like $=C$5$).
- 2) Imagine you are calculating the price of 10 different toys. Each toy has a different price (in Column B), but they all have the same Tax Rate written in one single cell (Cell $=E$1$).
 - a) Why is it a better idea to use an absolute reference for the Tax Rate instead of a Relative one?
 - b) What would happen if you forgot to use the \$ signs.

4.3. Working with many worksheets

Activity 4.3

You have a class of 15 students in Year 2. You have also the marks for those students in three separate sheets where each sheet contains marks for each term.

- 1) Discuss how you can combine the data from the three sheets to make the yearly total.
- 2) Explain how you can copy text from Sheet1 to Sheet2

In Spreadsheets, it is possible to work with many worksheets such as using formula between many worksheets, copying and pasting data and formula, etc.

a) Referencing another worksheet

Consider the example below which are data from two different sheets named Sheet1 and

Sheet2. These sheets contain marks for ICT and for Biology. You want to make totals for each student for the two subjects and keep those totals in a separate sheet "Sheet3".

Sheet1			Sheet2			Sheet3 containing totals for the two subjects for each student		
1	ICT MARKS		1	BIOLOGY MARKS		1	TOTAL ICT AND BIOLOGY MARKS	
2	SN	NAMES	2	SN	NAMES	2	SN	NAMES
3	1	MUCYO	3	1	MUCYO	3	1	MUCYO
4	2	NSHUTI	4	2	NSHUTI	4	2	NSHUTI
5	3	UWASE	5	3	UWASE	5	3	UWASE
6	4	MUTONI	6	4	MUTONI	6	4	MUTONI
7	5	AKIMANA	7	5	AKIMANA	7	5	AKIMANA
8	6	AKAYEZU	8	6	AKAYEZU	8	6	AKAYEZU

Picture 4. 5. Different Excel sheets containing data

To do this, go to the table in Sheet3, where the totals from the two sheets will be calculated. In the TOT column, click cell C3 and enter the following formula:

=Sheet1!C3 + Sheet2!C3

This formula adds the value in cell C3 of Sheet1 to the value in cell C3 of Sheet2.

You can enter this formula either by typing it directly or by following these steps:

- Type the equal sign (=) in the cell where the total will appear.
- Click the cell containing the first value to be added then type the plus sign (+).
- Go to the sheet containing the second value and click the appropriate cell.
- Press Enter to complete the formula.

The formulas used to calculate the totals in the example above are shown in the image below.

TOTAL ICT AND BIOLOGY MARKS		
2	SN	NAMES
3	1	MUCYO
4	2	NSHUTI
5	3	UWASE
6	4	MUTONI
7	5	AKIMANA
8	6	AKAYEZU

Picture 4. 4. Formula to be used to add data from different sheets

Consider another example in which two sheets namely Sheet1 and Sheet2 contains data

(score) on different subjects. The average of marks contained in the two sheets is going to be calculated and kept in Sheet1. The formula used is **=AVERAGE(C2:C10,Sheet2!C2:C10)**

b) Copy and paste option for text

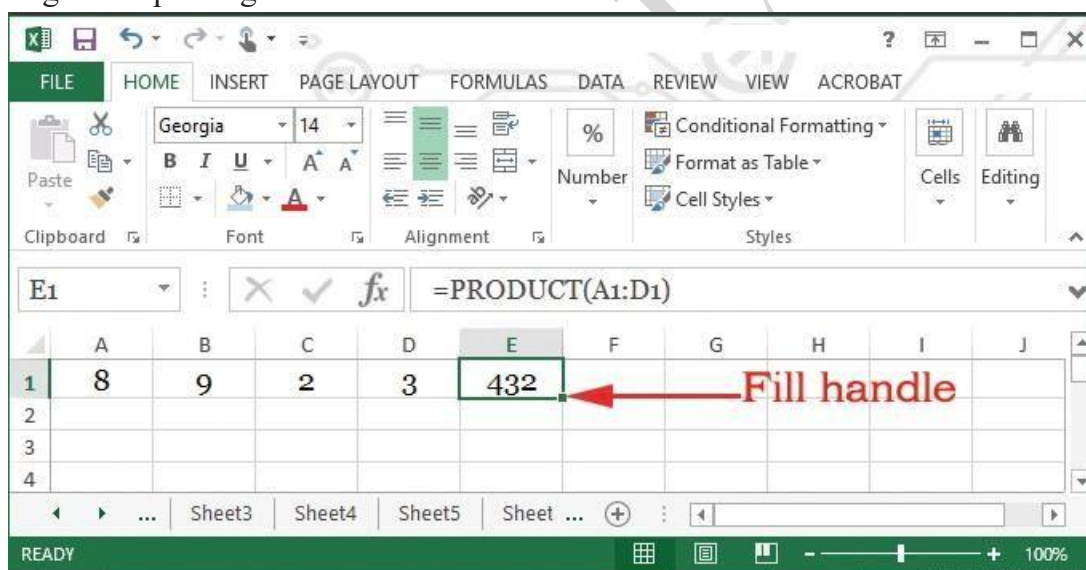
To copy data from one worksheet to another, do the following:

- Highlight the data to be copied.
- Click **Home** tab, select **Copy** command from the **Clipboard** group.
- Click on the name of the sheet where the data is to be copied, for example, Sheet1.
- Click on the position where the data is to be pasted.
- Click **Home** tab, select **Paste** command from the **Clipboard** group.

c) Copying formula

Once a formula is entered in a cell, it can be copied to other cells within the worksheet. When a formula is copied, the cell references are automatically adjusted depending on the type of reference used in the original formula. To copy a formula, do the following:

- Click on the cell that contains the formula.
- Move the cursor to the fill handle of the cell selector. Ensure that the cursor changes to a plus sign.



Picture 4. 6: Copying a formula to the desired location

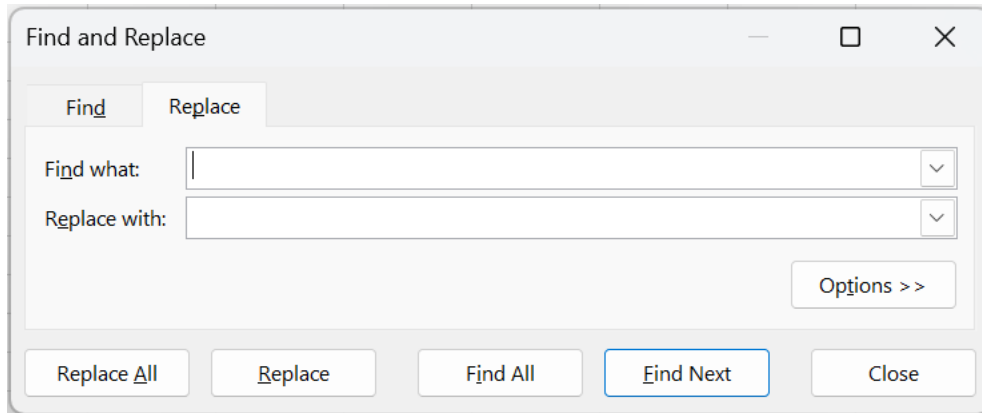
- Click and drag the cursor either across the row or column. Release the button once all the cells where the formula is to be copied are selected.

Note: The formula can also be copied by right-clicking on it then selecting **Copy** command from the pop-up menu that appears. Right-click on the cell where the formula is to be copied, then select **Paste**

When a formula is copied to another worksheet, relative cell referencing is applied. To prevent adjusting the cell references in a formula automatically, do the following:

- Select the range of cells to be copied.

- Click **Home** tab from the menu, select the **Find & Select** icon in the **Editing** group. A drop down menu appears.
- Select **Replace** command from the drop down menu that appears. The **Find and Replace** dialog box appears as in the picture below.



Picture 4. 5. Find and replace dialog box

- In the **Find What** box type "=", and in the **Replace With** box type "#".
- Click **Replace All**, to replace all occurrence of "=" with "#" in the range then close the dialog box. The formula is changed to a label. For example "=A1*B1" becomes "#A1*B1".
- Copy and paste the formula to the desired location in another worksheet.
- To change it back to formula, simply replace all occurrence "#" with "=".

Application activity 4.3

- 1) What does the formula =Sheet1!C3 + Sheet2!C3 do in Excel?
- 2) Using Excel, calculate the yearly total for a student if Term1 marks are in Sheet1, Term2 in Sheet2, and Term3 in Sheet3. Which formula would you use in Sheet4 to sum all three terms?
- 3) Why is it important to double-check formulas and references when working with multiple worksheets?

4.4 Functions

4.4.1. Mathematical function

Activity 4.4.1

After receiving their final examination results, four learners from the Mathematics and Science pathway sat together to analyze their performance. Their final exam results are shown below:

Learner	Math	Chemistry	Biology	ICT
---------	------	-----------	---------	-----

Aline	68	72	70	81
Brian	64	75	69	79
Claire	73	71	74	83
David	70	78	72	85

- 1) How did Excel calculate the total marks without adding them one by one?
- 2) Which Excel function could be responsible for finding the average score?
- 3) How did Excel identify the highest and lowest marks automatically?

Mathematical functions are used to perform common mathematical operations. They include: SUM, AVERAGE, ODD, INT, ROUND, EXP, SQRT, POWER, MOD, MAX, and PRODUCT.

- a) SUM:** This function **adds** all the values in a specified range of cells. The general syntax is =SUM (Number1, Number2...Number N).

Follow the following steps to use them:

- Open Microsoft Excel and write the following values which are savings of one student in his/her Piggy Bank in cells A3 to L3: 1500, 1600, 1100, 2000, 2100, 1850, 2200, 1800, 1700, 900, 4000, 2300.
- Use the **SUM** Function to calculate the total of all the savings from January to December.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	MONEY SAVED IN IN MY PIGGY BANK												
2	January	February	March	April	May	June	July	August	September	October	November	December	Total
3	1500	1600	1100	2000	2100	1850	2200	1800	1700	900	4000	2300	=SUM(A3:L3)

- After using the function =SUM (A3:L3) now realize that the total savings become 23050.
- b) AVERAGE:** It is used to calculate the mean of values specified in a selected range of cells. Empty cells within the range are ignored while those that have zero values are included. The syntax is: =Average (data range). For the example on the savings data above, the average function becomes =AVERAGE(A3:L3) and the result obtained is 1921.
- c) SQRT:** This function returns a positive square root of a number. The general syntax is =SQRT(number) where number is the value for which the square root is to be obtained. If number is negative, this function returns #NUM! error.
- d) POWER:** Return the result of a number raised to a power. The general syntax is =Power(number, power) where number is the base number which can be any real number while power is the exponent to which the base number is raised.

The ^ operator can be used instead of Power to indicate to what power the base is to be raised. In this case the function =POWER(5,2) is equivalent to =5^2.

- e) MAX:** The term Max refers to the function used for obtaining the maximum or largest

value in a selected range of cells. If the selected range of cells contains no value, it returns a zero. The syntax is: **=Max(data range)**.

By using the Piggy Bank savings data, you use the function **=MAX(A3:L3)** and you get 4000 as the result (highest saving).

f) PRODUCT: This function multiplies all the values within a specified range of cells. The general syntax is **=Product(Number1, Number2... NumberN)**.

Example: **=PRODUCT(A1:A2)** which multiplies the content of cell A1 with the content in cell A2.

Application activity 4.4.1

- 1) Write a function to add values in cells B2 to B10.
- 2) Which function would you use to find the square root of 144?
- 3) How do you find the product numbers in cells C1, C2, and C3?
- 4) If you want to raise 5 to the power of 3, what is the function?
- 5) True or False: Functions can be used without an equals sign
- 6) What happens if you use the PRODUCT function on a range containing text?
- 7) Write the function to calculate the total marks of 5 subjects located in cells D5, E5, F5, G5, and H5
- 8) If a cell displays #NAME?, what is likely wrong with your function?

4.4.2. Logical functions

Activity 4.4.2

I. Choose the correct answer for each question.

- 1) Which formula checks if the value in cell A1 is greater than 10 and shows “Large” if true and “Small” if false?
a) =IF(A1>10, "Large", "Small") b) =IF(A1<10, "Small", "Large")
c) =AND(A1>10, "Large") d) =OR(A1>10, "Small")
- 2) What is the result of the formula =AND (5>2, 1>3)?
a) TRUE b) FALSE c) 1 d) 0
- 3) Which formula checks if a student’s mark in cell B2 is greater than or equal to 50 and shows corresponding text?
a) =IF(B2>=50, "Pass", "Fail") b) =IF(B2<50, "Pass", "Fail")
c) =AND(B2>=50) d) =OR(B2>=50)
- 4) What is the result of the formula =OR(10<5, 2=2)?
a) TRUE b) FALSE c) 10 d) 5
- 5) Which function returns TRUE if at least one of its arguments is TRUE?
a) AND b) OR c) NOT d) IF

Logical function is a feature in Excel that allows Excel users to introduce automated decision making when executing formulas and functions. The role of functions in this is to check if a condition is true or false. It combines multiple conditions together and comes up with a result

and returns one value

)

ates to true and if so

the content of third

entered in the cell where

a) IF Function

The If function checks whether data in a cell meets a certain condition and returns one value which can be True or False

- **Syntax: = IF(Logical test, Value If True, Value If False)**

The If function takes as arguments the logical test, checks if it evaluates to true and if so returns as a result the content of the second argument and if false the content of third argument is returned.

Example: In the example below the If function with its arguments is entered in the cell where the result is to appear.

	A	B	C	D	E
1	Name1	Name2	Matching names ?		
2	Eserence Uwimana	Eserence Uwimana	=IF(A2=B2, "MATCH", "DON'T MATCH")		
3	Kanyamibwa Benjamin	Kanyamibwa Edouard	=IF(logical_test, [value_if_true], [value_if_false])		
4	Mussa Clenie	Clenie Mussa	DON'T MATCH		
5	Mugabo Idrissa	Mugabo Idrissa	MATCH		

The used If-function!

Picture 4. 6. The use of If function to compare names

To apply that function in other cells proceed like this:

- Place the cursor in the bottom corner of the cell
- Hold down the left key, scroll down to other cells and release the left button

The **If** function in the above examples checks if the two names are alike and if yes, the function writes MATCH in the cell, if not the function writes DON'T MATCH

Example: Considering the marks obtained by Irasubiza, Karenzi, Byukusenge and Shyaka in ICT, Maths and English. The If function checks if the marks are greater than 87 and gives to the candidate the **Very Good** note else the Good note is given.

G2 X ✓ fx =IF(F2>87,"Very Good", "Good")

	B	C	D	E	F	G	H
1	ICT/25	Maths/25	English/25	TOT/75	%	Testing	
2	19	18	24	61	81.33333	Good	
3	23	20	23	66	88	Very Good	
4	22	20	23	65	86.66667	Good	
5	24	21	23	68	90.66667	Very Good	

Testing

=IF(F2>87,"Very Good", "Good")

=IF(F3>87,"Very Good", "Good")

=IF(F4>87,"Very Good", "Good")

=IF(F5>87,"Very Good", "Good")

Here are formula used

Picture. 4. 7. Use of if function to award grades

b) AND Function

The Excel **AND** function is a logical function used to test if two or many conditions are true. The result is **TRUE** if all the conditions are true else the result is **FALSE**.

Syntax: =AND (Logical1, Logical2, logical3,...)

Example: In the table below the **And function** checks if people in the table studied Education and that their age is greater than 18

	A	B	C	D
1	NAME	EDUCATION	AGE	APPROVED
2	Izabayo	Chemistry	21	TRUE
3	Kagabo	Geography	20	FALSE
4	Azabe	Chemistry	19	TRUE
5	Twahirwa	Math	23	FALSE
6	Uwera	Biology	21	FALSE
7	Cyusa	Biology	18	FALSE

APPROVED

=AND(B2="Education",C2>18)

=AND(B3="Education",C3>18)

=AND(B4="Education",C4>18)

=AND(B5="Education",C5>18)

=AND(B6="Education",C6>18)

=AND(B7="Education",C7>18)

The used formula for this text

Picture 4. 8. Results of And function and on the left functions used

Note: The **And function** may have more than two arguments and for the results to be True all the arguments must evaluate to TRUE and if one of the arguments is false all the result is FALSE.

c) FALSE function

The FALSE function takes no arguments and generates the Boolean value FALSE. It is used to compare the results of a condition or function that either returns true or false. Its syntax is =False ()

Example: The False function used in the example below returns FALSE if the age entered in C2 is less than 20 (C2<10)

D2

=IF(C2<10,FALSE())

Formula Used

	A	B	C	D
1	NAME	EDUCATION	AGE	APPROVED
2	Izabayo	Chemistry	21	FALSE
3	Kagabo	Geography	20	FALSE
4	Azabe	Chemistry	19	FALSE
5	Twahirwa	Math	23	FALSE
6	Uwera	Biology	21	FALSE
7	Cyusa	Biology	18	FALSE

Picture 4. 9. The use of the False function in Excel

Interpretation: The used function “=IF(C2<10,FALSE())” will check if the C2 cell data is less than 10, if so it will return **False** as a result else it will return **True**. The same function will be applied to other cells by changing the cell position

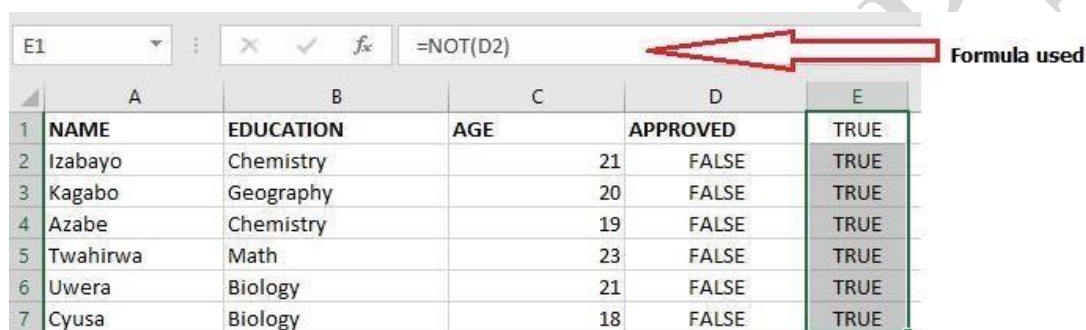
d) NOT Function

The Excel **NOT function** returns the opposite of a given logical or Boolean value. When

given TRUE, NOT returns FALSE. When given FALSE, NOT returns TRUE. Use the NOT function to reverse a logical value. Its syntax is **=NOT(Logical)**.

The NOT function takes one logical expression as an argument. It returns an error if more than one argument is used.

Example: The table below will have all its content changed to **True** by the use of the NOT function. If the results to reverse were got by using a formula the NOT function can be put in front of the function/formula to make the latter an argument of the NOT function.



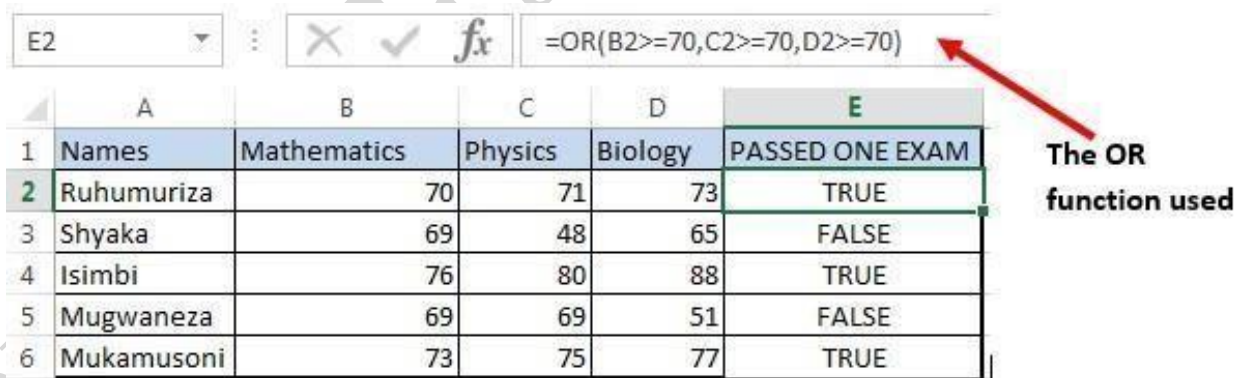
	A	B	C	D	E
1	NAME	EDUCATION	AGE	APPROVED	TRUE
2	Izabayo	Chemistry	21	FALSE	TRUE
3	Kagabo	Geography	20	FALSE	TRUE
4	Azabe	Chemistry	19	FALSE	TRUE
5	Twahirwa	Math	23	FALSE	TRUE
6	Uwera	Biology	21	FALSE	TRUE
7	Cyusa	Biology	18	FALSE	TRUE

Picture 4. 10. Not function and its results

e) The OR function

The OR function is a logical function to test multiple conditions at the same time. OR returns either TRUE or FALSE. Its syntax is **=OR(logical1, Logical2, ...)**

Example: The **OR function** in the screenshot below checks for students who got more than 70% as the Pass mark in any one of the three tests.



	A	B	C	D	E
1	Names	Mathematics	Physics	Biology	PASSED ONE EXAM
2	Ruhumuriza	70	71	73	TRUE
3	Shyaka	69	48	65	FALSE
4	Isimbi	76	80	88	TRUE
5	Mugwaneza	69	69	51	FALSE
6	Mukamusoni	73	75	77	TRUE

Picture 4. 11. The OR function to test if any of the arguments is true

Application activity 4.4.2

The ministry of Health is rewarding best performing students with marks between 80 and 100 in Biology and Chemistry. The reward is a laptop for those who succeeded in both subjects and a dictionary for those who succeeded in one subject.

	Marks in	Marks in	Rewarded	Rewarded	Not

	Biology	Chemistry	by a laptop	by a dictionary	rewarded
Kabera	70	92			
Mutoni	84	88			
Mugabo	60	75			
Isimbi	90	93			
Uwase	55	70			
Uwacu	70	81			
Gihozo	98	95			
Rurwiro	78	80			
Kamikaze	62	68			
Gakure	80	82			
Total laptops/ dictionary given					

Using Ms Excel answer the questions below:

- Write an Excel functions to find students who are to be rewarded with laptops, dictionaries and those not rewarded
- Determine the number of students that have not been rewarded?
- How many laptops and dictionaries will be given?
- In which subject does Kabera have more marks?
- In which subject does Umutoni have minimum marks
- Use Excel logical functions to fill the table above

4.4.3 Text functions

Activity 4.4.3

You are the school registrar and you have been given a list of 100 students. However, the list is not well organized because some students' names are in all lowercase letters, some cells are completely empty because the student hasn't submitted their phone number yet, some cells contain "N/A" or "Pending" instead of numbers. Also you need to change the school code prefix from "SCH-2023" to "KGL-2024" for every student.

- If you want to count only the students who have provided any kind of information in the "Contact" column, which function would you use?
- How can you automatically convert a list of 100 lowercase names into capital letters without retyping each one?
- If you need to find a specific word within a long ID code and swap it for a new one, which function is more appropriate: Search or Replace?

In the context of spreadsheet applications like Microsoft Excel, Text and Specialized Functions are pre-defined formulas used to manipulate strings of text and perform specific counting or searching tasks that standard mathematical formulas cannot handle.

a) Counting functions

- **COUNTA:** A function that counts the number of cells in a range that are not empty. It counts cells containing any type of information, including numbers, text, logical values, or error values. It is used for determining the total number of entries in a list regardless of whether they are names or figures. Its syntax is **=COUNTA(range)**
- **COUNTBLANK:** This function counts the number of empty cells in a specific range. It is used to identify missing data or gaps in a database. Its syntax is **=COUNTBLANK(range)**

These functions are designed to modify the appearance or case of text strings automatically, ensuring consistency across a document. Those functions are summarized in the table below:

Function	Its use	Example	Result of the example
UPPER	It converts every letter in a text string to uppercase.	=UPPER("rwanda")	RWANDA
LOWER	This function converts every letter in a text string to lowercase.	=LOWER("ICT")	ict

These functions allow you to locate specific characters within a cell or swap out parts of a text string for new data. Their use is summarized in the table below:

Function	Its use	Syntax
SEARCH	It finds one text string within another and returns the position (number) where it starts	=SEARCH (find_text, within_text)
REPLACE	A function that swaps a specific part of a text string with a different text string based on a starting position and character count is called REPLACE.	=REPLACE(old_text, start_num, num_chars, new_text)

1) Which function would you use to find out how many students have NOT yet submitted their email address?

- a) COUNTA b) COUNT
c) COUNTBLANK d) UPPER
- 2) If cell A1 contains "computer", what will =UPPER(A1) display?
a) Computer b) COMPUTER
c) computer d) C
- 3) What is the main difference between COUNTA and the standard COUNT function?
a) COUNTA only counts empty cells.

- b) COUNTA counts both text and numbers, while COUNT only counts numbers.
- c) COUNTA is used for uppercase letters. d) There is no difference.
- 4) A company wants to change its department codes from "DEPT-A" to "DIV-A". Which function should the secretary use to update the entire database at once?
- 5) You are checking a warehouse inventory list. You need to know how many items are missing a serial number (blank cells). Which function do you use?
- 6) You have a list of 500 names for a contract, but they are all in lowercase. How does the UPPER function save you hours of work?

4.5. Conditional formatting

Activity 4.5

A company has the following sales data for its sales team:

Salesperson	January sales
John	45,000
Mary	55,000
Ali	80,000
Grace	60,000
Peter	30,000

- 1) Which sales values are greater than 50,000?
- 2) Which salesperson(s) have sales above the average?
- 3) Which sales value(s) turn blue if the condition to change to blue is that the value must be above 60? What does this tell you about those sales?

Conditional formatting is a spreadsheet functionality which allows Excel users to specify conditions allowing specific answers on data in the worksheet that was otherwise not easy to analyze.

It can be applied to a range of cells or an entire worksheet and makes cells meeting specified criteria have relative formatting. If the formatting condition is true, the range of cells is formatted according to the preset criteria, else that range is not formatted.

Conditional formatting enables spreadsheet users to do a number of things. It calls attention to important data points such as deadlines. It can also make large data sets more understandable. Among the options provided by conditional formatting are: **highlight cell rules, top bottom rules, customized rules, etc**

4.5.1 Highlight cell rule

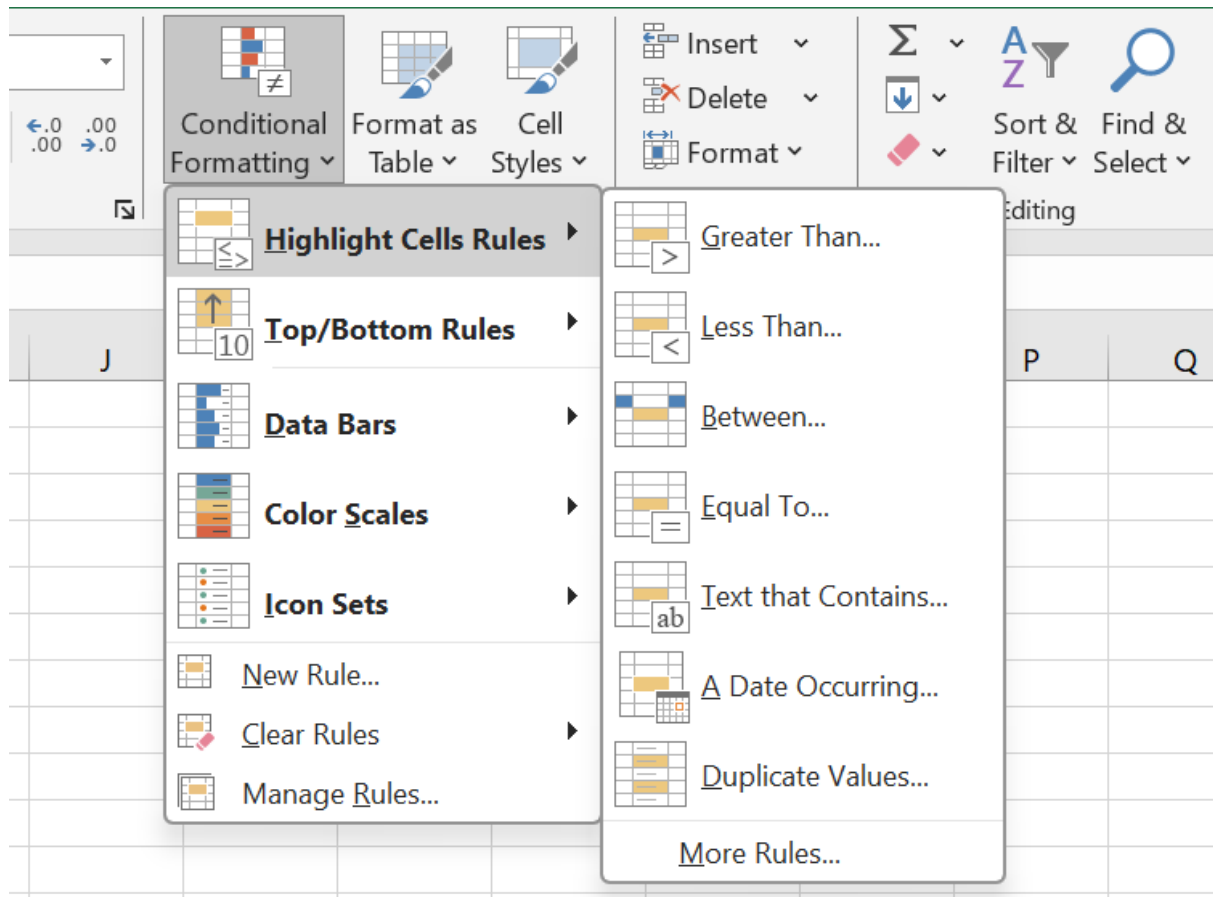
The conditional formatting statement under the **Highlight Cells Rules** category allow to highlight the cells whose values meet a specific condition. The rules that can be applied in Highlight Cell Rules are greater than, less than, equal to, between, text that contain, A date occurring, duplicate values, ...

Here are the steps to apply the highlight cell rule:

Step 1: Select the desired cells on which to apply the conditional formatting rule

Step 2: From the **Home** tab, click the **Conditional Formatting** command. A drop-down

menu will appear.

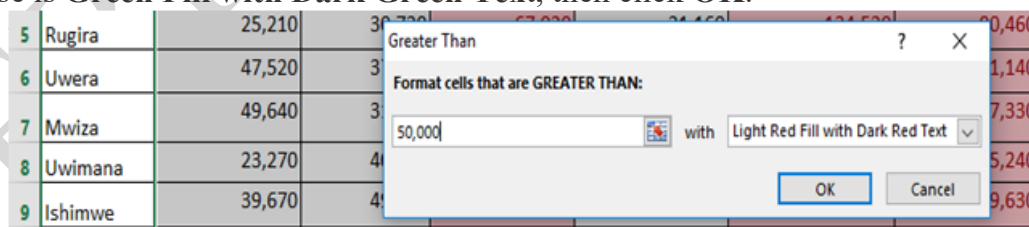


Picture 4. 12. Screenshots showing Home tab and Conditional formatting Command

Step 3: Hover the mouse over the desired conditional formatting type, and then select the desired rule from the menu that appears. In the next screenshot, cells that are greater than 50, 000 are to be highlighted.

Step 4: A dialog box will appear. Enter the **desired value(s)** into the blank field.

Step 5: Select a **formatting style** from the drop-down menu. In the Activity 3.1, the color to choose is **Green Fill with Dark Green Text**, then click **OK**.



Picture 4. 13. Setting the Highlight cell rule to Greater than 50000

Step 6: The conditional formatting will be applied to the selected cells. In this case, it's easy to see which salesperson reached the 50, 000 sales goals for each month.

	A	B	C	D	E	F	G
1		Burera Ltd Sales data					
2	Salesperson	May	June	July	August	September	October
3	Muhirwa	39,470	5,570	38,630	11,170	82,370	86,900
4	Mukamana	44,110	10,420	93,550	11,000	101,850	187,490
5	Rugira	25,210	30,720	67,020	21,160	134,520	80,460
6	Uwera	47,520	37,550	44,150	10,890	44,040	201,140
7	Mwiza	49,640	31,520	116,010	11,220	31,700	107,330
8	Uwimana	23,270	40,520	37,260	11,350	88,170	185,240
9	Ishimwe	39,670	49,060	90,070	21,130	130,900	139,630
10	Rurangwa	53,630	5,210	45,050	10,240	35,280	152,750
11	Umutesi	32,750	34,280	39,730	17,160	48,390	137,140
12	Kalisa	38,600	15,620	29,450	11,010	58,500	150,650

Picture 4. 14. Results of applying highlight cell rule (>50000)

Note: It is possible to apply multiple conditional formatting rules to a cell range or worksheet. This allows to visualize different trends and patterns in data.

4.5.2 Clear rule

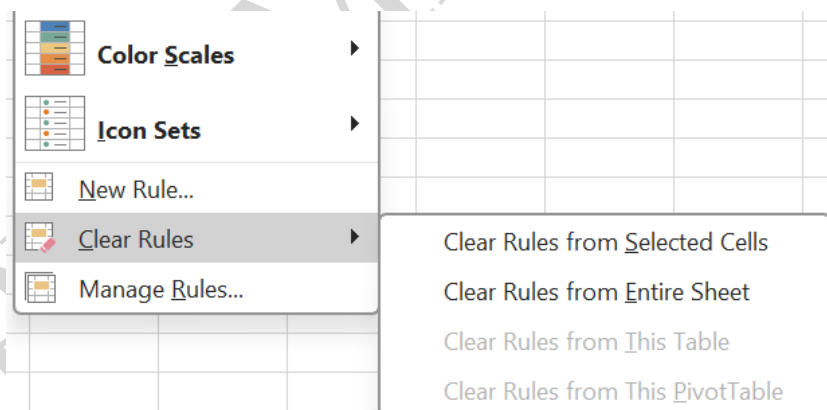
Rules that have been used to highlight cells that meet certain conditions can be removed and have text with no highlight when the rules are no longer needed or when there is a need to set new rules.

To clear rules, do the following:

Step 1: Select the desired cells for clearing the conditional formatting rule

Step 2: On the **Home** tab, in the **Styles** group, click **Conditional Formatting**

Step 3: Click **Clear Rules** then click **Clear Rules from Selected Cells**.



Picture 4. 15. Clear rule options

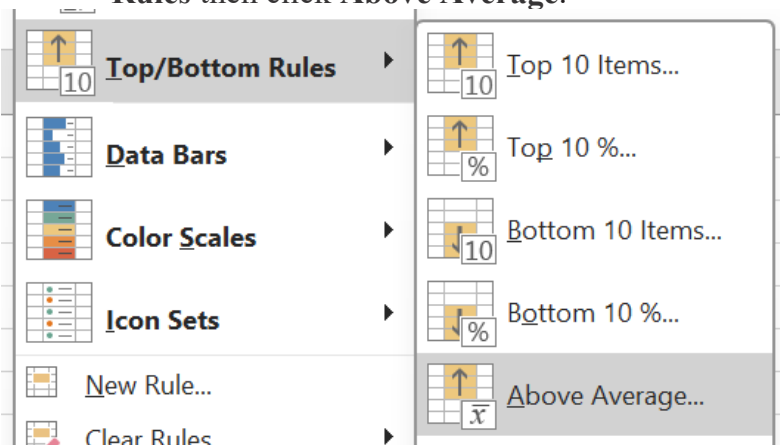
After clearing rules for selected cells, cells are no longer colored, there is no more formatting in cells.

4.5.3 Create Top/Bottom Rules

Top/Bottom rules are another useful Conditional Formatting in Excel. These rules allow users to call attention to the top or bottom range of cells, which can be specified by number, percentage, or average. The steps to create Top/Bottom rules are:

Step 1: Select the desired cells for the conditional formatting rule

Step 2: From the **Home** tab, Select **Conditional Formatting** then **Top/Bottom Rules** then click **Above Average**.



Picture 4. 16. Top/Bottom rules-Above average

Step 3: A dialog box will appear. Choose color format to use like “**Light Red Fill with Dark Red text**”.After choosing cells format, Press “**OK**”

	A	B	C	D	E	F	G
1	Burera Ltd Sales data						
2	Salesperson	May	June	July	August	September	October
3	Muhirwa	39,470	5,570	38,630	11,170	82,370	86,900
4	Mukamana	44,110	10,420	38,630	11,170	151,250	187,490
5	Rugira	25,210	30,720	38,630	11,170	151,250	80,460
6	Uwera	47,520	37,550	38,630	11,170	151,250	201,140
7	Mwiza	49,640	31,520	38,630	11,170	151,250	107,330
8	Uwimana	23,270	40,520	38,630	11,170	151,250	185,240
9	Ishimwe	39,670	49,060	90,070	21,130	130,900	139,630
10	Rurangwa	53,630	5,210	45,050	10,240	35,280	152,750
11	Umutesi	32,750	34,280	39,730	17,160	48,390	137,140
12	Kalisa	38,600	15,620	29,450	11,010	58,500	150,650

Picture 4.19: Above average rule dialog box

The sheet will now highlight the values above the average.

Application activity 4.5

- 1) List three types of conditional formatting.
- 2) Outline the steps followed to apply colour on data in a worksheet.
- 3) State the function of the Between option in conditional formatting.
- 4) Outline the steps followed to apply conditional formatting on data in a worksheet.

- 5) State one advantage of using conditional formatting.
- 6) What happens when conditional formatting rules are cleared?

4.6. End unit assessment

4.6.1. Project activity

Project Title: Students' reports automation

Guidance:

You are a class teacher and you have to make school reports for your 50 students

- Using Excel create a school report template and fill in the details which include student names and marks for each subject. Use formula to calculate totals of marks and average.
- Copy and paste that data in another sheet then change the student names and marks. Do this for all the remaining 48 students. Check if the new totals and average are automatically calculated
- Make a list of all the students with their marks and their average marks then arrange according to their marks.

Questions on the project

- 1) Which formulas did you use to calculate the totals and averages?
- 2) Explain the steps you went through in order to add other worksheets.
- 3) How did you create the school report lines?
- 4) a) How much time did it take to finish all the 50 reports.
b) Explain why you would prefer using Excel formula for calculations instead of using a calculator.

4.6.2. Assessment questions

- 1) Which of the following represents a complex formula?
a) =A1+B1 b) =SUM(A1:B10) c) =(A1+B1)*C1/2 d) =A1
- 2) Which symbol is used to create an absolute cell reference?
a) % b) # c) \$ d) &
- 3) Which function counts cells that contain any data and are not empty?
a) COUNT b) COUNTA c) COUNTBLANK d) COUNTIF
- 4) Which logical function returns TRUE only if all provided conditions are met?
a) OR b) IF c) NOT d) AND
- 5) Define "Conditional Formatting" and state one reason why a registrar might use it.
- 6) Explain the purpose of the REPLACE function.
- 7) Write a formula to calculate the average of cells B2 to B20, but only if you were using a manual complex formula instead of the AVERAGE function.
- 8) Write an IF function that displays "Low Stock" if the value in cell D5 is less than 10, and "Sufficient" if it is 10 or more.
- 9) If you have a name "jean paul" in cell A2, write the function to change it to "JEAN PAUL".
- 10) Why is it better to use cell references (like A1) in a formula instead of typing actual

numbers (like 100) into the formula?

EXPERIMENTAL VERSION

UNIT 5 NETWORK COMPONENTS AND SOCIAL MEDIA APPLICATIONS



Key Unit Competency:

Connect different computer network devices correctly according to their functions

Learning objectives

At the end of this unit, you will be able to:

Knowledge and understanding

- Explain the use of different computer networking components and their functioning
- Identify the social media platforms, their functions, benefits, risks, and the importance of responsible use.

Skills

- Connect and configure basic network devices, and demonstrate how to share resources such as files, printers, and storage.

Attitudes and Values

- Appreciate the importance of networking for communication, collaboration, and resource sharing.
- Demonstrate responsible, ethical, and safe behavior online, especially when using social media platforms.

5.0. Introduction

Due to technological developments, sharing of information is made possible through **internet connectivity**. The term **Global Village** is now commonly used to refer to the world. This is because the world is today considered as a single community where members are connected using **electronic communication**.

In this unit you will be familiar with networking devices used in getting connected to the internet, how they are used and ultimately use the internet to get connected to different social media platforms.

Introductory activity

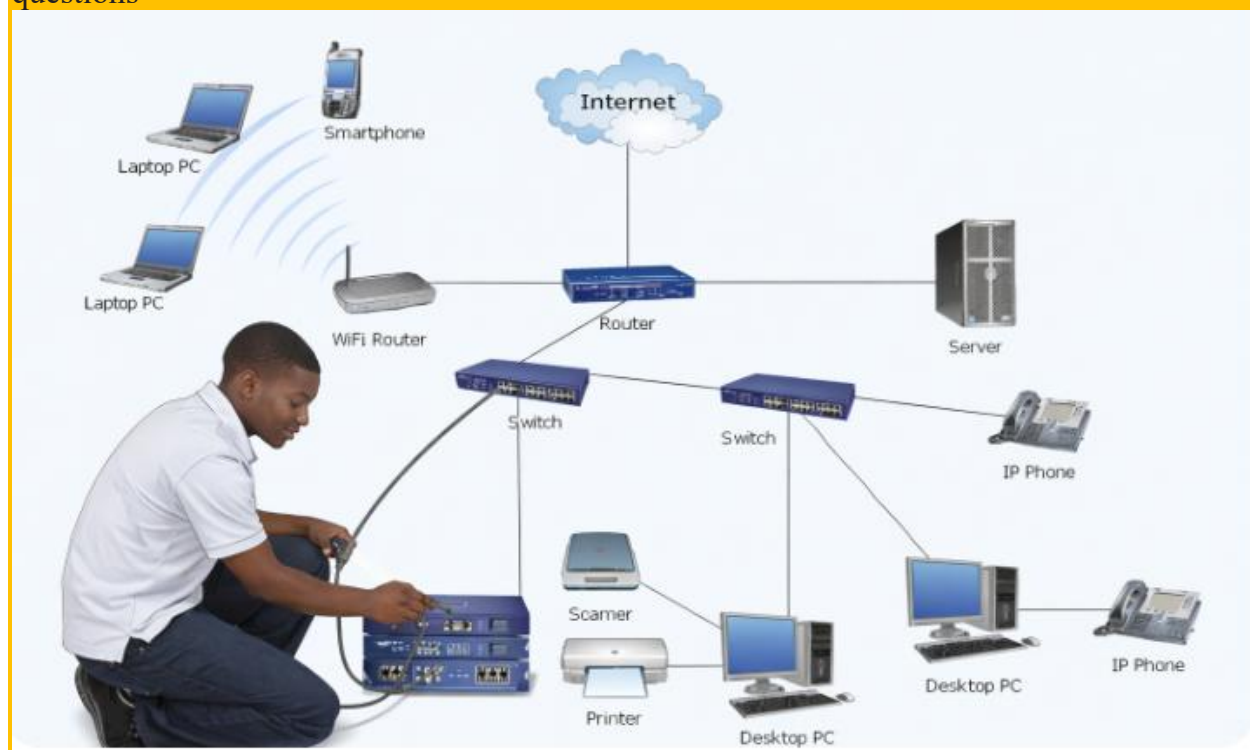
At school and at home, learners use computers, smartphones, and the internet to communicate, share information, print documents, and access online services such as social media. These activities require devices to be connected and to work together in a computer network.

- 1) Which devices do you use to communicate or access the internet?
- 2) How do you share information or files with friends or teachers?
- 3) Which social media applications do you know?

5.1 Network devices

Activity 5.1

Study the diagram of devices connected in a computer network below and answer the questions



- 1) Name three devices shown in the network diagram.
- 2) What cable is used to connect computers in a wired network?
- 3) What device allows computers and smartphones to connect to a network without using cables?
- 4) Mention resources that users can share in a computer network.

Network devices, also known as networking hardware or network equipment, are electronic or physical devices that are required for communication and interaction between all the hardware and software on a computer network.

A computer network is also defined as set of interconnected computing nodes that are organised in such a way that they can exchange data, information, or resources. A node is a

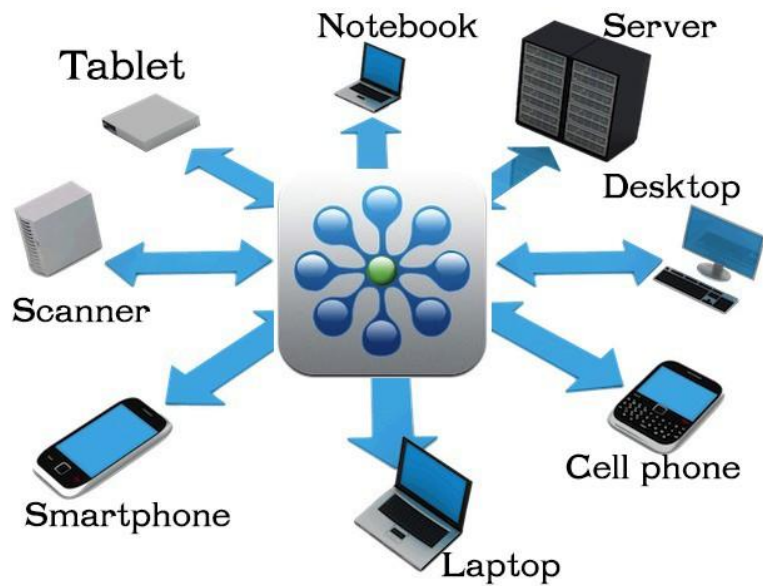
connection point in a computer network such as a laptop, a printer or scanner.

5.1.1 Devices that can be connected to a network

Computers in a network are connected through a transmission medium using a set of rules called **protocols**. In computing, a protocol or communication protocol refers to a set of rules in which computers communicate with each other.

Devices that can be connected to a network include desktops, laptops, printers, scanners, PDAs, smartphones and the server.

A **server** is an application program that provides services to other computer programs in a network. However the computer on which the server application program runs is also referred to as a server.

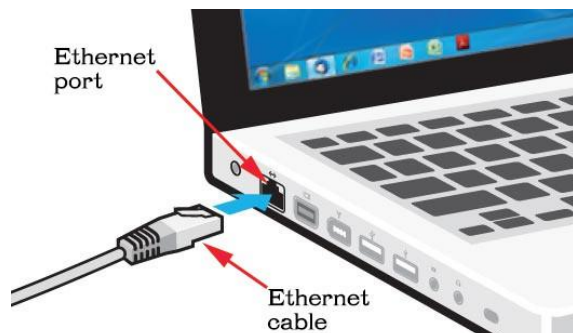


Picture 5. 1. Devices connected in a computer network

5.1.2. Connecting devices to a network

a. Connecting devices using cables

To connect devices in a computer network, you can use cables which are connected to the devices through an Ethernet port. An ethernet port is a socket on a computer that allows one to connect devices to a computer network. An ethernet port is usually found on networking devices, such as computers, routers, and television sets.



Picture 5. 2. An ethernet port and cable

b. Connecting devices using

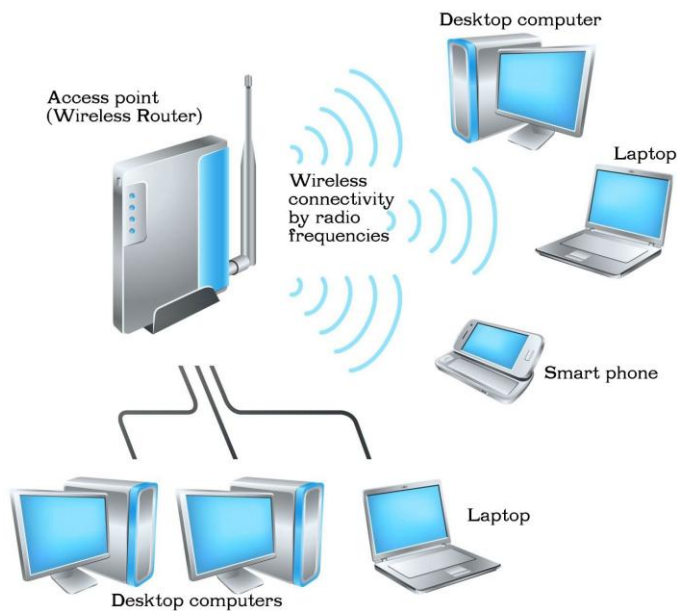
wireless media

As the word suggests wireless means without wires. Therefore a wireless computer network does not require a cable for connection. Sometimes the wireless network is also referred to as a **Wi-Fi network** which stands for **Wireless Fidelity**. It is a technology that allows computers, smart phones, or other devices to connect to the internet or communicate with one another in a wireless network.

❖ Components of a wireless network

- **Wireless Access Point (WAP):** It is a bridge between a wired and wireless network whereby it receives data via a cable and converts it into a radio signal that can be received without a cable
- **Wireless Network Interface Controller:** Every device on a network must have a **WNIC**. For electronic devices such as computer, it allows them to send and receive wireless signals.
- **Wireless router:** This is a physical device we can connect to. In a standard setup, the router acts as the brain whereby it manages the data traffic, assigns IP addresses, and often includes a built-in Access Point and a firewall for security.
- **Antenna:** Antennas are used to increase the range and strength of the radio signals. They can be internal (hidden inside a phone) or external.

Note in most cases, you don't need four separate physical boxes because they are sometimes combined into one single device.



Picture 5. 3. Devices connected in an wireless network

❖ Connecting to a wireless

A **wireless hotspot** is an area covered by a usable network signal that allows wireless connection to the internet or to some other computer network.

Some wireless connections are secured by passwords while others are open. To connect to a wireless connection:

- Click in the place where wireless connections are found on your device. For laptops, available wireless connections are found after clicking on the wireless icon  found at the bottom right of the screen. For Android phones, they are mostly found after swiping down from the top of the screen then tapping on the wireless icon.
- In the list of connections that will appear, click (tap) on one you want to connect to.



Picture 5. 4. A laptop screen showing wireless connections available

- After tapping the wireless connection to use, fill the password so that you get access to it

5.1.3 Network resources and services

- **Network resources**

Network resources or **shared resources** refer to forms of data, information, hardware devices, and software that can be accessed by a group of computers through the network. Network resources are important in work environments where collaboration is essential to success. The network resources include hardware such as computers, scanner, storages, data and files.

- **Network services**

Network services refer to software and connectivity tools that are installed in a network. Network services are managed by a central server and are then distributed to the networked computers. The connected computers are known as **client computers** or **workstations** while the computer from which data is got are known as servers.

The server may be located in a different room or building away from the workstations. The networked services are accessed through network cables or through a wireless network.

Some of the services offered by a network include user management, system administration, Email services, shared printing, File sharing, shared computer software, etc.

Application activity 5.1

Part A: Fill in the blank spaces with the correct answers

- 1) A system or a group of interconnected objects or people is known as ...
- 2) A set of rules that allow communication between computing nodes are called _____
- 3) Identify devices that can be connected in a computer network. (Look for the words in

this puzzle)

Q	A	D	E	S	K	T	O	P	S	D
W	H	E	N	E	V	Y	N	R	C	R
E	M	T	E	O	U	P	A	I	A	E
R	H	E	S	G	T	R	E	N	T	N
L	A	P	T	O	P	O	W	T	E	N
T	O	P	A	L	D	E	S	E	T	A
Y	C	A	N	N	A	K	T	R	O	C
S	M	A	R	T	P	H	O	N	E	S

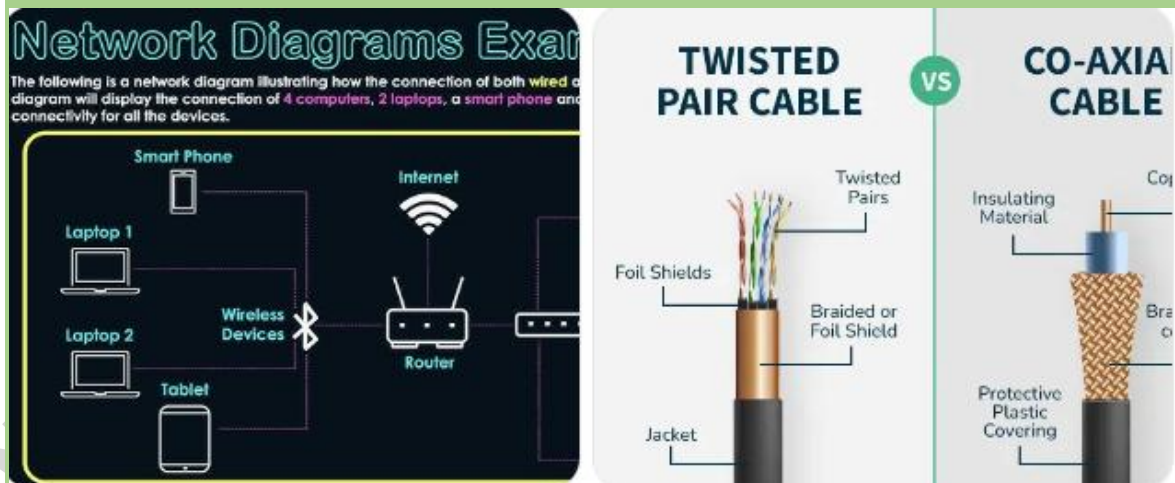
Part B: Answer the following questions

- 4) Differentiate between a laptop and a desktop computer as used in a network.
- 5) Explain two advantages of a computer network.

5.2 Network medium

Activity 5.2

Observe the image below and answer the related questions



- 1) List the devices that can be connected using cables.
- 2) List the devices that can be connected using wireless connection.
- 3) How many pairs of cables are in a twisted pair cable?
- 4) Where is a coaxial cable usually used?

Network medium is the physical or wireless pathway through which data is transmitted

from one device to another in a computer network. It is the channel that carries communication signals between network devices.

Network media are generally classified into two main categories: **Wired Network Media** and **Wireless Network media**.

5.2.1 Wired network media

Wired network media are physical communication channels that use **cables** to transmit data signals from one device to another in a computer network.

There are three types of bounded media, namely **twisted pair cable**, **coaxial cable**, and **fibre optic cable**.

i. Twisted-pair cable

This is a cable made up of pairs of insulated copper wires twisted along each other. The twists are done to protect the cable from crosstalk.

Crosstalk is interference that is caused by one or more wires. Crosstalk can cause errors, noise, or prevent cables from transmitting accurate data. They are usually packed as four sets of pairs in a cable. There are two types of twisted pair cables namely: **Unshielded Twisted Pair (UTP)** and **Shielded Twisted Pair (STP)**.

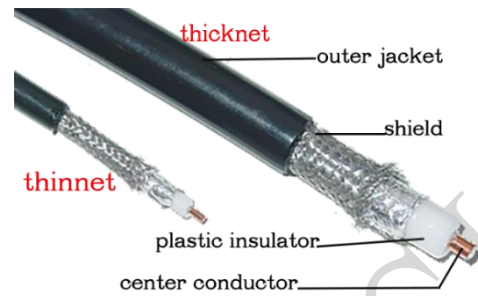
- ✓ **Unshielded Twisted Pair:** This is a type of twisted pair cable where each pair is not shielded from the others and hence the name unshielded twisted pair. This cable is vulnerable to noise whose source could be radio signals, radiation from spark plugs in motor vehicles and lightening sparks.
- ✓ **Shielded Twisted Pair:** Each twisted pair has a shield around it protecting the pair from others. The shielding minimizes electromagnetic interference, as well a cross-talk from neighboring pairs.



Picture 5. 5. Types of twisted cables

ii. Coaxial cable

This type of cable contains two conductors, namely copper wire which runs through the centre of the cable where the actual data travels and a shield made of wire mesh tube, metallic foil, or both.

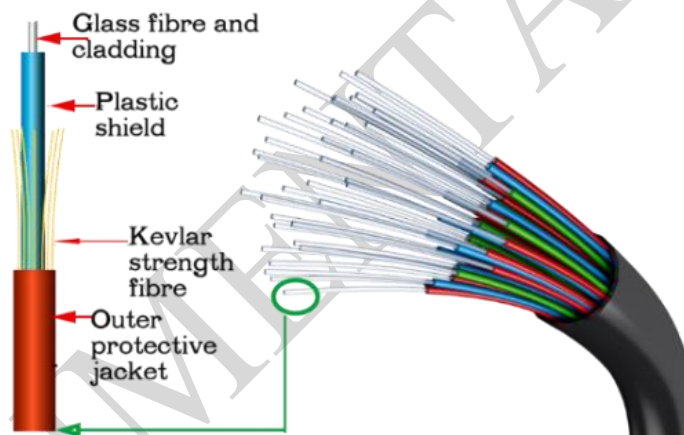


Picture 5. 6. Coaxial cables

The copper wire is surrounded by plastic insulator which is covered by a conducting shield that acts as a return path and also filters Electromagnetic Interference (EMI). The outer jacket which also acts as an insulator forms a protective covering for the cable. It resembles the cable used for connecting a television (TV) aerial to the TV set. It carries data in form of electrical signals.

iii. Fibre optic cable

This type of cable uses glass material strands or fibres which are as thin as a human hair. Fibre optic cables carry digital information over long distances.



Picture 5. 7. Fibre optic cable

Light is used to transmit data signals unlike in coaxial and twisted pair cables which use electrical signals. The electrical signal from the computing node is converted to a light signal using a Light Emitting Diode (LED) or lasers and then they are transmitted through the cable. At the receiver the light signals are converted back to electrical signals by a photosensitive device.

Fiber optics have an advantage of carrying more information at a quick speed. However they are expensive and their installation is a delicate process because the cables must be handled carefully.

5.2.2. Wireless Network

i. Infrared transmission

Infrared does not penetrate walls or solid objects hence security and interference problems are not encountered. There is no frequency allocation issue since no licence is required. This technology is used in TV remote control and wireless computer networking.

For transmission to take place, there must be a transceiver to aid in the communication. Transceivers must be in close range with each other, either directly or via reflection from a light-coloured surface such as the ceiling of a room.

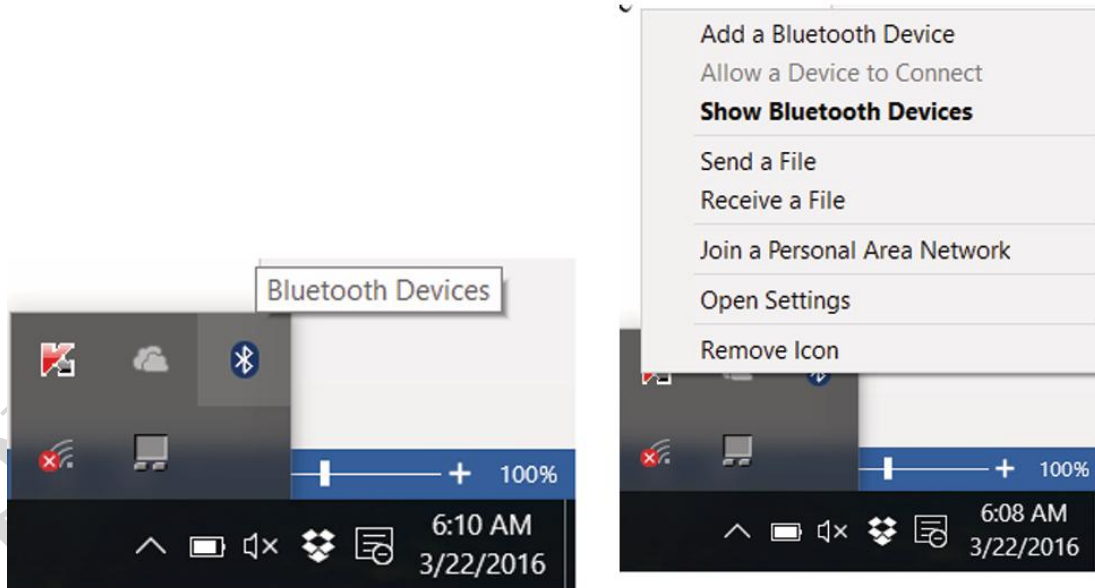
ii. Bluetooth

This is a technology that enables the connection of electronic devices over a short distance of up to about 10 metres creating a Personal Area Network (PAN) with a high level of security.

This technology can be used with cell phones, personal computers, laptops, printers, digital cameras and telephones among others. It is also an example of radio transmission technology. It is not licensed, therefore, communication takes place at no cost.

To connect a laptop to a Bluetooth network, do the following:

- Click on the **Bluetooth** icon on the task bar. Ensure that the Bluetooth feature in the other device is activated.
- Click on **Add a Bluetooth Device** for the system to automatically locate a nearby Bluetooth devices. Pair the devices then click **Join a Personal Area Network**.



Picture 5.8: Connecting via Bluetooth

iii. Wi-Fi (Wireless Fidelity)

Wi-Fi or Wireless Fidelity is a wireless networking technology that uses radio waves to provide high-speed internet and networking connections. Examples of WiFi-enabled devices include laptops, cell phones, tablets, iPads, peripheral devices such as printers and keyboards among others.

They communicate with a single computer which is fitted with a **WiFi adapter**.

One of its advantages is that enables communication in remote areas where the laying of cables is not possible while it has a disadvantage of being insecure as information can be tapped as it moves from one device to another.

Application activity 5.2

Part A: Fill in the blanks with the correct answers

1) Write the following abbreviations in full:

a) Wi-Fi _____ b) HF _____ c) LED _____ d) UTP _____

2) What is the other term for:

(a) Bounded media _____ (b) Unbounded media _____

Part B: Answer the following questions

3) Explain two advantages of each of the following media:

(a) Twisted pair _____ (b) Fibre optic cable _____

4) Give and explain three advantages of wireless communication.

5) Identify the various types of network connectivity.

6) Connect to a Bluetooth enabled network and share files of your choice.

5.3 Computer network devices

Activity 5.3

Your school has received a donation of 20 new computers and plans to open a new computer lab. The computers must be connected so that students can share files, access the internet, and use a shared printer. The school asks you and your classmates to help plan the network setup.

1) How can those computers share information with each other?

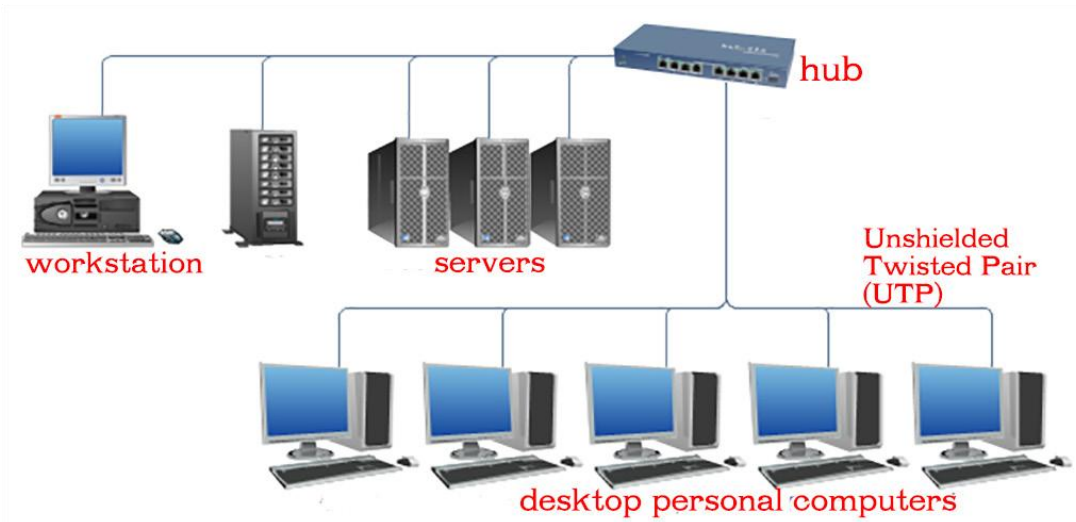
2) Can computers communicate directly without any extra equipment? Explain.

They are communication devices that are used in networking. They are used in between the DTE and transmission media to provide an interface or junction for the network to be fully operational.

They are also known as Data Circuit Terminating Equipment (DCE). Examples of computer network devices include switch, hub, Access Point, router and multifunctional device.

5.3.1 Hub

A hub is a connection point in a network. The hub connects many computing nodes together and is able to send signals from one node to another on the same network.

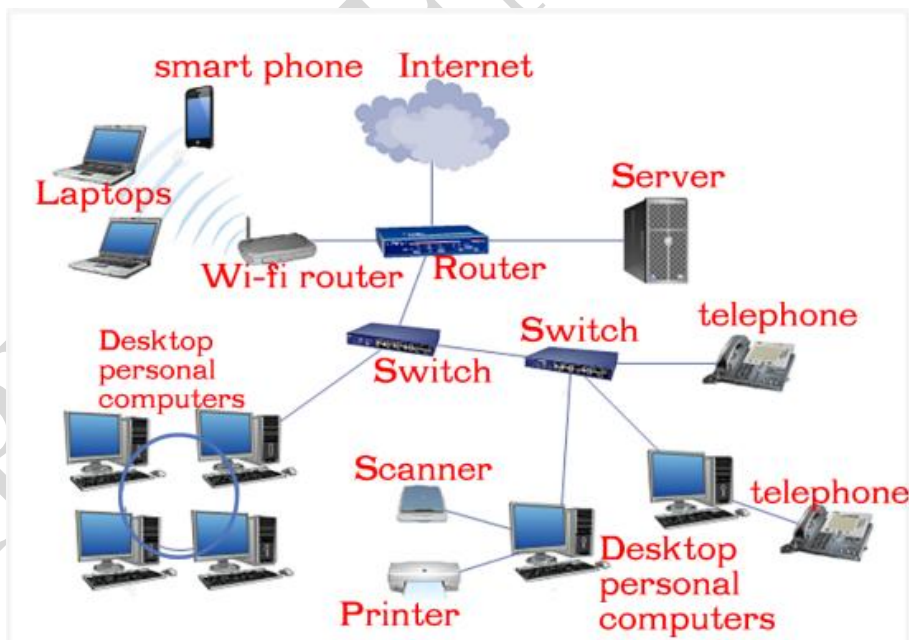


Picture 5. 8. A hub that has devices connected to it

A hub connects networks that have the same rules and it transmits signals by broadcasting them to all the computing nodes in the network.

5.3.2 Switch

A switch is similar to a hub in physical structure except that it forwards data directly to its destination address without broadcasting it to all the other nodes in the network.



Picture 5. 9. Computers connected to a network switch

In order for a switch to be able to perform its function, it has to keep records of the addresses of all the nodes in the network.

When a data signal is received, it goes through the records to confirm the address and transmits the signal directly to the device whose address matches the one in the signal.

5.3.3 Wireless Access Points (WAP)

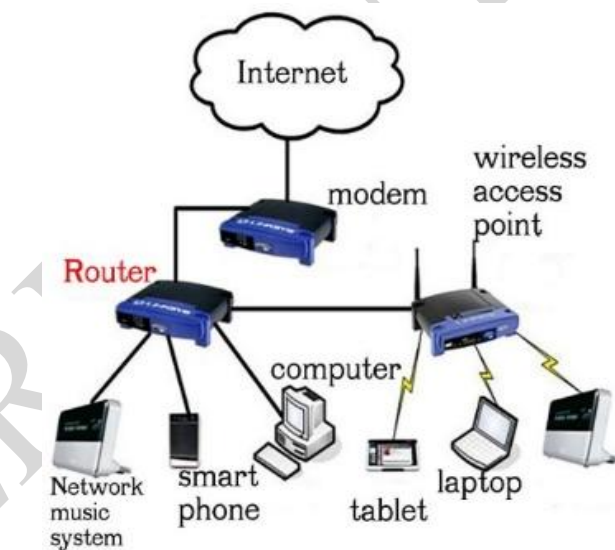
In a wireless computer network, a **wireless access point** (WAP) is a hardware device that allows a Wi-Fi enabled devices such as tablets, laptops, and smartphones to connect to a wired network.

5.3.4 Router

It is a device that connects two different networks and sends data signals from one network to another. For example, it can connect a Local Area Network to a Wide Area Network.

A router performs a routing function which is the process of moving data signals across a network from the source node to the destination node.

A router performs its function by listening to the entire network to determine the best path to route a data signal based on the destination address and the source.



Picture 5. 10. A Router

5.3.4 Network peripherals

Networking peripherals refer to hardware equipment, which enables computers to

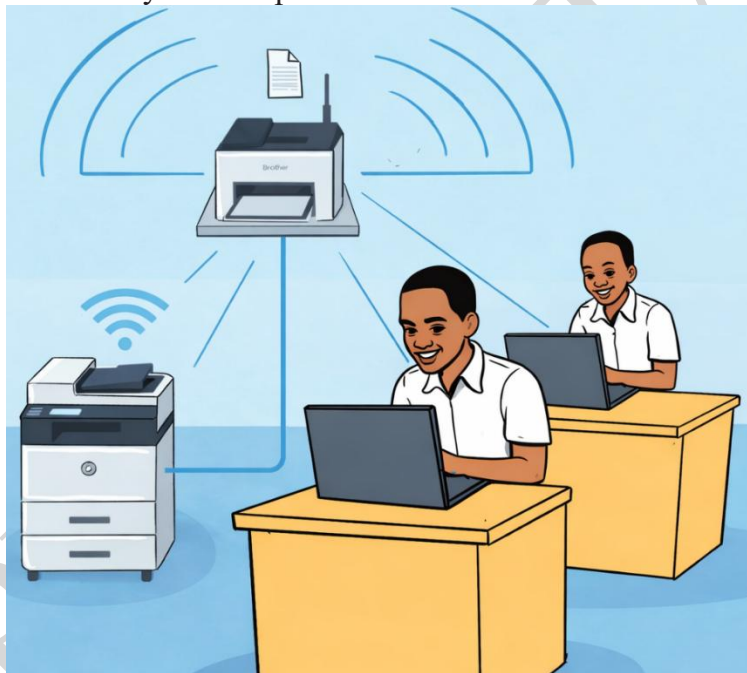
network.

Examples include hardware such as routers, switches, access points, network interface cards, and other hardware devices.

Hubs, switches, and routers are all devices that allow computers to be connected to other computers, other network devices, or even other networks. Each of the above peripherals has two or more connectors called ports. In connecting devices, the connecting cables are plugged in through the ports.

A network peripheral can also be a device that can be connected to a computer in the network either through the ports using data interface cables or using wireless media. Examples are: printers and scanners.

- **Scanners:** These are peripheral devices that input data by capturing data directly from the source to the computer and convert it into a digital format. They are used for making turnaround documents. A turnaround document is a hardcopy output that has been taken back to softcopy for further processing. A scanner in a network can be used by several computers in a network, if it is shared.
- **Printers:** A printer is an output device that produces hardcopy output usually on a physical print medium such as paper. A printer in a network can serve many computers which submit their print jobs to it. If more than one computer sends a job to the printer at the same time, the submitted jobs are placed in a queue. The jobs in the queue are printed one after the other in the order in which they are admitted in the queue until they are completed



Picture 5.12. Shared printer using wireless connection

Application activity 5.3

Part A: Fill in the blanks with the correct answers

- 1) A network device that forwards data directly to its destination address without

broadcasting is known as a _____.

2) Wireless access points are used in a _____ network.

3) A _____ is a device that connects a Local Area Network to a Wide Area Network.

Part B: Answer the following questions

4) State the function of a router in a network.

5) Differentiate between a hub and a switch.

Part C: Connect various devices in a network

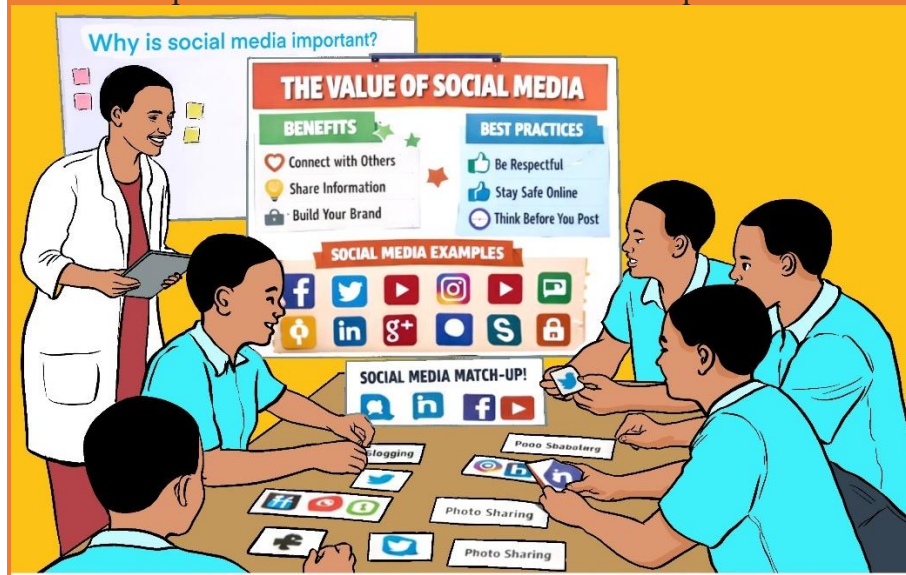
6) You can connect computers in a school computer lab using a switch or hub.

Identify other devices in your environment that can be connected using a wireless access point or a router.

5.4 Social media

Activity 5.4:

Look at the picture below and answer the related questions:



1) a) Which social media illustrated above do you prefer to use?

b) Why do you prefer that social media?

2) List the social media platforms shown in the picture and their uses.

Social media refers to internet-based applications that enable users to create and exchange information, career interest ideas and other content such as images or videos. They enable people to participate in group networking.

5.4.1 Value of social media

Clarity of information: The information should be clear and easy to understand in order to be very helpful.

Effectiveness: Only write or talk about something when there is solid evidence that it is true.

Interesting: One should not be boring. Ensure that the information is engaging and interesting.

Respectful: One should be careful to be respectful and to appreciate diverse points of view with openness and inclusion.

5.4.2 Social media best practices

The following are some of the social media best practices:

- **Transparency:** Being open builds trust among the social media members. It creates links to connect us with one another by always keeping in touch.
- **Be genuine:** Be the real person the members would have the pleasure to know.
- **Be interested in others:** Be a good listener. This shows that one takes an interest in other people.
- **Follow back and interact:** The interacting members should always be in contact. This creates a healthy goodwill for chatting with each other.
- **Do not overshare:** Always focus on sharing items that are of value to members.
- **Respond to positive and negative comments:** Do not only respond to positive comments; pay attention to negative comments as well and respond with maturity. Use good judgement to determine which ones to ignore.
- **Keep it short:** Some social media platforms, for example, Twitter limit the number of characters in a message. Others such as Facebook and Google+ do not have this limit. Keep your messages short.

5.4.3 Examples of social media

Some examples of social media include Facebook, Twitter, Google+, LinkedIn, YouTube, Blogging platforms, and Skype among others.

a) Facebook

This is a popular free online social networking website that allows registered users to create profiles, share information such as messages, games, photos and video and keep in touch with friends, family and colleagues.

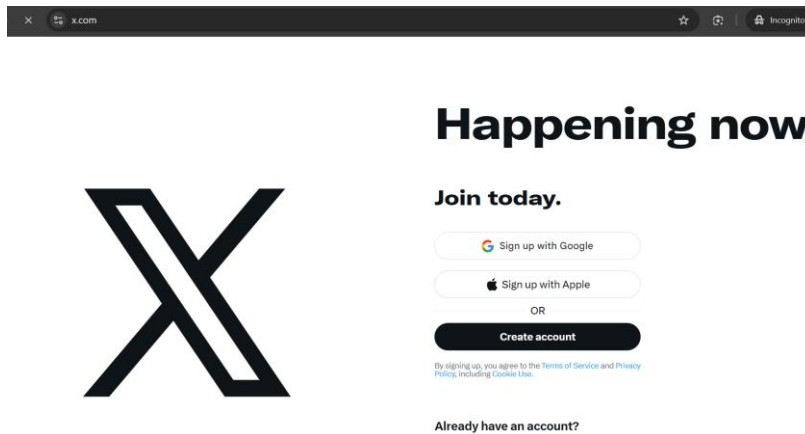
Practice activity: Signup to Facebook

Create a new facebook account by first starting your browser (such as Google Chrome).

- Write **facebook.com** in the address bar
- Follow the guidance to create your account starting with filling your personal details
- Search for friends on facebook

b) X

X was formerly known as Twitter. It is a free online social networking service that allows registered members to send short messages or short posts referred to as tweets.

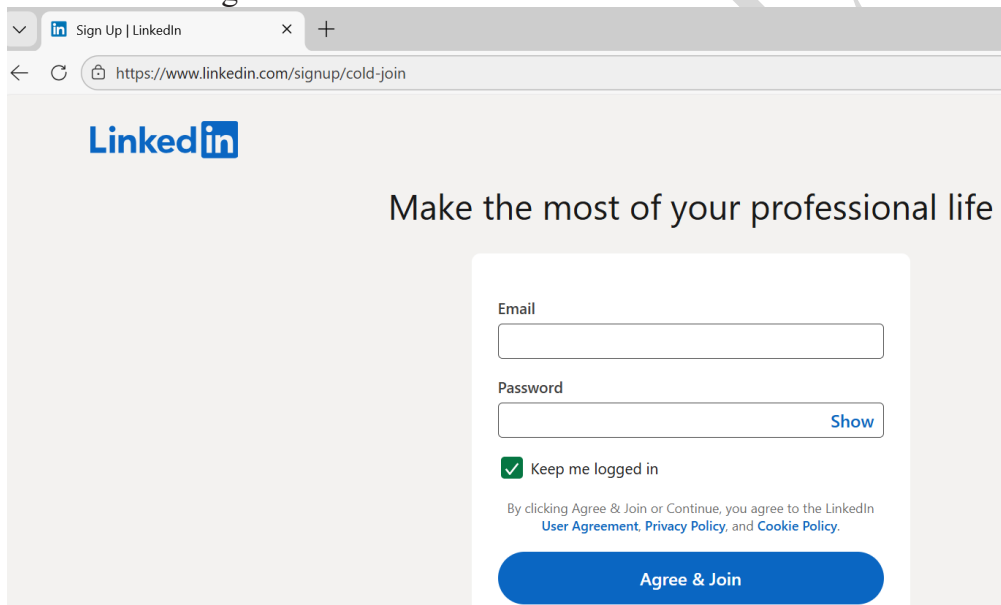


Picture 5. 11. X first window

c) LinkedIn

This is a social networking site designed for professional networking. It allows registered members to establish and document networks of people they know and trust professionally.

To log in to LinkedIn or sign up for an account, search “LinkedIn” on Google, and follow the instructions given.

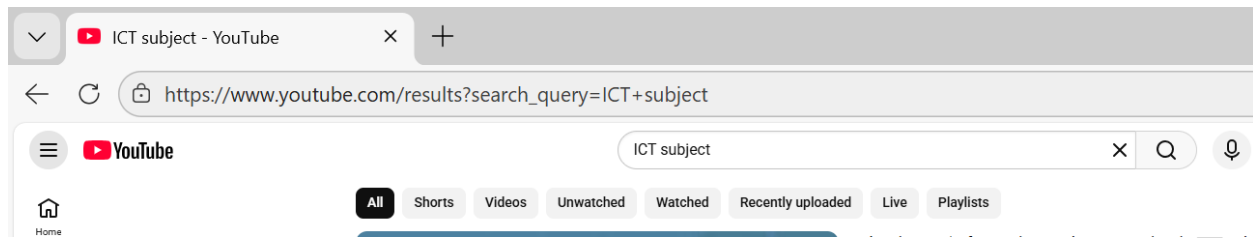


Picture 5. 12. LinkedIn window for user account registration

d) YouTube

This is a free social media site that allows a user to upload, share or watch video content posted from other users.

To access YouTube or write youtube.com in the address bar of your search box then write the title of the video you want to watch.



Picture 5. 13. Part of the youtube window

e) Blogging

A blog is a website that contains online personal content such as opinions, reflections, comments and experiences provided by the writer. It usually has images and links to other websites.

Blogging is the act of posting content on a blog or posting content on another person's blog. Examples of blogging platforms are Wordpress, Blogspot, Tumblr, Weebly, LiveJournal and Blogger.

The best blogging website is WordPress and can be to create professional sites that include service pages, contact forms, and appointment bookings. It can also be used in e-commerce through a plugin called WooCommerce.

Do the following to create a WordPress account for starting using it:

- Go to WordPress.com and click on the **"Get Started"** button in the top left corner.
- Provide your details including your email address. Choose a username and create a strong password. Alternatively, you can sign up using a Google or Apple account.
- Choose a Domain Name: You will be asked to enter a name for your site. If you want a free option, look for the option that includes wordpress.com in the URL.
- Identify Your Goals by specifying the main focus of your website
- Check your inbox for a confirmation email. You must click the link inside to verify your account before you can officially launch your site for others to see.

5.4.4. Impact of social media

Social media has some positive and negative impact on individuals and society. Those impacts are depend on how it's used, how often, and by whom.

a. Positive impact

- **Global connectivity:** Platforms like Facebook, Instagram, and X (Twitter) let people communicate instantly across the world.
- **Information:** Social media spreads news quickly in a short time.
- **Business and career opportunities:** It helps small businesses reach customers easily and allows individuals to build personal brands, find jobs, or work as influencers.
- **Learning and creativity:** Users can learn new skills, share ideas, and express themselves through videos, posts, and discussions.

b. Negative impact

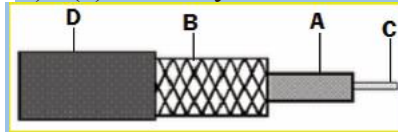
- Mental health issues: Heavy use is linked to anxiety, depression, and low self-esteem due to comparison with people you see on social media who display lavish life they don't really live.
- Addiction and time waste: Apps are designed to keep users scrolling, which can reduce productivity.
- Cyberbullying and harassment
- Misinformation: False or misleading content spreads quickly, influencing public opinion.

Application activity 5.4

- 1) List three examples of social media platforms and their uses.
- 2) State three best practices for using social media safely.
- 3) What are the potential risks associated with social media use?
- 4) Mention two benefits of social media in an institutional setting.

End unit assessment 5

- 1) Explain three benefits that an institution would gain by installing a computer network.
- 2) Name two types of twisted cables.
- 3) (a) Identify the transmission media shown in this diagram.



- (b) Name the parts labelled A to D.
- 4) Apart from printers and scanners, give two other network peripheral devices.
- 5) Social media is today becoming very popular in communication. List three examples of social media platforms.
- 6) State three best practices of social media.
- 7) Mention two values of social media.
- 8) Explain how you would behave while using social media platforms

UNIT6. GAME PROGRAMMING



Key Unit Competency:

By the end of this unit, you should be able to create interactive digital games by programming sprite interactions, managing game logic, and implementing progressive difficulty levels

.Learning Objectives

Knowledge and understanding

- Explain the basic components of a computer game
- Identify the role of each type of sprite in game development.
- Explain the function of different block categories to set up a game

Skills

- Create a new game project using a programming tool
- Set up interactions between sprites using control and sensing blocks
- Manage different backgrounds for different game levels

Attitudes and values

- Express the desire to produce more interesting games using Scratch commands.
- Appreciate the creativity involved in game design and programming.

6.0. Introduction

Game programming is the process of writing code that controls how a video game works, including how characters move, how players interact, and how the game responds to actions. It involves taking the design specifications of the game designs, combining the art and

sound and translating it into a playable game.

In this unit you will acquire the necessary programming logic necessary for creating a game and you will apply this in Scratch.

Introductory activity

You have played a video game at least once in your life. Answer the following questions about that video game.

- 1) Mention the name of that game you once played.
- 2) In the game you mentioned what do you control?
- 3) What makes the game end?
- 4) What makes someone win the game?

6.1. Game concepts

Activity 6.1

The "Catch the Falling Apple" game is a game in which an apple falls from the top. A basket at the bottom must move left and right to catch it. Each time the basket catches the apple, the score goes up. If the apple touches the ground, the game is over.

- 1) Write down "If... Then..." statements for the Falling Apple game.
- 2) What makes the game end?
- 3) What makes the player's score go up?

To create a basic game, have first an idea of the game name, list of sprites, sprite interaction, backgrounds, the counter rule and level advancement rule. The steps to create a game include designing the game, creating game sprites, adding game rules to game sprite(s), changing background when the sprite moves and managing scores and levels.



Picture 6. 1: A group of learners planning together a game

6.1.1 Game name

When creating a game, it is important to assign a unique name to it. This is done in order to be able to retrieve and play the game easily. It is a good practice to have the name of the game related to the nature of the game.

For example when creating a Cat and Mouse game a favourable name would be **Cat_Mouse game**.

6.1.2 List of sprites

a) Designing a game

This involves creating a basic table that can be filled in as shown below:

Game name	Desired game name, for example, Cat_Mouse Game
Sprites	List of sprites in the game (Sprite name, Costumes, Sounds, Movements)
Sprite Interaction	Relationship between sprites in the game.
Backgrounds	The area that is behind the sprites, forming the setting for it. For example, the Blue sky available in the Backdrop Library
Scores and Levels	The score counter rule and level of advancement rule used to determine the result of the game.

Table 6. 1: Aspects of a game

After creating a basic table, it can be expanded by placing each item to its own table as shown in the **Cat-Mouse-Cake Game** example below:

Example:

- The Cat moves back and forth across the screen and keeps moving in one direction. When it hits the edge of the stage, it bounces back. When the Cat touches the Mouse, it produces the sound, Meow2 and the thought bubble Meow!!! is displayed. The background switches to Brick Wall1 when the Cat eats the Mouse.
- The Mouse reappears after four seconds.
- The Mouse chases the mouse-pointer in steps and when it touches the Cat sprite, it disappears. The Mouse reappears after a certain period of time.
- The Cake disappears when it is touched by the Mouse because it is eaten. The background switches to Garden Rock when the Cake is eaten and the Cake keeps changing between Cake-a and Cake-b costumes.

(i) Sprites table

The table below provides a list of the sprites in a project with details about the name, costumes, sounds and movements

	Name	Costumes	Sounds	Movements
--	------	----------	--------	-----------

Sprites	Cat	None	Meow 2	Run
	Mouse	None	Pop	Run
	Cake	Cake-a, Cake-b	None	None

Table 6. 2: List of sprites

in the Cat_Mouse_Cake game

ii) Interaction table

The table below lists all the relevant interactions between various sprites in the project.

Sprite 1	Sprite 2	Interaction
Cat	Mouse	If the Cat touches the Mouse, the Mouse disappears.
Mouse	Cake	If the Mouse touches the Cake, the Cake disappears.

Table 6. 3: Interaction of sprites in the Cat_Mouse_Cake game

(iii) Backgrounds table

The table below lists the relevant backgrounds in the project.

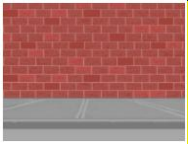
Name	Description
 Level 1_1	Start of the game, all sprites displayed.
 Level 1_2	Cat and Mouse as enemies.
 Level 2_1	Mouse and Cake as enemies.
Level 2_2	Garden rock

Table 6. 4: Backgrounds table

(iv) levels table

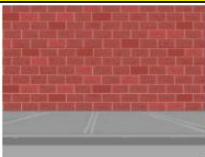
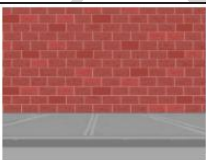
Level	Requirement for this level	Starting background	Ending background
1	When the game is started	 <i>Doily Level 1_1</i>	 or  <i>Brick wall Level 1_2</i> or <i>Garden rock Level 2_2</i>
2	When the cat catches the mouse	 <i>Doily Level 1_1</i>	 <i>Brick wall Level 1_2</i>
3	When the Mouse eats the Cake.	<i>Doily Level 1_1</i> or <i>Brick wall Level 1_2</i>	 <i>Garden rock Level 2_2</i>

Table 6. 5: Levels table

b) Costumes

Sprites can have different costumes. Once a sprite has been inserted in a game, its appearance can change to any of the available costumes. The game designer can also draw a costume for a sprite.

However, some sprites do not have costumes. In the **Cat_Mouse_Cake** game, the cake has two costumes, namely cake-a and cake-b. The mouse and cat do not have costumes. To switch between costumes, do the following:

- Select the sprite if already inserted, otherwise insert the sprite.
- Click on the costume tab.
- Select the desired costume from the available thumbnails displayed.



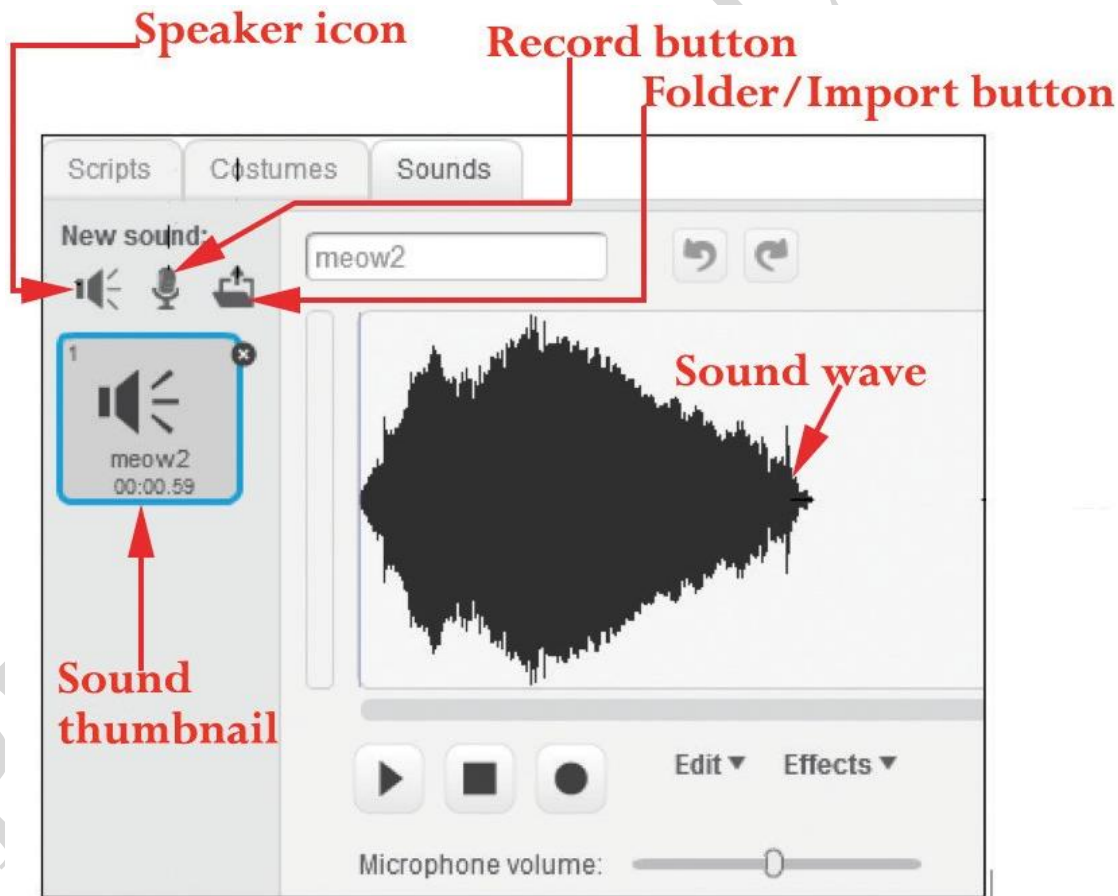
c) Sounds

A sprite existing in a game can be assigned a sound. By default, some sprites do not have sounds assigned to them.

The designer can choose sound from the sounds library, record new sound or upload a sound from file. In the **Cat_Mouse_Cake** game described above, the Cat has the **Meow** sound.

To assign a sound to a sprite, do the following:

- Select the **Sprite** then click on the **Sounds** tab.
- Select the sound from the thumbnail displayed. If no sound is displayed in the thumbnail area, click on the **Speaker** icon to choose sound from library, click the **Record** button icon to record new sound or click on **Folder/Import** button to select the sound file from the computer.



Picture 6. 3. The available Cat sound

c) Movement

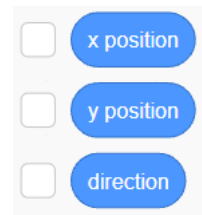
Sprite movement is also referred to as sprite action. The **Motion** block palette contains the commands for assigning controlling movement to a sprite. There are seventeen Motion blocks categorized into two namely **Motion Stack** blocks and **Motion Reporter** blocks with fourteen and three blocks respectively

Move () steps: Moves the sprite in the direction it is facing the amount of steps specified.

- **Turn () degrees:** Turns the sprite either clockwise or anti-clockwise by the specified amount of degrees.
- **Point in direction ():** Points the sprite in the direction specified.
- **Point towards ():** Points the sprite towards the mouse-pointer or the selected sprite.
- **Go to x: () y: ():** Moves the sprite to the specified x and y positions.
- **Go to ():** Moves the sprite to the mouse-pointer or the selected sprite.
- **Glide () secs to x: () y: ():** Moves a sprite smoothly to an x y position on the stage by the amount of time specified.
- **Change x by ():** Changes the x position of the sprite by the amount specified.
- **Set x to ():** Sets the x position of the sprite by the amount specified.
- **Change y by ():** Changes the y position of the sprite by the amount specified.
- **Set y to ():** Sets the y position of the sprite by the amount specified.
- **If on edge, bounce:** This means that if the sprite is touching the edge of the screen, its direction turns over quickly or reverses.
- **Set rotation style ():** Determines the manner in which the sprite rotates.

The following are **Motion Reporter** blocks:

- **X position:** Displays the X position of the sprite.
- **Y position:** Displays the Y position of the sprite.
- **Direction:** Displays the direction of the sprite.



Picture 6. 4: Motion Reporter blocks

6.1.3 How sprites interact in this Game

In a game different sprites interact with each other for the success of the game. In the interaction class, there are different commands which are used when creating a game. In the example of **Cat_Mouse_Cake** game, the sprites named Cat, Mouse, and Cake relate with one another.

6.1.4 List of backgrounds

The term background refers to those things that can be seen behind the main things. There are

various backgrounds in Scratch and can be accessed through the backdrop pane. The user can choose a backdrop from library, paint a new backdrop, upload a backdrop from file or get a new backdrop from camera.

To apply a background in a game, do the following:

- Click on the icons under **New backdrop** and select the desired option.
- To choose backdrop from library, click the **Choose backdrop from library** icon. This displays the **Backdrop Library**.
 - Scroll to view all backdrops available in Scratch or click on each folder under the Backdrop Library pane to view the backgrounds underneath.
 - Select the desired background.
 - Click **OK** to apply.

The example below shows how in a Scratch program the background can be changed.

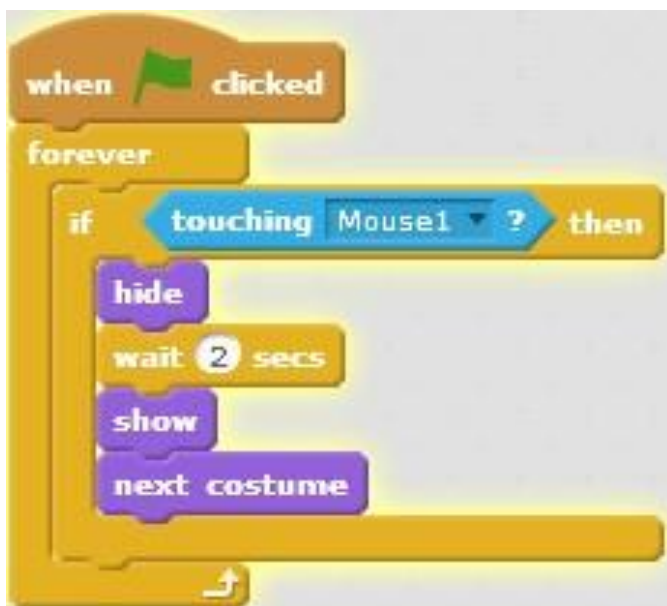
	• The starting background switches to Doily when the Green flag is clicked. • The Cat moves back and forth across the screen and keeps moving in one direction and when it hits the edge of the stage it bounces back. • When the Cat touches the mouse it produces sound Meow2 and thought bubble Meow!!! is displayed. • The background switches to Brick wall1.
--	--

Picture 6. 5. Script for the Cat sprite

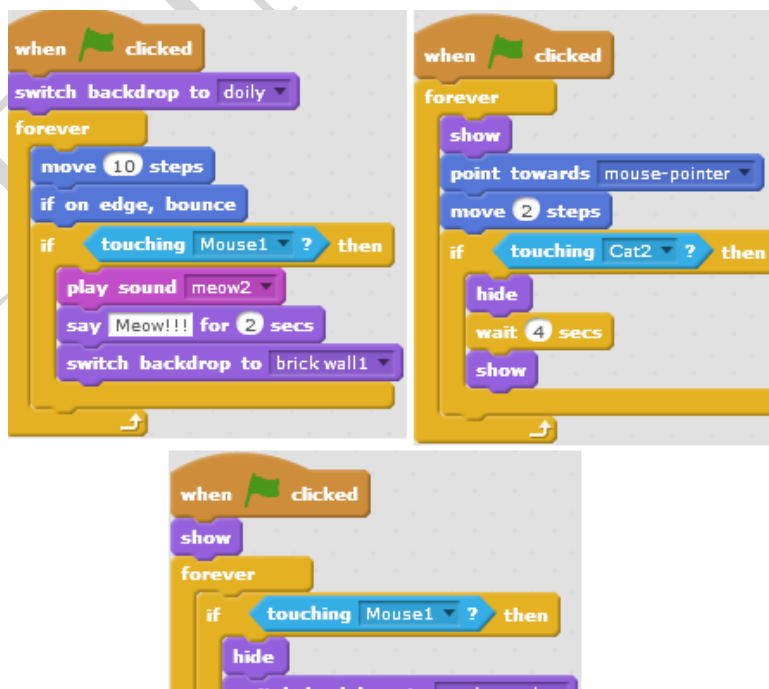
In the picture below, the mouse chases the mouse-pointer in steps and when it touches the Cat sprite, it disappears; that is, it is eaten. The Mouse reappears after four seconds.



From the picture below, the **Cake** disappears when it is touched by the **Mouse**; that is, it is eaten and reappears after two seconds. The background switches to **Garden rock** when the **Cake** is eaten and the **Cake** keeps changing between **Cake-a** and **Cake-b** costumes.



Picture 6. 7. Script for the Cake sprite





Picture 6. 9. Cat Mouse Cake game

Application activity 6.1

Part A: Choose the correct answer from the options given.

- 1) Sprite movement is also referred to as _____.
 a) Motion b) Programming c) Designing d) Drawing
- 2) Which block palette contains the movement commands?
 a) Sound b) Looks c) Motion d) Events
- 3) Motion blocks can be divided into ... and ... blocks.
 a) Sound and Looks b) Stack and Reporter
 c) Control and Sensing d) Events and Operators
- 4) Sound can be assigned to a sprite by clicking on the ... icon ... button, or ... button.
 a) Costumes, Move, Delete b) Sound, Choose a Sound, upload
 c) Events, Start, Stop d) Motion, Turn, Glide

Part B: Answer the following questions

- 1) Give the three Motion Reporter blocks.
- 2) Which motion block turns the sprite either clockwise or anti-clockwise by the specified amount of degrees?
- 3) Give the steps for adding sound to a sprite.

4) What is the procedure for applying a background in a game?

6.2 Collect detailed data in the table

Activity 6.2:

The Fruit Catcher's inventory

You are building a game where a character named Anna collects different types of fruits. Instead of just showing a score, the game needs to keep a record of exactly which fruits she has picked up so she can sell them later in the game. You decide to use a list named My Basket to act as a table for this data.

- 1) If Anna picks up an Apple then a Banana, what will the MyBasket list look like?
- 2) Which Scratch block would you use to clear the table at the start of a new game?

In game programming, collecting and managing data is essential for tracking progress and defining rules. In Scratch, this is primarily done using **Variables** and **Lists**. While a variable holds a single piece of information like a score, a List can hold multiple items of data simultaneously.

In Scratch game programming, the transition from using **Variables** to **Lists** is the step that allows you to move from simple games to complex ones, data-driven applications.

Here are the more technical and creative details on how collecting detailed data in a table works:

a) The anatomy of a Scratch list

Unlike a variable which stores one value, a **list** in Scratch acts like a numbered spreadsheet or a table with a single column.

Index: Every item in the list has a specific position (1, 2, 3...). This allows you to retrieve specific data points.

Dynamic Size: You can add or delete rows while the game is running, meaning your table can grow or shrink based on player actions.

b) Practical uses for data collection in games

Collecting data in tables allows for advanced game mechanics:

- **High score leaderboards:** Instead of just showing the current score, a list can store the top 10 scores and the names of the players who achieved them.
- **Inventory systems:** In adventure games, a list acts as a table of items the player has collected.
- **Level sequencing:** You can store the coordinates of where enemies should appear in a table, and have the game read from that table.
- **Saving progress:** You can collect data about a player's choices or achievements to determine which ending they get in a story based game.

c) Essential table management blocks

To collect and manage this data, Scratch provides specialized blocks found in the **Variables** palette:

- **add [thing] to [list]:** Appends a new piece of data to the end of a table.
- **delete (1) of [list]:** Removes specific data.
- **item (index) of [list]:** It is used to read the data by specifying the index (place) at which it is located.
- **contains [thing]?:** A sensing block that checks if a specific piece of data exists anywhere in a table.

c) Data Initialization

It is the process of setting the starting values for your game's variables and state before the player begins.

- **Clearing the table:** At the very start of a game programmers usually use the delete all of [list] block which ensures that the data from the previous player does not carry over into the new game.
- **Setting Defaults:** If your game relies on a table of "Level Data," you might use a script to automatically fill the list with default values at the start.

d) Why use a Table (List) instead of many Variables?

Imagine a game where you collect 50 coins. Creating 50 different variables (Coin1, Coin2, etc.) would be difficult and impossible to manage. By using a List, you have one single table that manages all 50 items efficiently, making the code more understandable.

Application activity 6.2

- 1) What is the difference between a variable and a list in Scratch?
- 2) How do you create a data table (List) in Scratch?
- 3) Which block is used to increase a player's score by 5 points?
- 4) Why is it important to initialize variables at the start of a game?
- 5) Can a list be used to create a high score leaderboard? Explain.

6.3. Command to set game rules

Activity 6.3

- 1) Which block category would you use to change the color or size of a sprite?
a) Motion b) Looks c) Sensing d) Operators
- 2) If you want a sprite to move forward forever, which block should you put the "move" command inside?
a) Wait until b) If then c) Forever d) Repeat 10
- 3) Which block is used to pick a random number for a game like when rolling a dice?
a) Round b) Pick random () to () c) Length of d) Mod
- 4) a) Explain the difference between the If then block and the If then else block.
b) In what situation would a game need the else section?

6.3.1 Looks block palette

The Looks block palette is the category of blocks that controls the visual appearance of sprites and the stage. This block manages what the player actually sees, such as costumes, speech bubbles, size, and graphic effects. They can be used for 4 purposes as detailed below:

a) Providing feedback

These blocks allow the sprite to display text to the player like for example giving "Game Over" warnings.

- **Say [Hello!] for [2] secs:** Displays a speech bubble for a set time (best for temporary messages like "Congratulations!").
- **Say [Hello!]:** Displays a speech bubble that stays until changed (best for constant instructions).
- **Think [Hmm...] for [2] secs:** Displays a thought bubble (cloud-shaped) for a set time.
- **Think [Hmm...]:** Displays a permanent thought bubble.

b) Creating animation

It is like when a brick is made to appear cracked by switching different costumes.

- **Switch costume to [costume name]:** Instantly changes the sprite to a specific look (e.g., changing Anna to her "Jump" costume).
- **Next costume:** Moves to the very next look in the list (perfect for walking animations inside a forever loop).
- **Switch backdrop to [backdrop name]:** Changes the entire stage background.
- **Next backdrop:** Cycles to the next background image.

c) Indicating game state

These blocks manage the presence of objects. It includes the blocks below:

- **Show:** Makes the sprite visible on the stage.
- **Hide:** Makes the sprite completely invisible (though it can still run scripts in the background).
- **Go to front layer:** Ensures the sprite appears on top of all other objects.
- **Go back [1] layers:** Sends the sprite behind other objects.

d) Visual effects and sizing

These blocks allow the sprite to change its size and color.

- **Change size by [10]:** Increases or decreases the sprite's size.
- **Set size to [100] %:** Instantly sets the sprite to a specific scale on a range from 1% to 100%.
- **Change [color] effect by [25]:** Applies a visual filter.
- **Set [color] effect to [0]:** Fixes the effect at a specific intensity.
- **Clear graphic effects:** Instantly removes all visual filters and returns the sprite to its original appearance.

6.3.2 Control block palette

a) Timing blocks:

These manage the speed of the program.

- **Wait [1] seconds:** It pauses the program for a specified time. It is useful for making animations look natural rather than instant.
- **Wait until <condition>:** It pauses the script until a specific sensor is triggered

b) Loop blocks

These allow to repeat actions without dragging the same blocks again.

- **Repeat [10]:** Runs the code for a certain number of times specified.
- **Forever:** Runs the code endlessly until the "Stop" button is clicked.
- **Repeat until <condition>:** This is a loop that stops once a specific requirement is met.

c) Conditional blocks

These allow the program to react differently depending on what is happening.

- **If <condition> then:** If the condition is true, the code inside runs once.
- **If <condition> then/else:** This provides two paths. If the condition is true, do the first thing; if not, do the second thing.

d) Cloning blocks

These allow a sprite to make copies of itself while the project is running.

- **When I start as a clone:** A starting block for the copy sprite.
- **Create clone of [myself]:** The command to make the copy.
- **Delete this clone:** Removes the copy to keep the project running smoothly.

e) Termination Blocks

- **Stop [all]:** Ends the program or specific parts of it.

6.3.3 Sensing block

a) Touch and distance

These blocks help a sprite know where it is in relation to other objects on the stage.

- **Touching [mouse-pointer]?:** This block checks for contact.
- **Touching color []?:** Detects if the sprite is touching a specific color in the backdrop or another sprite.
- **Color [] is touching []?:** Checks if a specific part of a sprite (identified by color) hits another color.
- **Distance to [mouse-pointer]:** Reports the distance between the sprite and another object.

b) User input

These blocks allow the person playing the game or using the app to interact with the code.

- **Ask [What's your name?] and wait:** Opens a text box for the user to type an answer.
- **Answer:** A variable block that stores whatever the user typed in the "Ask" box.
- **Key [space] pressed?:** Checks if a specific keyboard key is being held down.
- **Mouse down?:** Checks if the user is clicking the mouse or tapping the screen.
- **Mouse x and Mouse y:** Reports the exact coordinates of the mouse cursor.

c) Environment and variables

These blocks get data from the computer's system or other parts of the project.

- **Loudness:** Detects the volume level from the computer's microphone.
- **Timer:** A stopwatch that starts the moment the project begins.
- **[Backdrop #] of [Stage]:** A powerful block that lets one sprite peek at the properties of another.
- **Current [minute/hour/day]:** Pulls the real-world time from the computer.

6.3.4 Motion block

The motion blocks program the sprite to move from one place to another on the stage which is 480 units wide and 360 units high.

a) Instant motion

These blocks change the sprite's position or orientation immediately with no visible travel time.

- **Go to x: [] y: []**: Sets the sprite to a specific coordinate.
- **Go to [random position]**: moves the sprite to a random spot.
- **Set x to [] / Set y to []**: Changes only one axis while leaving the other unchanged.
- **Point in direction []**: Instantly snaps the sprite to face a specific angle.

b) Continuous motion

These blocks are used for creating animations where the user can see the sprite move across the screen.

- **Move [] steps**: Moves the sprite forward based on its current direction.
- **Glide [] secs to x: [] y: []**: Moves the sprite from point A to point B over a set duration.
- **Change x by [] / Change y by []**: Shifts the sprite relative to its current spot.

c) Rotation and turning

These blocks deal specifically with the sprite's direction value.

- **Turn right/left [] degrees**: Rotates the sprite relative to its current angle.
- **Point towards [mouse-pointer]**: Constantly updates the angle to face a target.
- **Set rotation style [left-right / don't rotate / all around]**: Controls how the costume looks when it rotates. This is used to prevent the upside-down effect.

d) Boundary and physics logic

These blocks help the sprite interact with the walls of the stage.

- **If on edge, bounce**: Automatically changes the sprite's direction when it hits the stage boundary.

e) Data reporters

These are not action blocks instead they report the current state of the sprite. They can be used inside other blocks to create complex logic.

- **x position**: The current horizontal value.
- **y position**: The current vertical value.
- **direction**: The current angle

Application activity 6.3

1) Which control block runs the blocks inside a specified number of times?

- a) Forever b) If () then c) Repeat () d). Repeat until ()

2) What is the name given to a block that checks if a sprite is touching a specific colour?

- a) Colour () is touching () b) Touching colour ()
c) Touching () d) Current ()

3) Which blocks can be used to control how the sprite appears?

- a) Switch costume to (), show, hide, switch backdrop to ().
b) Rest for () beats, play note () for () beats, set pen colour to (), change pen shade by ().
c) Set () to (), change () by (), add () to (), show list ().

d) Switch costume to (), go to (), hide, set y to ().

4) Explain any 3 Sensing Reporter blocks.

6.4. Creating a game

Activity 6.4

1) Which tool in the Scratch backdrop pane should you use to draw the black road?

a) The Eraser tool b) The Brush tool c) The Fill tool d) The Text tool

2) Answer by True or False:

a) The “Move 1 step” block should be placed inside a “Forever” or “Repeat” loop to make the car drive continually.

b) Using the “move 10 steps” block will always move a sprite to the right, regardless of which way it is pointing.

c) What happens to Y-coordinates when the sprite isn't touching the ground?

Building a game requires turning an idea into a technical plan. In this project, you will build a complete system using Scratch blocks.

The steps below are developed to make a racing car game.

a) Collect detailed data in the table

The following are the two relevant tables that can be associated with this game.

• Game name table

Game name	Car Racing game
How to play	• The car is driven along a road by manipulating the arrow keys. • Clicking the Green flag takes the car to the beginning of the track.
How to win	The car should not leave the track or touch the grass, and should get to the finish line; that is, touch the Red line.

Table 6. 6. Game name table

• Sprites table:

There is only one sprite for this game, that is, the Car.

Sprites	Name	Costume	Sounds	Movements
	Car	Car-bug	None	Drive

Table 6. 7. Car sprite table

• Background table

Name	Description
Level 1_1	Green ground, game stops when car touches the green colour.

Level 1 2	Track for the car to move
Level 1 End	Red line

Table 6. 8: Background table

- **Score management table**

How	How many points
Car off the track	Percentage of the distance before car getting off the track
Car touching red line	Total distance =100%

Table 6. 9: Scores table

- **Sprite interaction table**

The sprite interacts with the backgrounds; that is, the Track (black colour) and the Grass (green colour). This is shown in the table below:

Sprite	Background	Interaction
Car	Track (Black)	The game continues provided the Car does not leave the Track.
Car	Grass (Green)	If the Car touches Grass, the game stops.

Table 6.10: Sprite interaction table

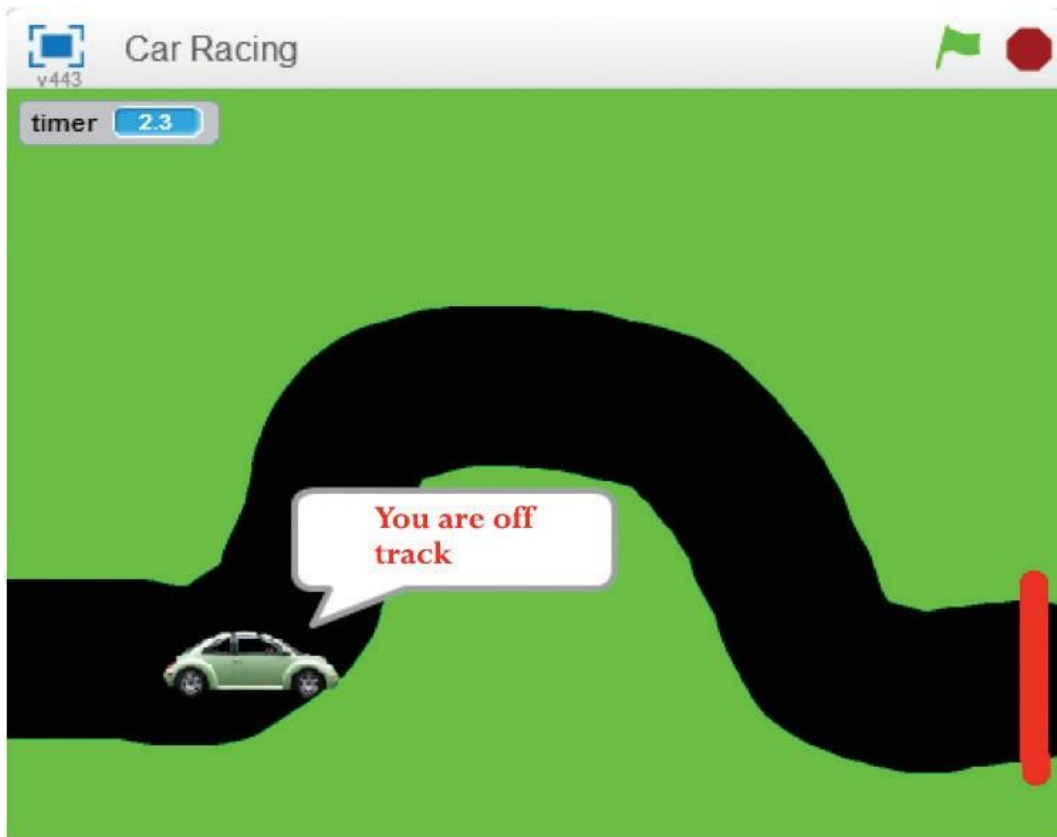
b) Game rules

- The car goes back to the beginning of the track after 2 seconds if it touches the grass and a message “You are off track” is displayed
- If the car gets to the end of the track by touching the Red line, a message “Congratulations” is displayed for 2 seconds followed by a timer showing how long it took for the car to get to the finish line.

To create the game:

- Delete the Cat sprite.
- **Create background.** Go to the Stage backdrop pane and click **Fill with colour**. Click the Scripts area to apply the preferred background colour. The background colour for this game is green to signify green grass.
- Paint a road running across the screen and draw a coloured line at one end to mark the finish line. To paint a road and line, use the **Brush** and **Line** icons respectively. The road is black and the finish line is red.
- **Add a new sprite.** The sprite for this game is a **Car**. Go to **Choose sprite from library** and click **Transportation**. Select **Car-Bug**.
- Reduce the size of the sprite to fit on the road and to enable the car to go around the bends using the **Shrink** button . To do this, click on the **Shrink** button then click on the car.

- Set game rules



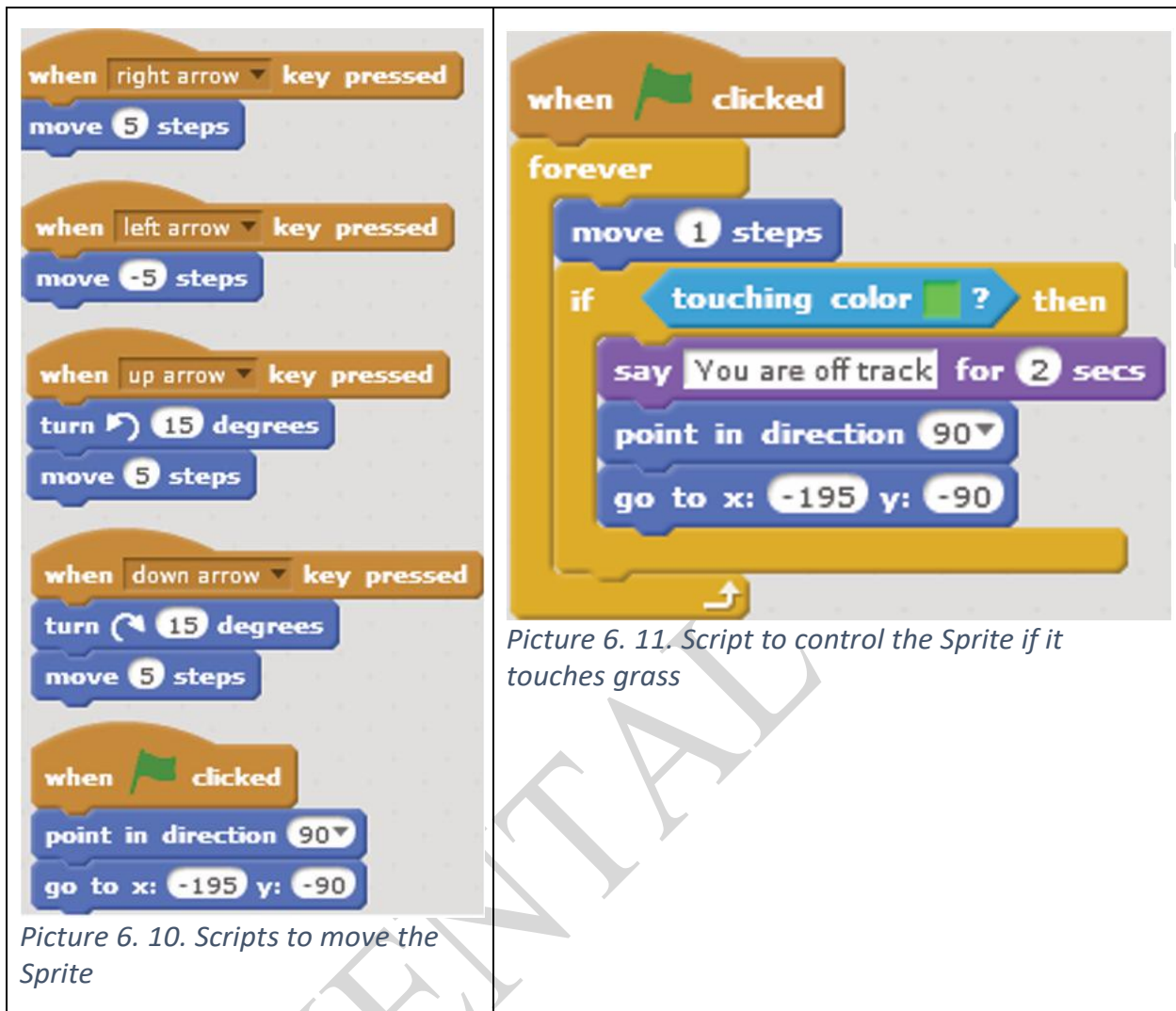
Picture 6.10. Stage area showing the Car Racing Game

Program the arrow keys to move the car using the following scripts:

(i) Rule: If the car touches the grass

The Car continually moves 1 step when the Green flag is clicked. If the **Car** touches green colour (grass), a message “You are off track” will be displayed for two seconds and it goes back to the starting point. The game is won if the **Car** gets to the finish line.

Note: To get the correct colour, click in the colour box of the block. The arrow turns into an eyedropper. Move the eyedropper over the area of the stage containing the desired colour and click to reflect the new colour.



Note: To get the correct colour hold the dropper over the area of the stage containing the desired colour.

(ii) Rule: If the car gets to the end of the road.

A message “Congratulations” is displayed for 2 seconds each time the car touches the Red line, followed by a timer that takes 10 seconds showing how long it took for the car to get to the finish line.

(iii) Combine the if conditions



Picture 6. 12. Script to control the Sprite to the end of the road



Picture 6. 13. Script to control the Game

Final Script for the Car racing game

The above scripts can be combined using the if () then block as shown below:



Picture 6. 14. Final Script for the Car racing game

Application activity 6.4

Improve the Car Racing Game to include the following:

- 1) Add more bends to the road to make the game more challenging.
- 2) Paint the car.
- 3) Add sound effects to different parts of the game.
- 4) Add a variable called timer and, at the end of the game, broadcast a message saying how long it took for the car to get to the finish line.

6.5. End of unit assessment

6.5.1. Project activity

Project Title: Fruit Catcher Adventure

Guidance:

- Set up the stage with a backdrop.
- Add the main character sprite Anna.
- Add fruit sprites: Apple, Banana, Orange.
- Program Anna to move left and right using arrow keys.
- Program fruits to fall from the top of the screen.
- Make Anna collect fruits when she touches them.
- Keep track of collected fruits using a list called My Basket.
- Initialize the game by clearing the list and setting the score to 0.
- Add rules for missed fruits and level advancement.
- Add optional enhancements like sounds, messages and more fruits.
- Test the game regularly to ensure everything works.

Questions on the Project

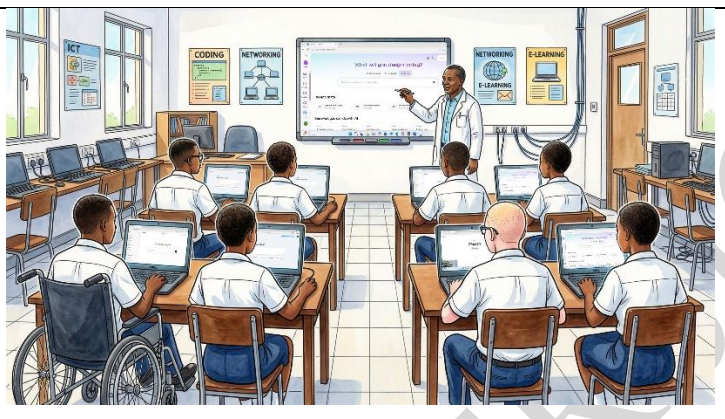
- 1) Explain how the sprites Anna and the fruits interact in the game.
- 2) Describe how the score and the MyBasket list are used to track the player's progress.
- 3) What happens when a fruit reaches the bottom without being caught, and how is this programmed?
- 4) How is Level 2 activated in the game and what changes when the level increases?
- 5) List the Scratch block categories used in this project and the role of each.

6.5.2. End of unit assessment questions

- 1) What is the difference between a variable and a list in game programming
- 2) Which block palette contains commands for sprite movement?
- 3) What are the three Motion Reporter blocks?
- 4) How do you implement a Game Over rule in Scratch?
- 5) What is the function of the "if on edge, bounce" block?

- 6) Which category of blocks is used to change a sprite's appearance or show speech bubbles?
- 7) What is initialization of variables?
- 8) How can you add randomness to a game to make it more challenging?

UNIT 7: AI APPLICATIONS



Key unit competence:

Use AI applications and tools responsibly for learning, content creation and research.

Learning objectives:

At the end of this unit, students will be able to:

Knowledge and understanding

- Explain how AI optimizes professional tasks and communication.
- Identify platforms for automated text, media, and educational resources.

Skills:

- Use tools like ChatGPT and Gemini to produce high-quality digital content.
- Apply AI to automate research, quizzes, and study activities.

Attitudes & Values

- Appreciate AI's impact on creativity while staying curious about new technologies.
- Prioritize data privacy, originality and ethics in AI use.

7.0. Introduction

There are various AI tools that are currently in use and their popularity varies depending on people's preferences. On the other hand, there may be new AI tools in the future while the current ones may lose popularity and become obsolete.

AI applications change how we use technology to work, research, and communicate. By learning the basics and using smart tools, anyone can quickly generate text, images, and videos. This helps turn difficult tasks into simple steps, making work faster and more creative.

Success with AI requires both technical skills and strong ethics. In this unit you will learn how to use different AI tools to generate text, images, etc.

Introductory activity

Your school is hosting a Tech and Science Festival, and your team must create a presentation, a map poster, and a quiz in just 15 minutes. Normally, this would take days of work. However, your teacher introduces AI tools like ChatGPT, Canva, and Gemini to help you research, write, and design everything in a few minutes.

- 1) a) Which AI tools mentioned in the story could help you finish the poster and the quiz

quickly?

b) List one tool for writing and one for design.

2) a) If the AI creates the quiz and the poster for you, does that make you the author or the editor? Explain.

b) Why is it important to check the facts before showing them to the visitors?

3) a) If you use AI to do all the work and tell your teacher you did it all by yourself, is that ethical?

b) How can you use AI and still be honest about your own work?

7.1. Introduction to AI applications

Activity 7.1

MUCYO is stuck on his space homework. He uses an AI application called "Robo-Brain", a smart program that understands and answers questions. First, it gives him three funny story ideas; then, it explains the meaning of the word "astronaut." With these hints, MUCYO writes the rest of the story by himself.

1) a) Based on the story, what is an AI application?

b) Name two ways Robo-Brain helped MUCYO.

2) a) Did AI do all the work?

b) How did MUCYO use the tool to learn something new and finish his project?

3) a) If MUCYO asked AI to write the entire story and told his teacher he did it alone, would that be honest? Explain.

b) Why is it better to use AI only as a helper?

7.1.1. What are AI applications

An AI application is a smart computer program designed to perform specific tasks that usually require human intelligence. Unlike a regular calculator that only follows simple math rules, AI applications can understand language, recognize images, and learn from information. Below are four common types of AI tools:

a. Generative AI tools (Generators)

These are AI tools that create new content such as text, images, or music based on instructions given by the user.

Examples: Canva Magic Media (for images), DALL-E, or AI story writers.

b. Chatbots

These are a type of generative AI tool that you can talk to by typing messages. They answer questions, explain ideas, and help with tasks through conversation.

Examples: ChatGPT, Google Gemini, Microsoft Copilot.

c. Translators

AI Translators change text or speech from one language to another instantly. Modern AI translators are very good because they understand the meaning of sentences, not just single words. They help people who speak different languages communicate.

Examples: Google Translate, DeepL.

d. Summarizers

Summarizer is a tool that takes a long piece of writing and makes it shorter by keeping only the most important points. It saves time by highlighting the main ideas of a long article or

book chapter.

Examples: QuillBot, SummarizeBot.

7.1.2. Learning with AI

AI supports learning in many ways including the following:

- **Personal tutor:** If you don't understand a math problem, you can ask an AI to explain it step by step. It provides instant help when a teacher or parent is not available.
- **Research assistant:** AI can help you find information quickly. Instead of searching through hundreds of websites, you can ask for a list of facts about a specific topic.
- **Study materials:** You can use AI to create practice quizzes to test your knowledge before an exam.

7.1.3. Responsible use of AI

While using AI tools and as a digital citizen, follow three main rules: Ethics, bias awareness, and data privacy.

a. Ethics

Ethics means doing the right thing.

- **Avoid plagiarism:** Do not ask an AI tool to write your whole assignment and then say you did it. This is cheating.
- **The helper rule:** Always use AI as a helper, not a replacement. If AI helps you, it is good practice to mention it to your teacher.

b. Bias

It is when AI shows a preference for one group of people over another or provides one sided information. This happens because AI is trained on information written by humans which may result in AI tool bias if there are faults in the information used for training the AI tools. Therefore always check the information provided by AI. If an AI says something that sounds unfair or incorrect, look it up in a trusted textbook.

c. Data privacy

AI applications remember what you tell them. Therefore protect your identity and never give an AI tool your full name, home address, phone number, or passwords. Also assume that everything you type into a chatbot is being recorded.

Applications activity 7.1

Part A: Match AI tool in column A with its purpose in column B.

Column A

- 1) Chatbot
- 2) Generator
- 3) Summarizer
- 4) Translator

Column B

- a) Makes a long story short and clear.
- b) Changes English text into Kinyarwanda.
- c) Answers questions like a digital assistant.
- d) Creates a new image from a text description.

Part B: Answer by True or False

- 1) It is ethical to let an AI write your entire essay and put your name on it.
- 2) AI can sometimes show bias by favoring one group of people over another.

- 3) You should never share your school password or home address with an AI chatbot.
- 4) AI always gives 100% correct facts, so you don't need to check a textbook.
- 5) Summarizers tool quickly identify the main points of long reports.

Part C: Answer the following questions:

- 1) Give one example of how a student can use a Chatbot to study better for a Science exam.

7.2 AI for content creation

Activity 7.2:

Think about the last time you had to write a long essay or prepare a presentation for class.

- 1) How long did it take to write that essay?
- 2) If you had a digital assistant that could write it for you, how would that change the way you work?

7.2.1 AI text tools

AI text tools use Generative AI to help you write, fix, or shorten text. These tools are like smart editors that understand human language.

- **ChatGPT** and **Gemini**: They are used for brainstorming ideas, writing draft emails, or explaining difficult science topics in simple words.
- **Diffit**: A special tool that takes any article and turns it into a summary, a vocabulary list, or a quiz.

- **Practical demonstration: Summarizing a story**

Step 1: Open Gemini or ChatGPT (<https://chatgpt.com/>)

Step 2: Type “Summarize the history of the Rwandan Kings in three short bullet points for a Grade 7 student”

Step 3: Observe how the AI picks the most important facts instantly.

7.2.2 AI in presentation creation

Creating slides used to take hours of picking colors and layouts. AI tools now help you build the skeleton of a presentation so you can focus on the content to fill in.

- **Canva**: Uses magic design to suggest beautiful layouts and images based on your presentation title.
- **SlidesAI**: It creates a presentation automatically from a paragraph of text.
- **SlideSpeak**: Allows you to upload a long document and automatically turns it into a slide.

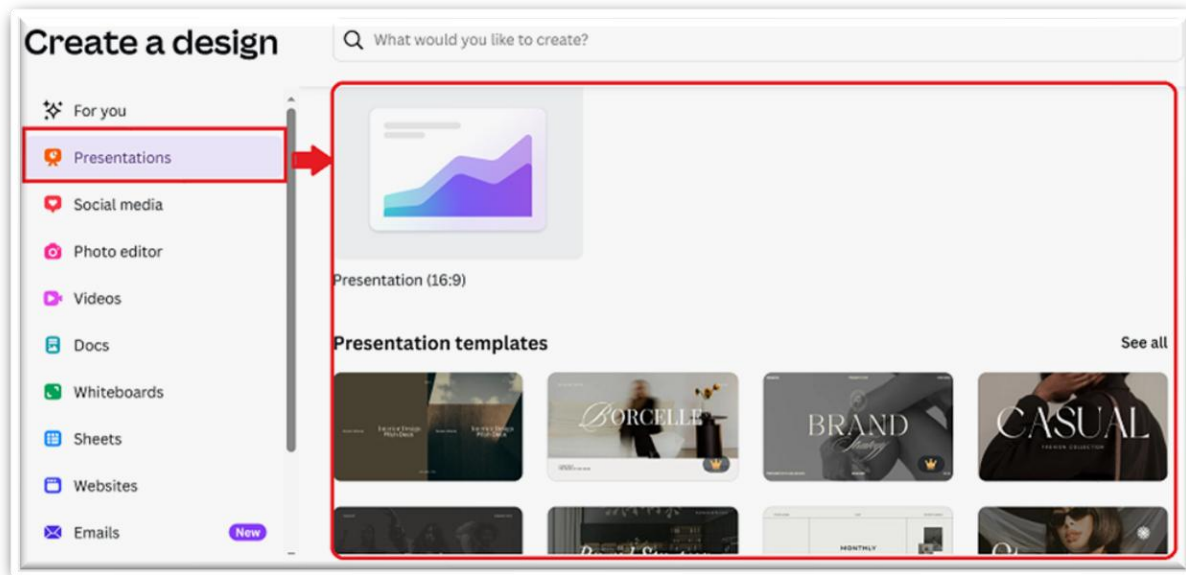
Practical demonstration: Instant Slide Design

Step 1: Open **Canva** (www.canva.com/ai/) and click on **Create**. You may be required to sign up or login.

Step 2: Click on **Presentations**, then choose Presentation Template you desire.

Step 3: Click on **Design**, then describe your idea by typing for example “A5 slide presentation about the benefits of planting trees in Rwanda” then click on submit button.

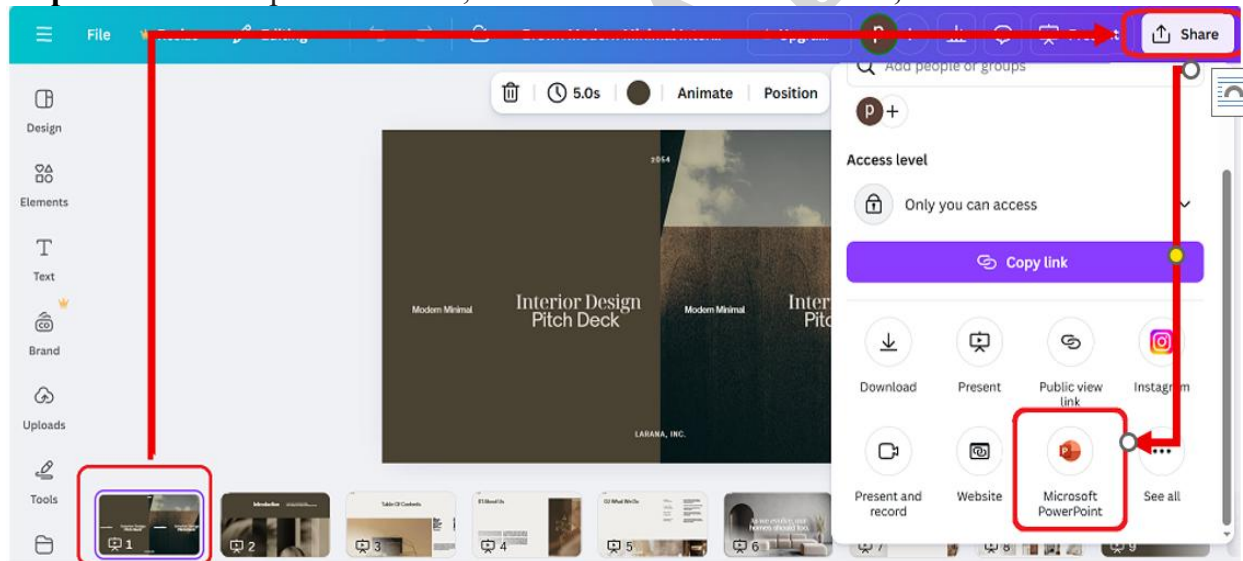
Step 4: Click on **Generate Design**. AI will generate a theme, images, and text for each slide.



Picture 7. 1. Selecting a template

Step 5: Double clicks on the produced slide, then click on **Share**.

Step 6: On the drop down **Menu**, Select **Microsoft Power Point**, then click on **download**



Picture 7. 2. sharing and downloading a presentation

Step 7: Save the file then click on **Open**

Application activity 7.2

- 1) Use an AI tool to write a 3-sentence email to the headteacher requesting permission to start a Coding Club at school.
- 2) If you had to make a presentation about AI in Agriculture, which tool would you choose and why?
- 3) Identify one piece of information in your AI-generated email that you must double-check for accuracy.
- 4) Write a procedure for producing a PowerPoint presentation for this unit using SlideSpeak.

7.3 AI image and video tools

Activity 7.3

You are writing a story about a futuristic city in Rwanda with flying cars and vertical gardens.

- 1) If you cannot draw well, how would you show what this city looks like?
- 2) If you had to make a one-minute movie about this city, what would be the hardest part to do by yourself?
- 3) Can a computer imagine a picture or edit a video if you just give it instructions?

7.3.1 Creating educational images with AI

AI image tools use text to image technology. You type a description and AI generates a completely new image based on your words. Such tools are Canva Magic Media, Bing Image Creator, and DALL-E.

Below are the steps to go through to generate educational images with AI:

- Step 1:** Open **Canva** then click on **Create** then in the **Create Design** pane choose a category such as Education then in this category choose a type by clicking on it.
- Step 2:** Click on **Magic Media** from the **Apps tab** then click on any of the types you want such as Images, Graphics.
- Step 3:** In the prompt box, type: “A realistic 3D diagram of a plant cell with labeled parts like nucleus and vacuole, bright colors”
- Step 4:** AI will provide four different images. Pick the most accurate one for your homework.

7.3.2 Video editing with AI

AI video tools help you create and edit videos faster. They can automatically add subtitles, remove background noise, or even generate video clips from text. The common tools for this are **CapCut (AI features)**, **Pictory**, and **InVideo**.

AI tools that can edit videos have the following features:

- **Auto-Captions:** The AI listens to your voice and writes the subtitles for you.
- **Background remover:** Removes the background behind a person.
- **Text-to-Video:** Turns a written script into a video with stock footage and music.

Follow these steps to add subtitles to a video:

- Step 1:** Upload the video to CapCut.
- Step 2:** Select the **text** menu and click **Auto captions**.
- Step 3:** The AI tool will process the audio and place text on the screen perfectly timed with your voice. You can then change the font to make it look professional.

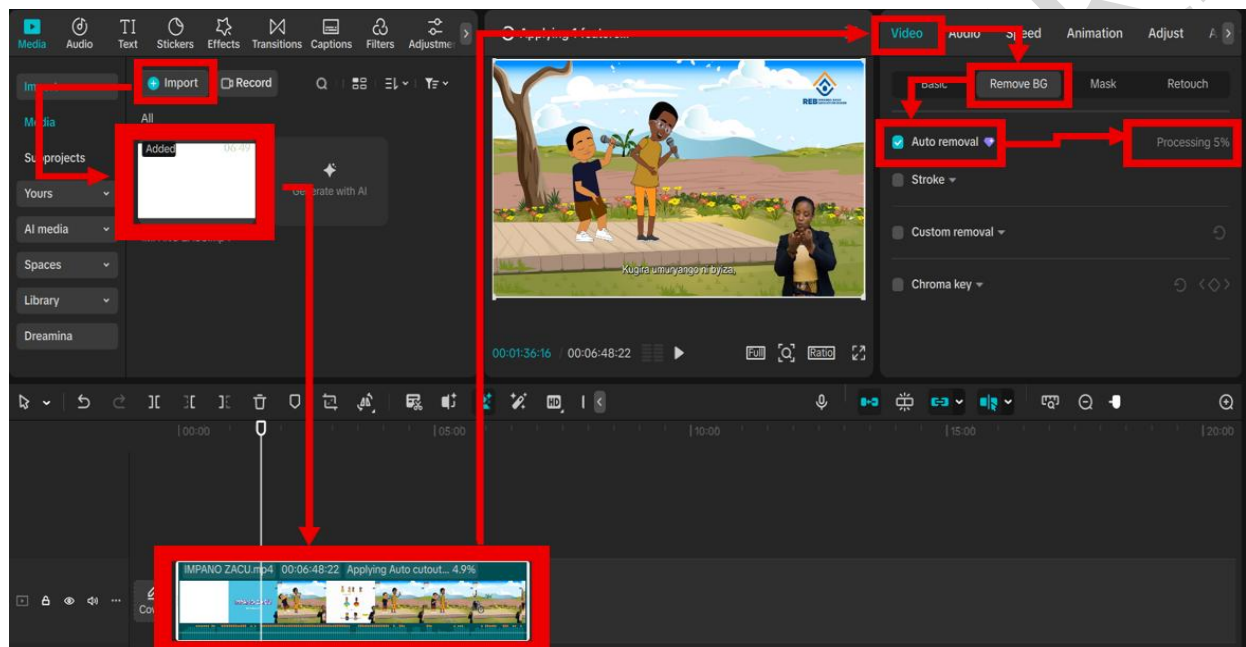
Follow these steps to remove video background in CapCut:

CapCut is mainly a **video editing application**, but it includes several **AI features** that use artificial intelligence to make editing easier and faster. Download capcut on [capcut.com](https://www.capcut.com) and install it on your computer, or use online version.

Step 1: Upload your video first by opening capcut online (capcut.com) or the offline application. In that application, click **Create New Project** then click **Import** to bring your video into the editor.

Step 2: Drag it onto the timeline at the bottom. **the AI Removal Tool**

- Click on your video clip in the timeline so it is highlighted.
- On the right side settings panel, go to **Video**, then click on **Remove BG**.
- Select **Auto removal**.
- The AI tool will now process the video. You will see a percentage bar; wait for it to reach 100%.

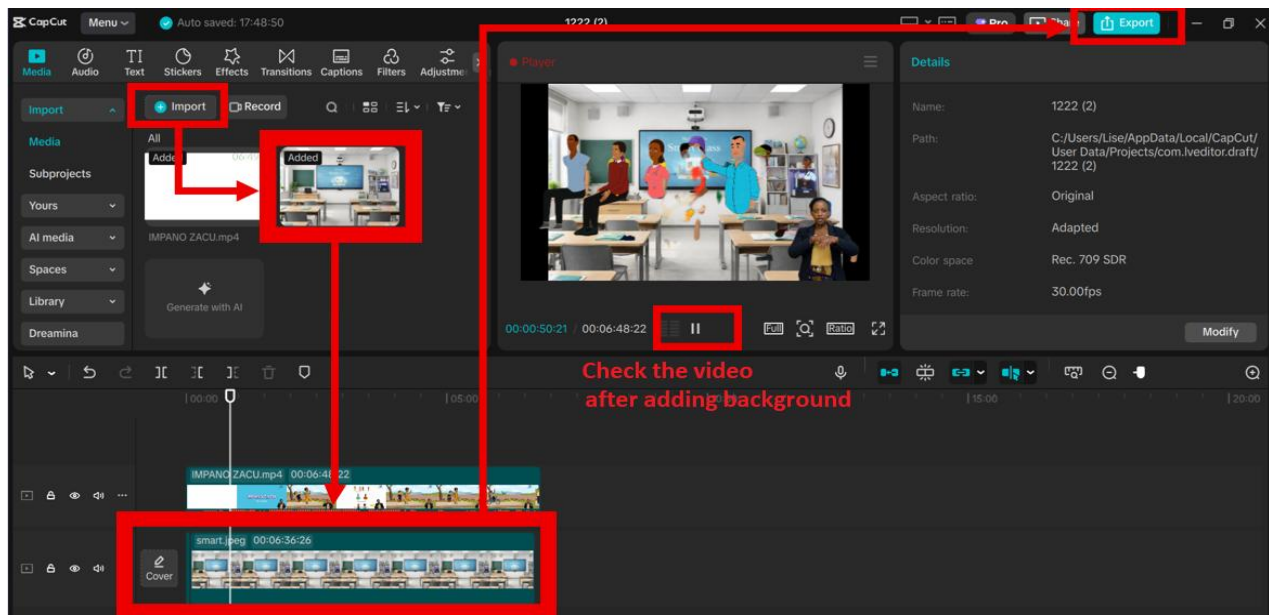


Picture 7. 3. Removing background of video

Step 3: Add a new background

- Once the background is gone, your video will be against a black screen. To add a new background, import a new image or video.
- Drag the new background to the timeline below your original video layer.

Step 4: Export your video by clicking the **Export** button in the top right corner. Choose the resolution and click Export again to save it to your computer.



Picture 7. 4. Adding a new background on Video in Figure 7.3

Application activity 7.3

- 1) Use Canva Magic Media to generate an image of "A clean, modern Kigali city in the year 2050 with solar panels and high-tech buses." Save your best result.
- 2) You are asked to make a video about "The Importance of Handwashing."
 - a) List two AI features you would use to make the video better
 - b) Explain why you chose those features
- 3) Look at the image you generated in question 1.
 - a) Does it have any issues such as extra fingers on a person or strange-looking text?
 - b) Why is it important to check AI images before using them in a project?

7.4. AI research and study tools

Learning activity 7.4

You are given a research topic: "The impact of the volcanic eruptions on local agriculture"

- 1) If you search for this on a traditional search engine, you get millions of links. How do you know which one to trust?
- 2) How much time do you spend reading through long articles to find one specific answer?
- 3) What if you could talk to a document and ask it questions directly?

7.4.1. AI research tools

Unlike traditional search engines that give you a list of websites, AI research tools read through sources and provide a summarized answer with citations which tell you exactly where the information came from.

Here are some AI tools for research:

- **Perplexity AI:** Acts like a research assistant. It searches the internet and writes a cited report.

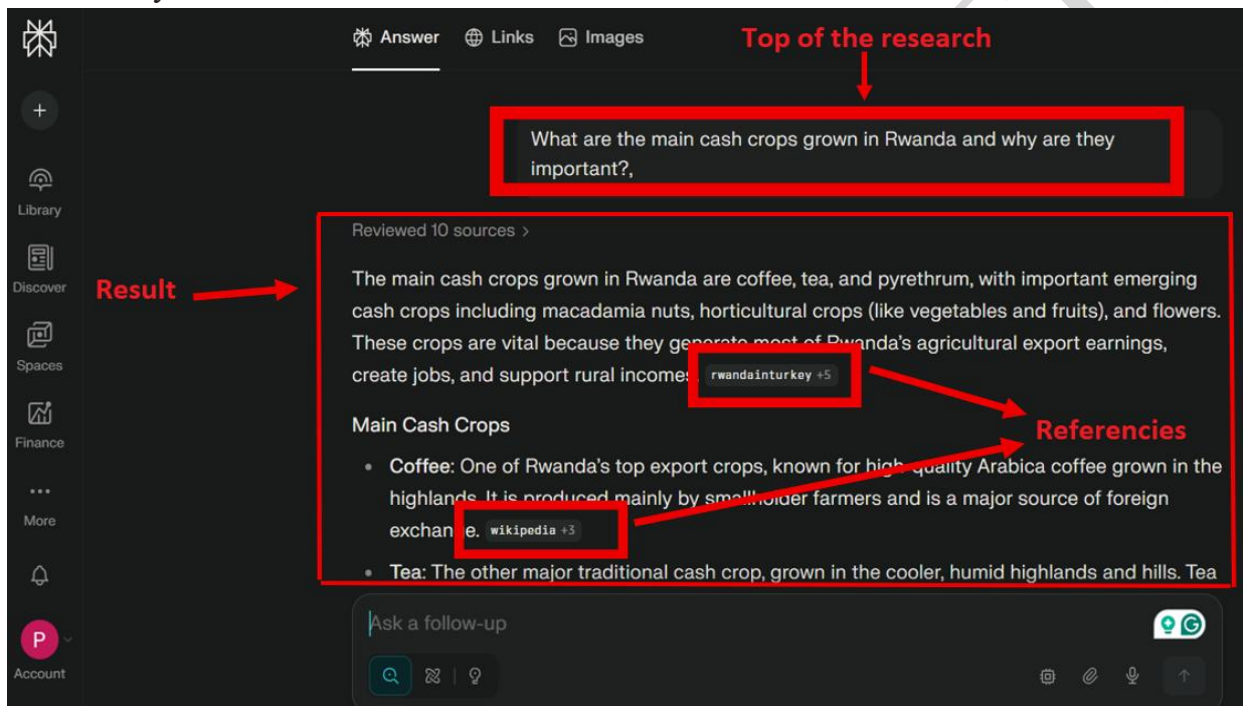
- **Consensus:** It is a tool that searches through scientific research papers to give you evidence-based answers.
- **ChatPDF:** Allows you to upload a long PDF document like a chapter from your Geography textbook and ask it questions.

Follow these steps to use Perplexity AI

Step 1: Open **Perplexity AI**.

Step 2: Type some search terms such as "What are the main cash crops grown in Rwanda, and why are they important?"

Notice how AI provides a summary and includes small numbers as reference like [1], [2] that link directly to the websites it used.



Picture 7. 5: Result of searched topic in perplexity AI tool

7.4.2. AI quiz generation

Studying is more effective when you test yourself. AI tools can now take lesson notes and automatically turn them into practice quizzes.

Here are some AI tools for generating quizzes:

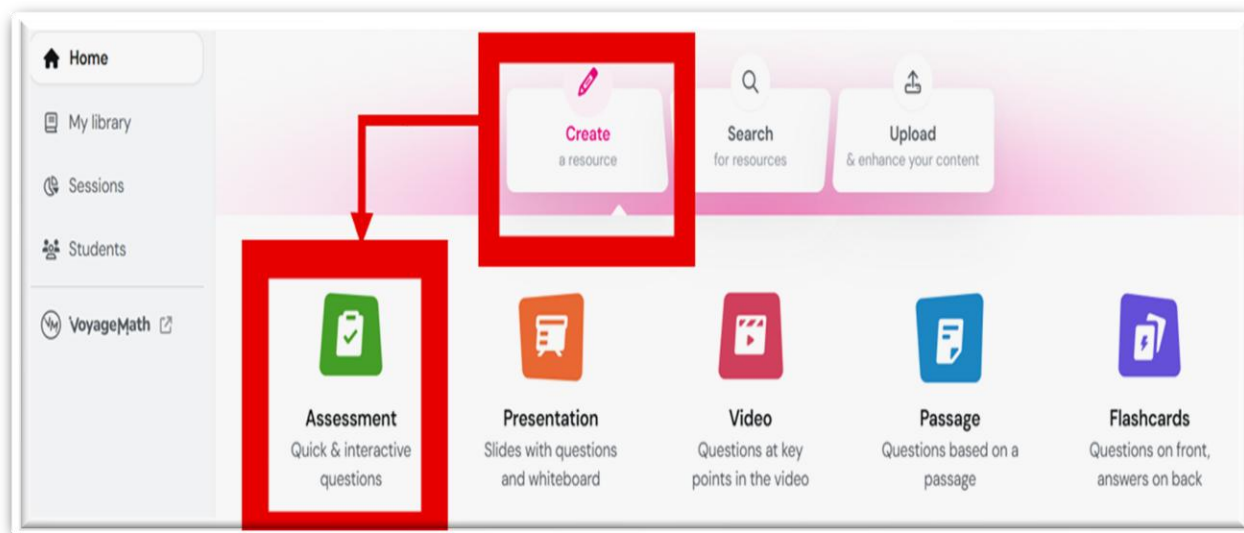
- **Quizizz AI:** You can paste text or a link to a YouTube video and it will instantly create a multiple-choice quiz.
- **Conker:** A tool specifically designed for students and teachers to create quick assessments on any topic.

Follow these steps to create a self study quiz with Wayground AI

Step 1. Sign In and Start:

- Go to [Wayground.com](https://wayground.com) then log in with your school Google account.

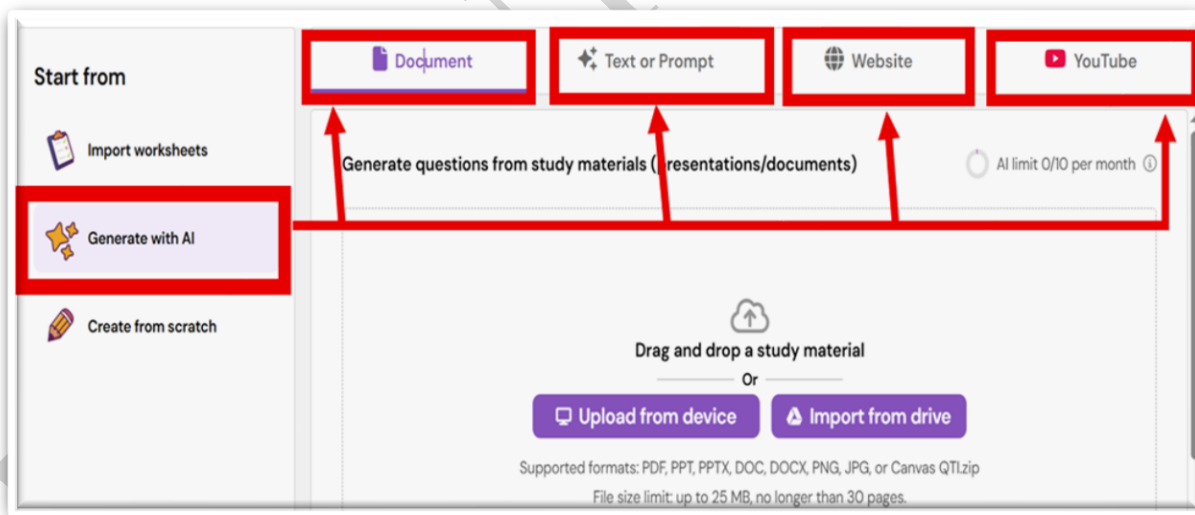
- Click the **Create** button on the left menu and select **Assessment** or **Quiz**.



Picture 7. 6: Starting the assessment creation.

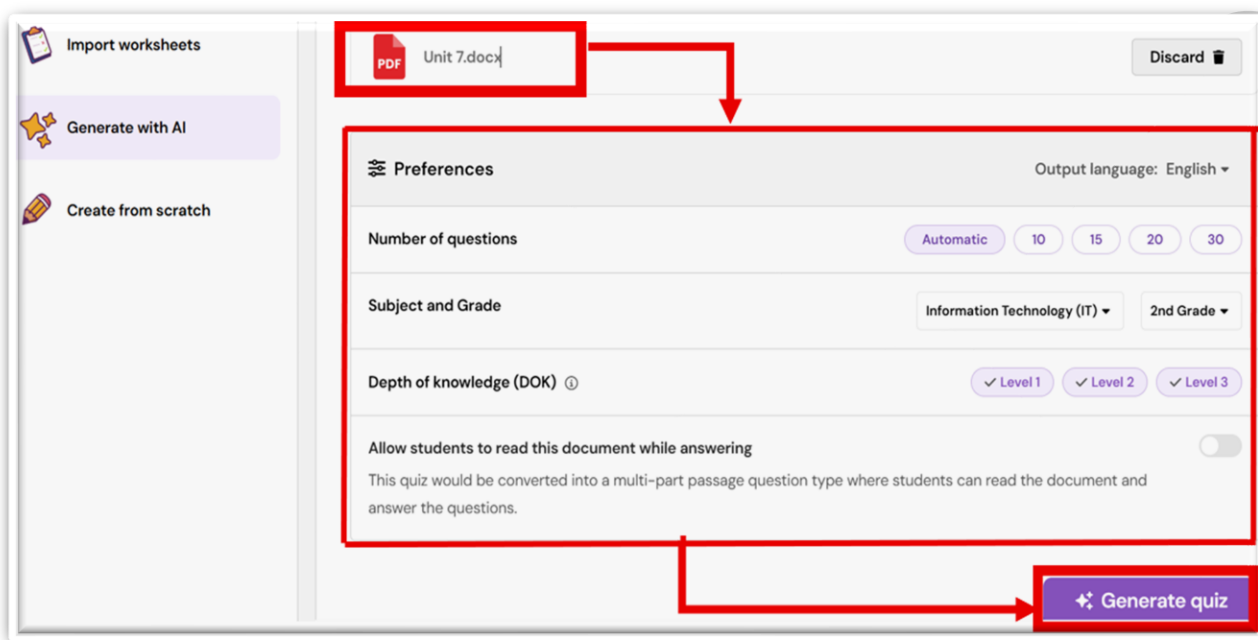
Step 2. Choose your AI method. You don't have to type questions manually. Choose one of these AI options after clicking on **Generate with AI**:

- **Text or prompt:** Type or paste a paragraph from any subject.
- **Upload a document:** Upload a PDF or a photo of your handwritten notes.
- **YouTube link:** Paste a link of an educational video such as a video on the water Cycle.



Picture 7. 7: Choosing the content to assess

Step 3. Set your preferences: Tell the AI tool how many questions you want then select your Grade Level (Senior 2) and click **Generate**.



Picture 7. 8: Uploading, setting and Generating Quiz

Step 4. Review and edit: Check the facts by reading through the questions and if there are mistakes, click **Edit** to fix the text or the correct answer.

Step 5. Play and practice: Click **Publish** in the top right corner. To study by yourself, click **Practice** or **Solo play**.

Application activity 7.4

- 1) Using Perplexity AI, find out how AI is being used to help farmers in Rwanda. Copy one fact and include the name of the website where the information was found.
- 2) Go to Conker or Quizizz AI and generate a 3-question quiz about the water cycle.
- 3) Explain why it is better to use a tool like Perplexity for a school report instead of only using a regular chatbot like ChatGPT.

7.5. End unit assessment

7.5.1. Project activity

Project Title: Creating text and video using AI tools

Guidance:

Use AI tools to create text content and a movie on one unit of any subject.

- Choose one topic in any subject and use ChatGPT or Gemini to generate short notes in simple language.
- Read the AI generated text carefully and rewrite it in your own words to ensure it is your own work and not copied.
- To illustrate your text better use an AI design tool (such as Canva) to create images to add to your text

